# Chemical Bonding

It helps making water “liquid”(Hydrogen Bonding)



Can you think how life would be if water was in gaseous state?

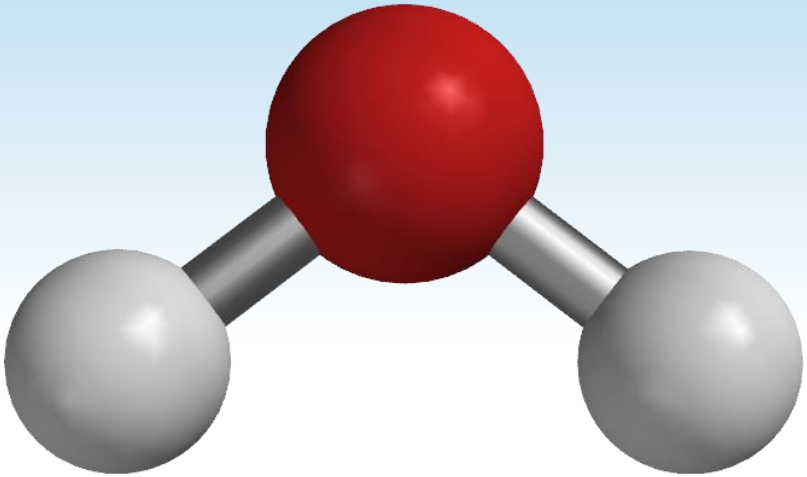
It makes the existence of air possible

(Van der waals force of attraction between different Constituent of air)

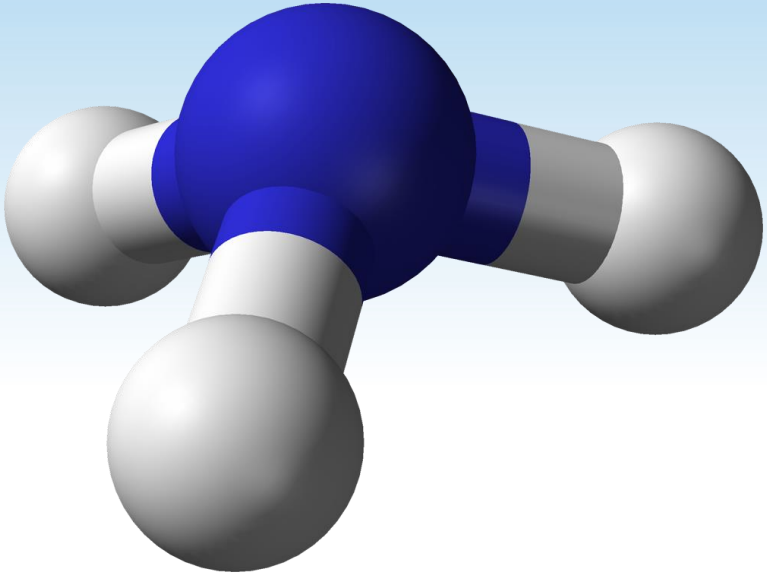


Can you consider life without atmosphere?

# Elements in Combined state



$H_2O$ Molecule



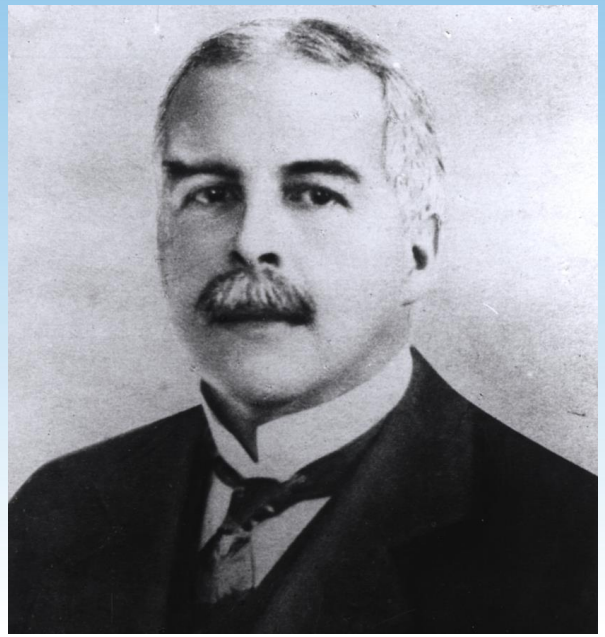
$NH_3$ Molecule

# Elements in free state

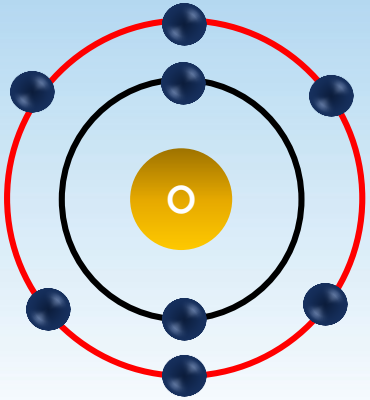
| | |
|---|---|
| 2 He | $1s^2$ |
| 10 Ne | $1s^2\ 2s^2\ 2p^6$ |
| 18 Ar | $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ |
| 36 Kr | $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6$ |
| 54 Xe | $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 5s^2\ 5p^6$ |
| 86 Rn | $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 4f^{14}\ 5s^2\ 5p^6\ 5d^{10}\ 6s^2\ 6p^6$ |

Stable electronic configuration

# Lewis Symbols



G.N. Lewis

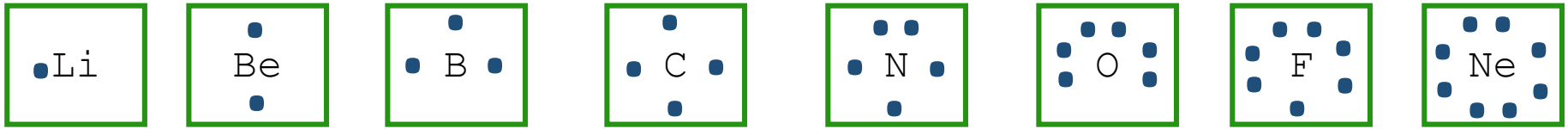



outer shell electrons

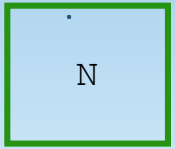
or

valence electrons

Take part in Chemical reaction



# Significance of Lewis Symbols

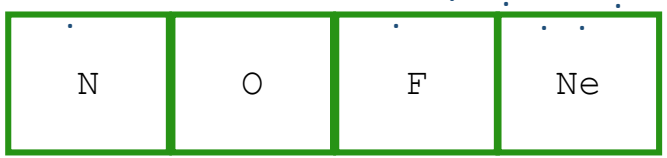



Number of dots = number of valence electron

It helps to calculate the common or group valence




Group valence = Number of valence electrons




Group valence = **8** – Number of valence electrons

**Q.** Draw the Lewis structure for the molecule $NO_2^-$ ?

**Sol.** Step-1: skeletal structure of $NO_2^-$ : O N O

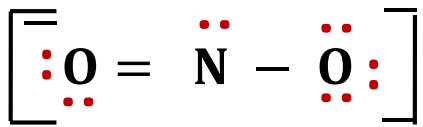
Step-2 :



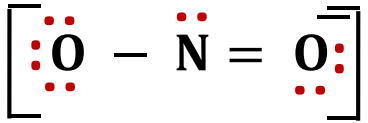
Step-2 :



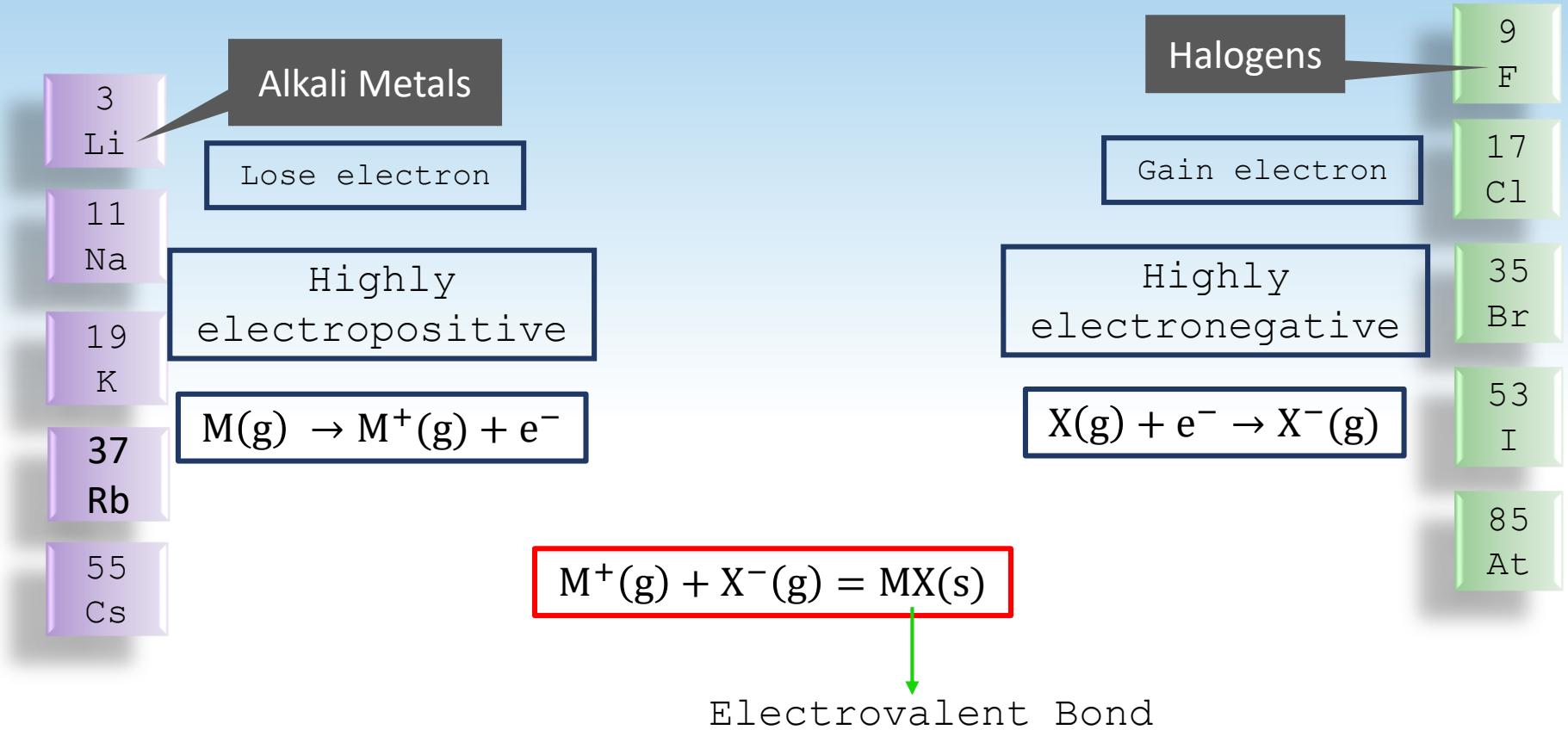
Step-3 :



OR

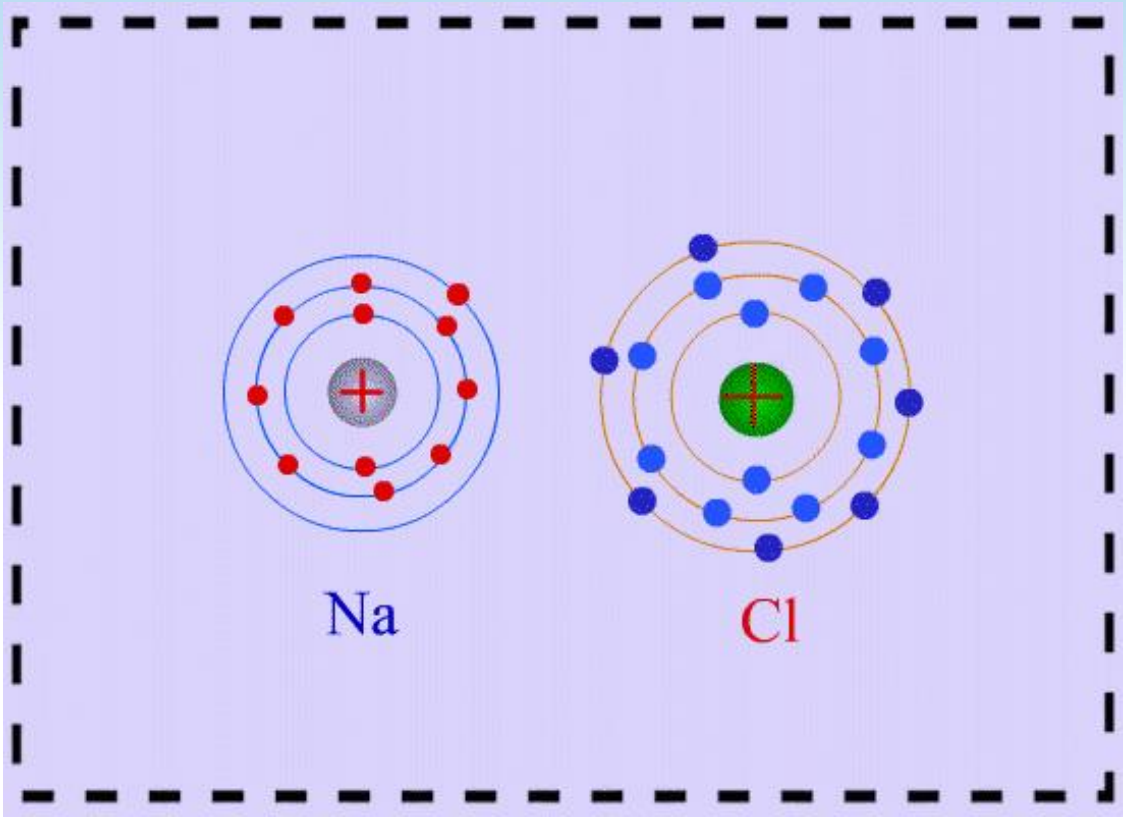


# Kössel Approach to chemical bond



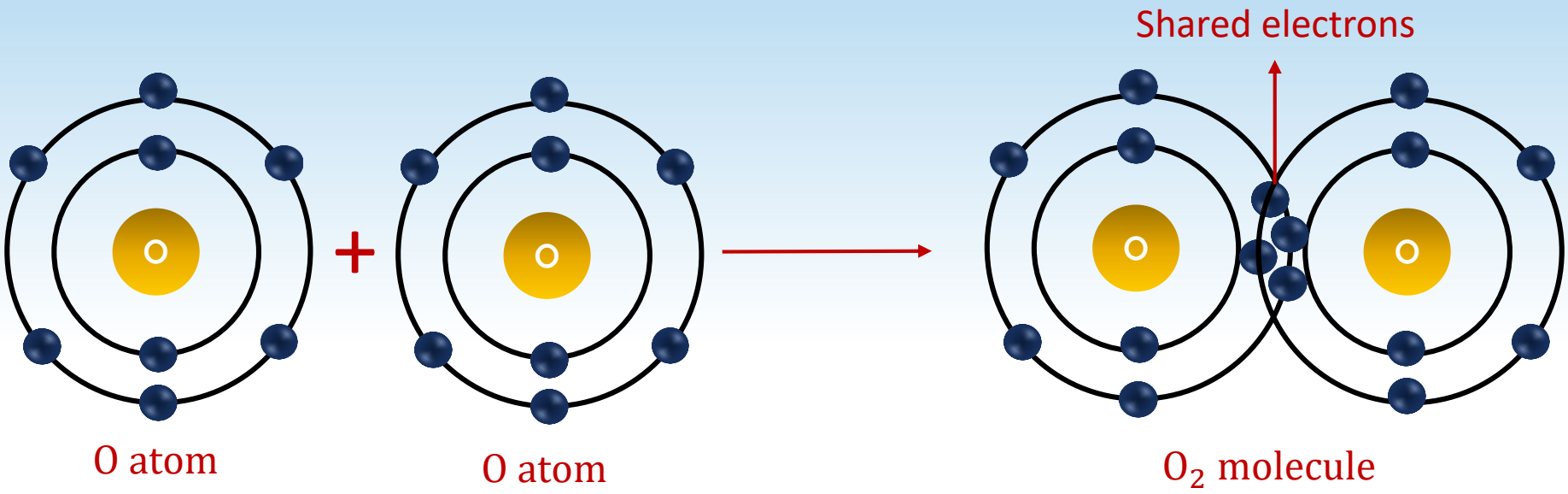
# Octet Rule

Atoms can combine  by transfer of valence electrons from one atom to another (gaining or losing)



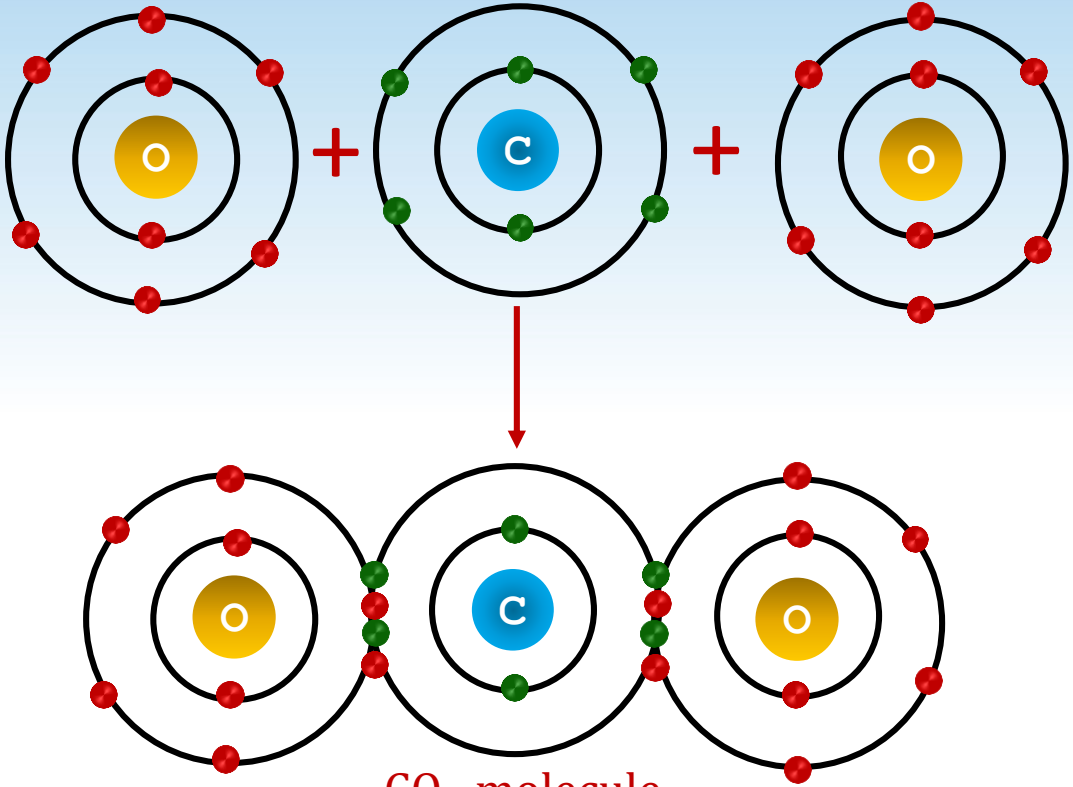
# Octet Rule

Atoms can combine by sharing of electrons



# Octet Rule

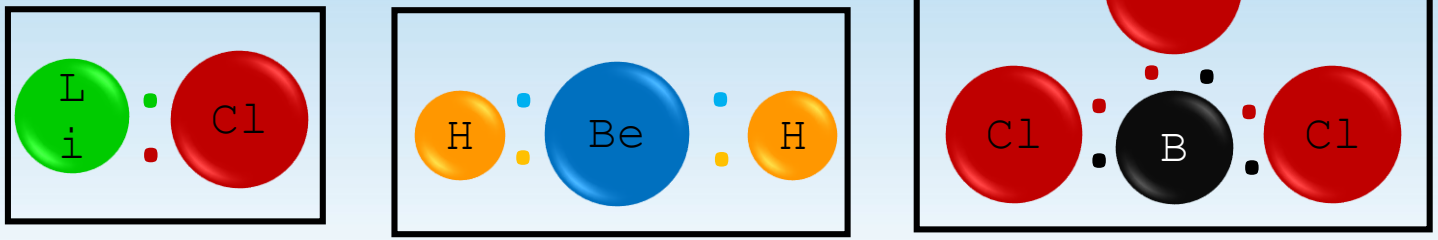
Atoms can combine by sharing of electrons




$CO_2$ molecule

# Limitations of Octet Rule

The incomplete octet of the central atom



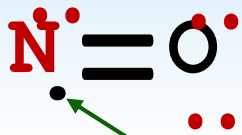

Valence electrons : 1 2 3

- In some compounds, the number of electrons surrounding the central atom is less than eight
- This is specially the case with elements having less than four valence electrons

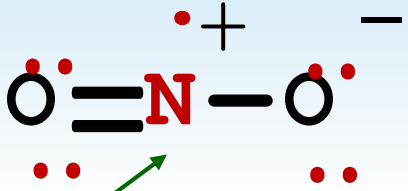
# Limitations of Octet Rule

Odd-electron molecules

Nitric oxide (NO)



Nitrous oxide ($NO_2$)

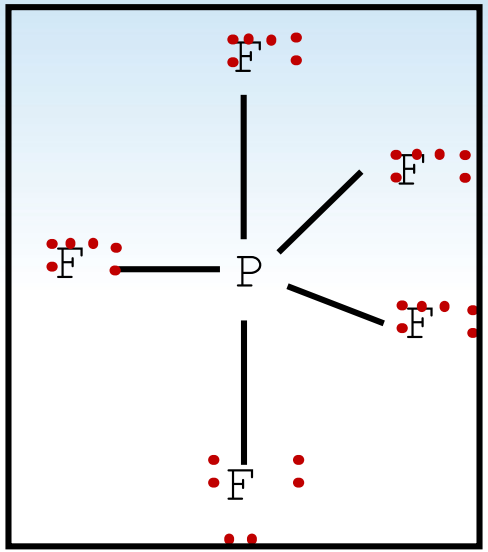


Nitrogen has Odd number of electrons

# Limitations of Octet Rule

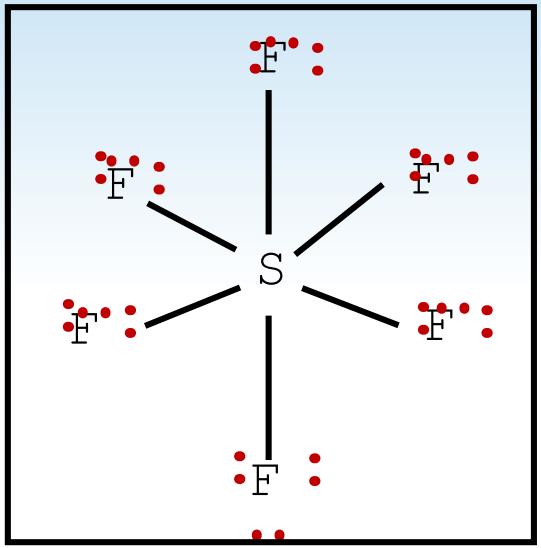
## The expanded octet

Elements beyond third period of periodic table have apart from 3s and 3p orbitals, 3d orbitals also available for bonding



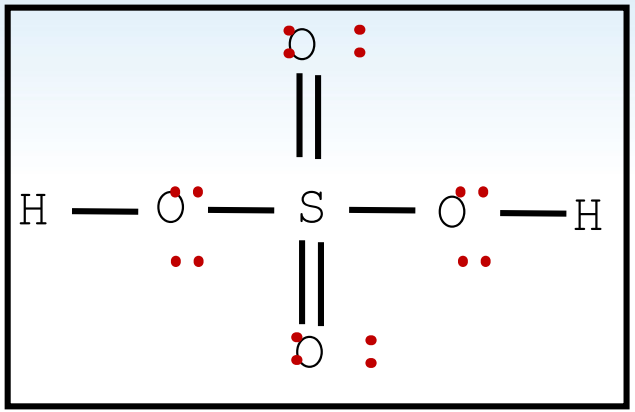

$PF_5$

10 electrons around the P atom




$SF_6$

12 electrons around the S atom




$H_2SO_4$

12 electrons around the S atom

# Limitations of Octet Rule

## The expanded octet

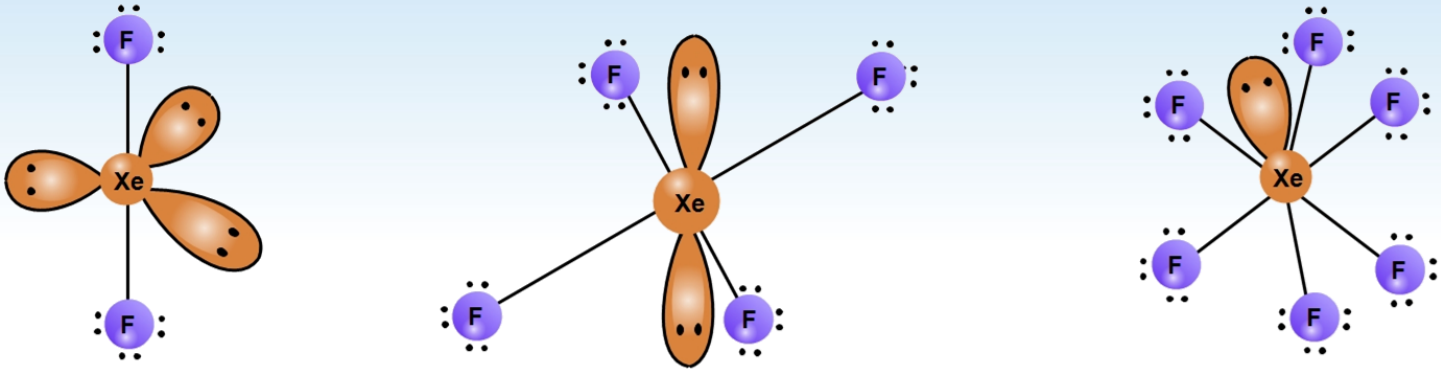
Sulphur also forms compounds in which octet rule is obeyed



# Other drawbacks of Octet Theory

Octet rule is based upon the chemical inertness of noble gases

However, some noble gases also combine with oxygen and fluorine to form a number of compounds

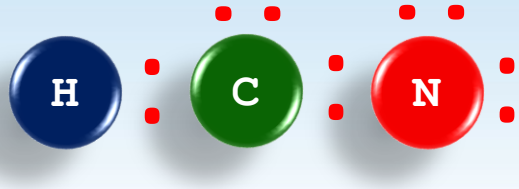



This theory does not account for the shape, stability and energy of the molecule

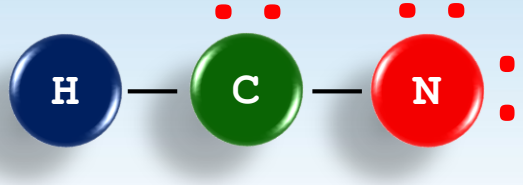
**Q.** Draw the Lewis dot structure of HCN molecule?

**Sol.** Step-1: skeletal structure of HCN : H C N

Step-2:



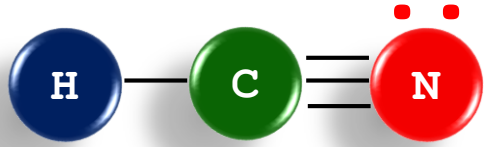

OR




Step-3:




OR



# Ionic Bond

An ionic bond is formed by transfer of electrons from a metal to a non-metal in order to obtain a full valence shell for both atoms.



# Ionic Bond

## Key Factors for the formation of Ionic bond

1. Ease of formation of positive or negative ions from the respective neutral atoms

2. Arrangement of positive or negative ions in the ionic solid

$$M(g) \rightarrow M^{+}(g) + e^{-};$$

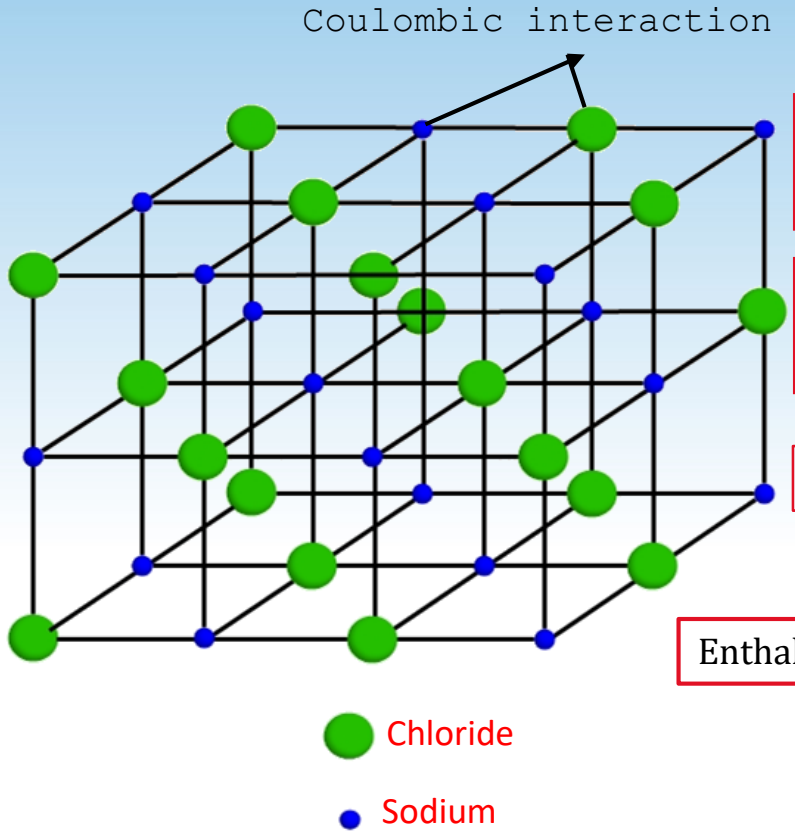
Ionization energy ⟶ (Always endothermic)

$$X(g) + e^{-} \rightarrow X^{-}(g);$$

Electron gain enthalpy ⟶ (Exothermic or Endothermic)

$$M^{+} + X^{-}(g) + e^{-} \rightarrow MX(s)$$

# Ionic Bond




$Na \rightarrow Na^{+} + e^{-}$

(Ionization enthalpy $= 495.8$ KJ $mol^{-}$)

$Cl + e^{-} \rightarrow Cl^{-}$

(Electron gain enthalpy $= -348.7$ KJ $mol^{-1}$)

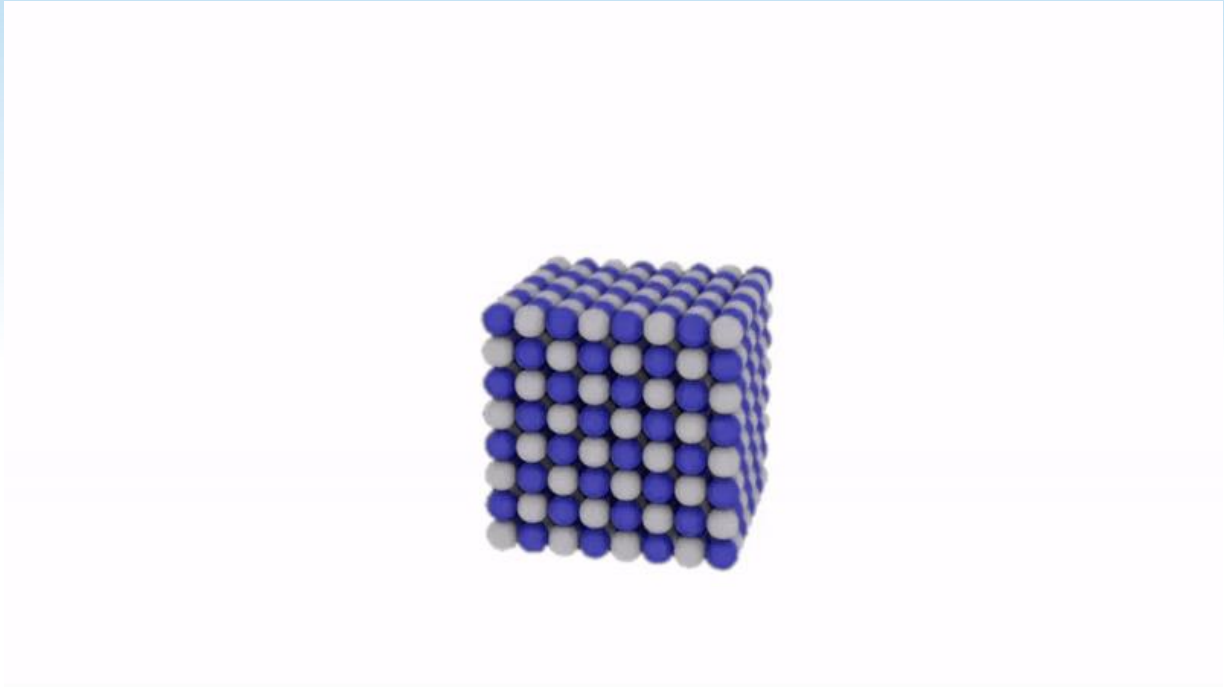
Ionization enthalpy + Electron gain enthalpy $= 147.1$ KJ $mol^{-1}$

Enthalpy of lattice formation of $NaCl(s) = -788$KJ $mol^{-1}$

Crystal structure gets stabilized due to the energy released in the formation of the crystal lattice.

# Lattice Enthalpy

Energy required to completely separate one mole of a solid ionic compound into gaseous constituent



**Q.** Write the favorable factors for the formation of ionic bond

**Sol.** i. Low ionization enthalpy of the metal atom

ii. High electron gain enthalpy of the non-metal atom

iii. High lattice enthalpy of the compound formed

# Covalent Bond

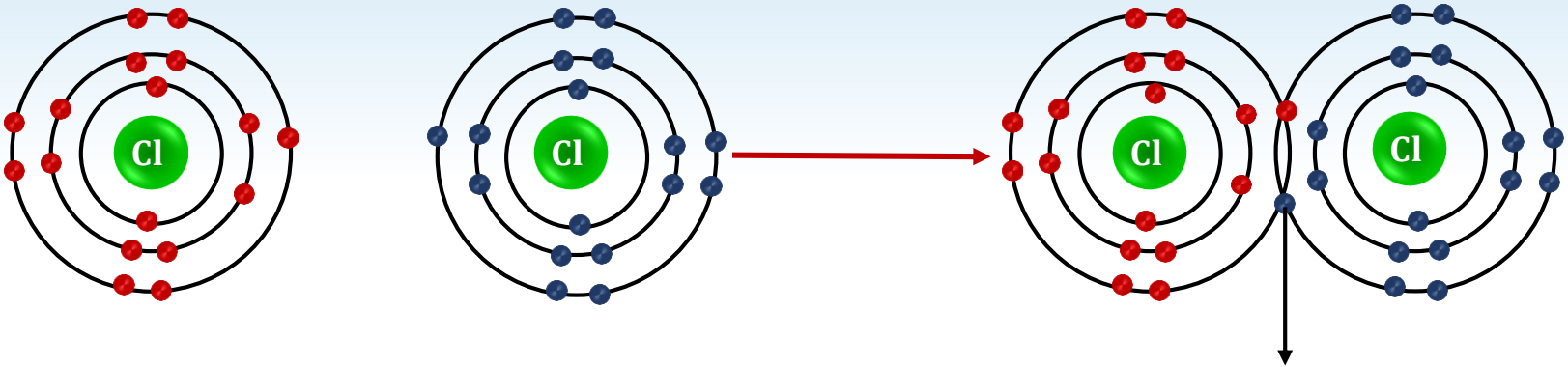
It involves sharing of electron pairs between atoms

Cl atom= $[Ne]3s^2 3p^5$ one electron short of the argon Configuration.

Chlorine atom, Cl

Chlorine atom, Cl

Chlorine molecule, $Cl_2$ (stable)

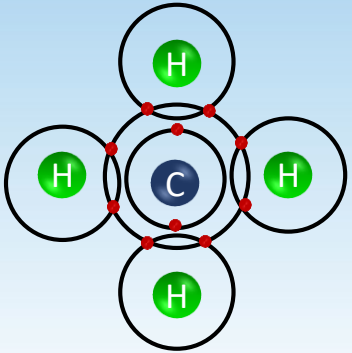


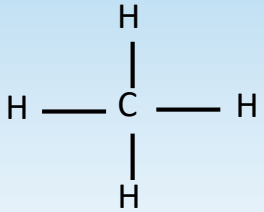

If two atoms share one pair of electron, the covalent bond between them is called single bond

# Covalent Bond

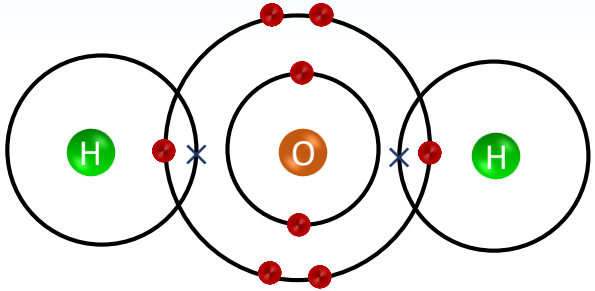
Conditions:

1. Each bond is formed as a result of sharing of an electron pair between the atoms



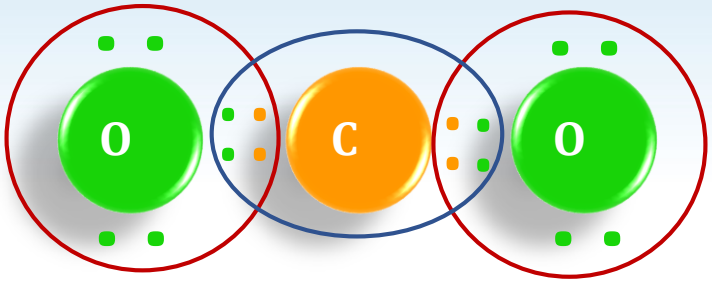

2. Each combining atom contribute at least one electron to the shared pair



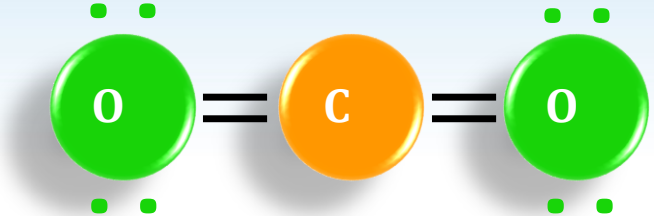

3. The combining atoms attain the outer shell noble gas configurations as a result of the sharing of electrons.

# Covalent Bond

If two atoms share two pairs of electron, the covalent bond between them is called a double bond

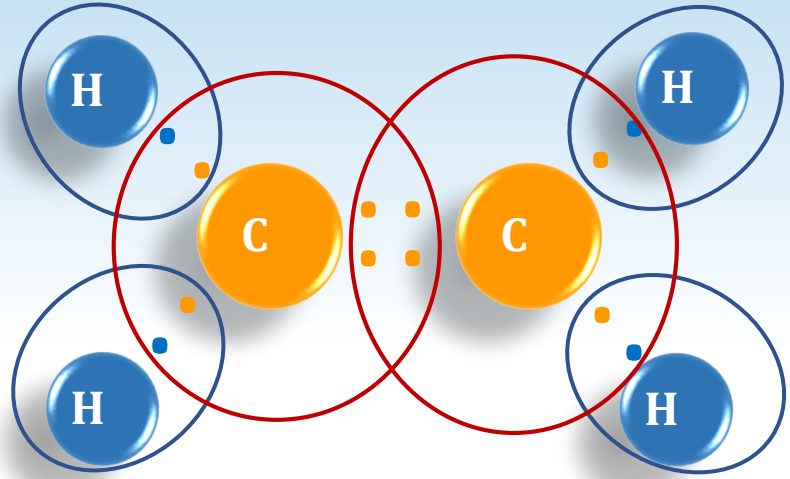



OR

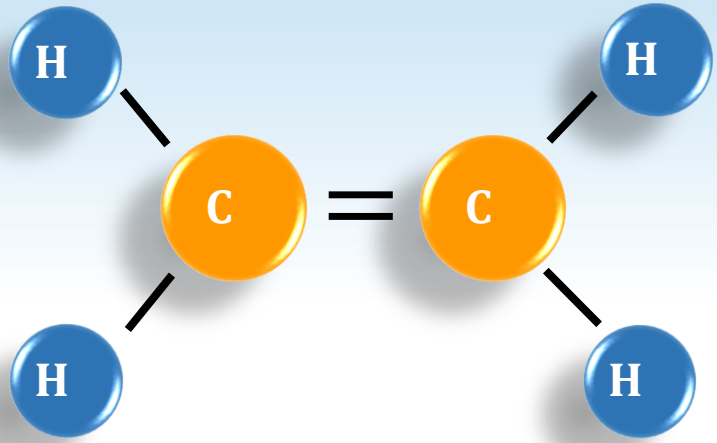


# Covalent Bond

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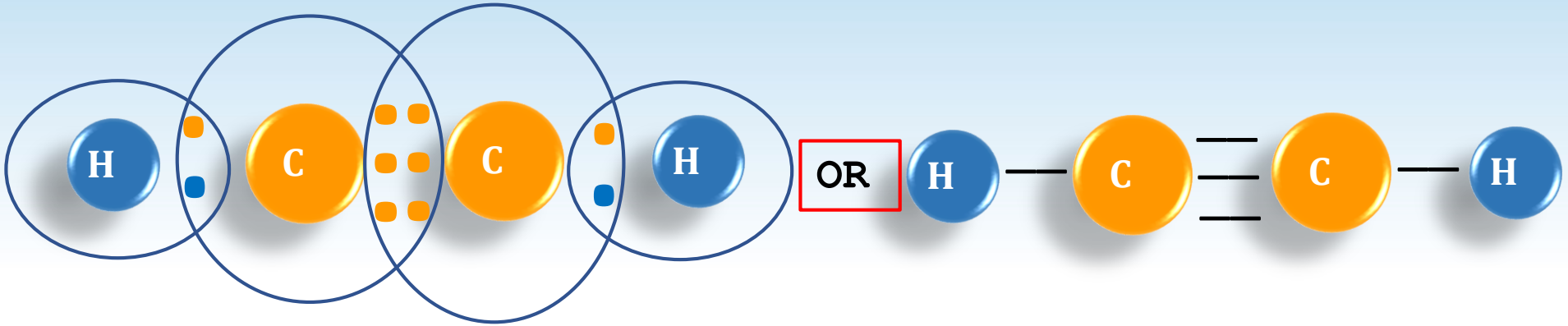



OR



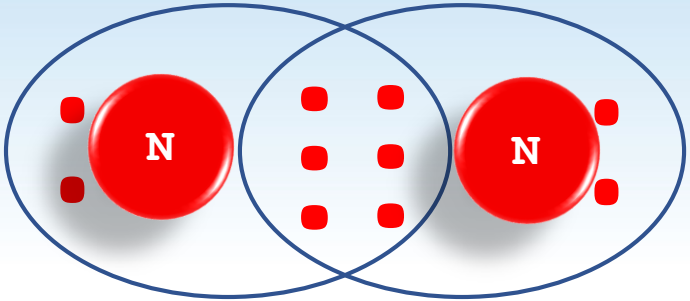
# Covalent Bond

A triple bond is formed when combining atom share three electron pairs

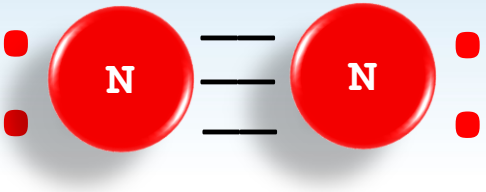


# Covalent Bond

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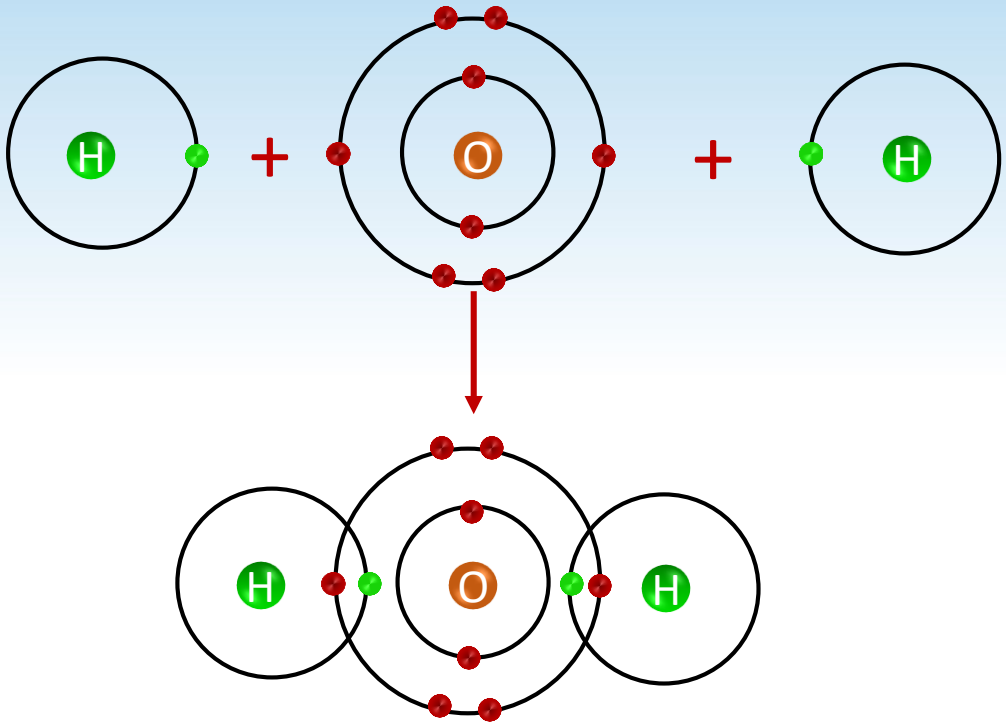



OR



**Q.** How do atoms form covalent bond?

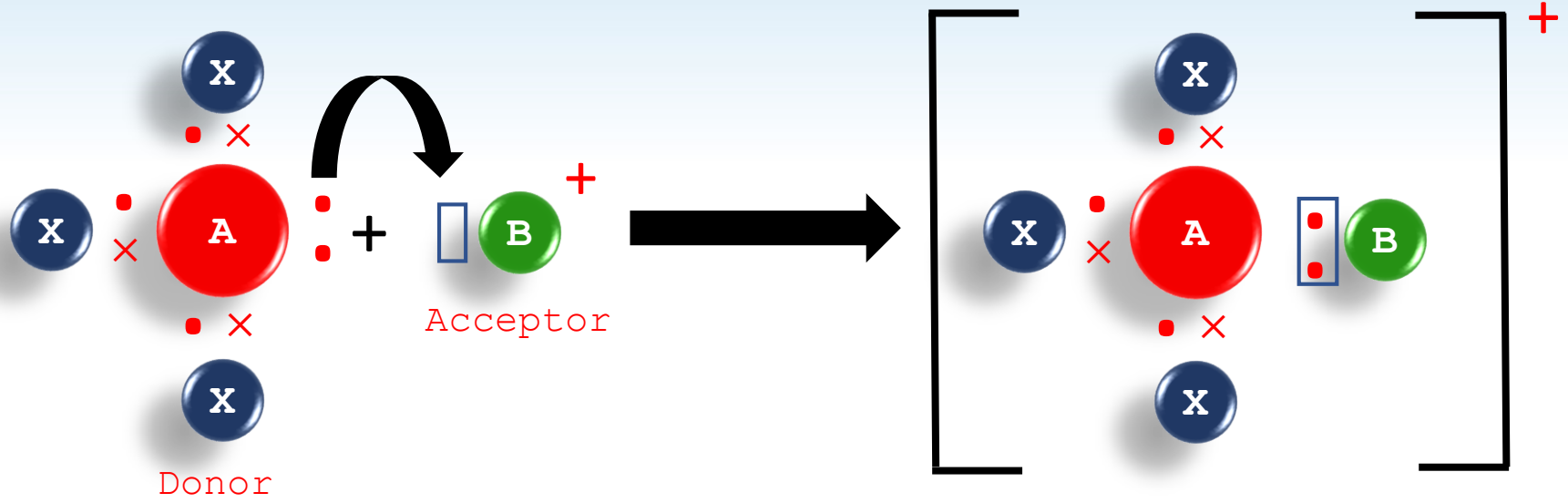
**Sol.** Covalent bond is formed by sharing of electrons.



# Co-ordinate Bond

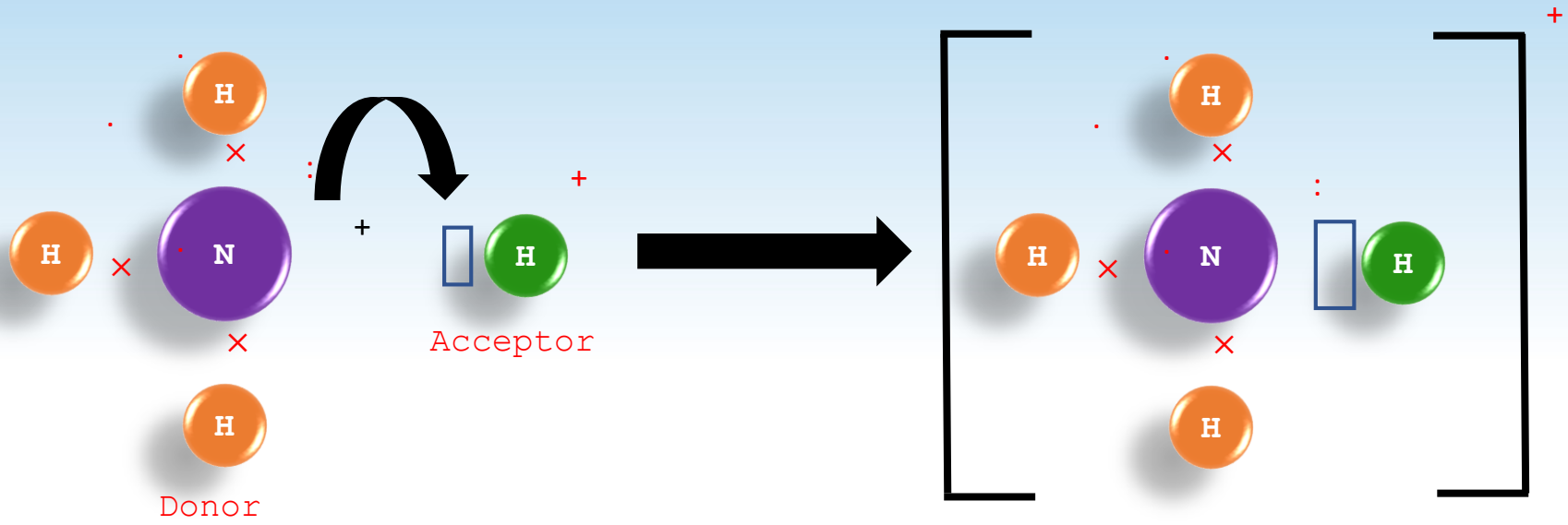
Also known as dative bond

It is  a kind of 2-center, 2-electron covalent bond in which the two electrons derive from the same atom



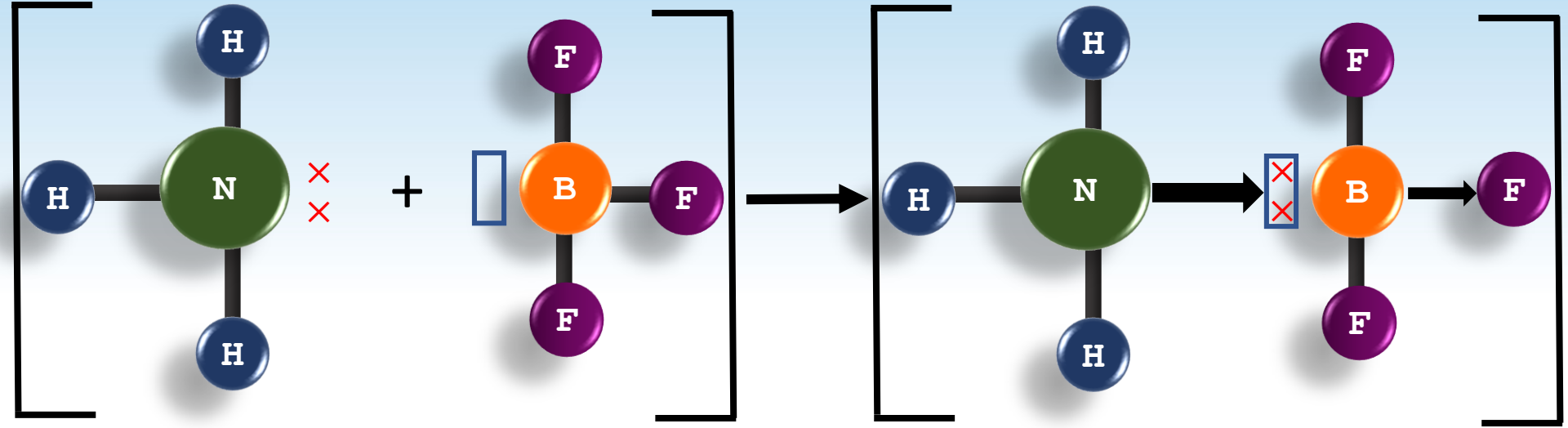
# Co-ordinate Bond

Nitrogen has Atomic number of N = 7= $1s^2\ 2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^1$



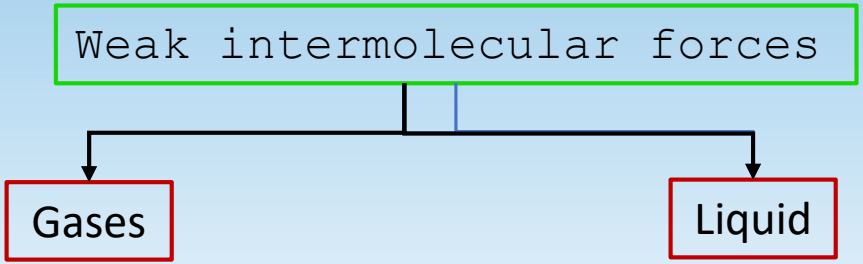
# Co-ordinate Bond

Ammonia Boron Trifluoride



# Co-ordinate Bond

Properties of compounds with Co-ordinate bond:




- M.P. or B.P. = Ionic compound > Co-ordinate bond > Covalent bond
- Only soluble in non-polar solvents like benzene
- Exhibit isomerism due to directional nature of bond

## Summary

- The Lewis dot symbols show the valence electrons of the atoms of a given element and Lewis dot structure show pictorial representations of bonding in molecules

- The crystal lattice is stabilized by the enthalpy of lattice formation

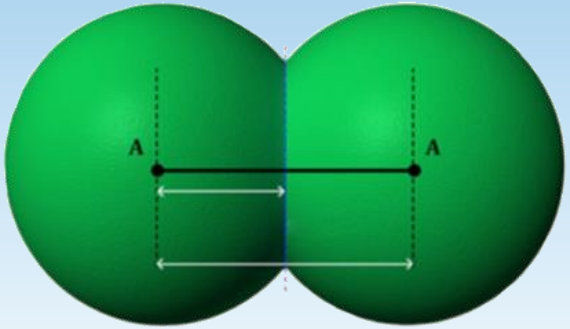
- Covalent bond is formed by sharing of an electron pair between two atoms

- An ionic bond is formed by transfer of electrons from a metal to a non-metal in order to obtain a full valence shell for both atoms.

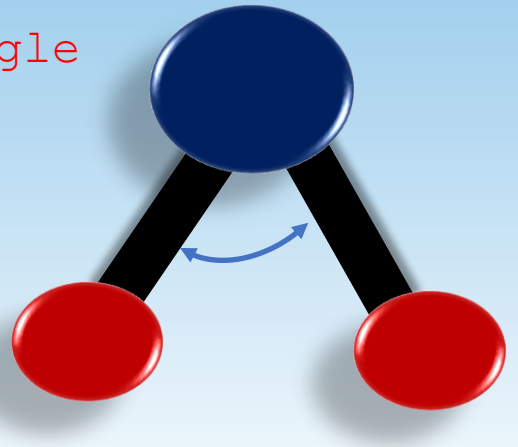
- A coordinate bond is a type of covalent bond where both of the electrons that form the bond originate from the same atom

# Bond Parameters

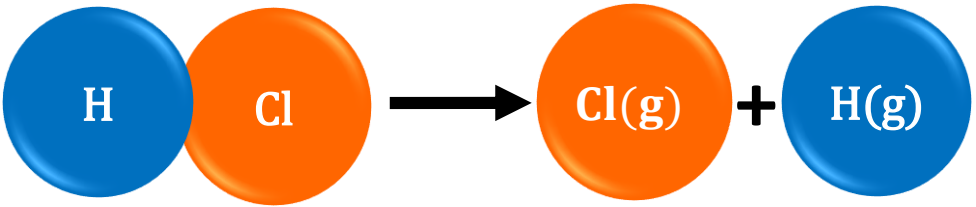
## Bond length



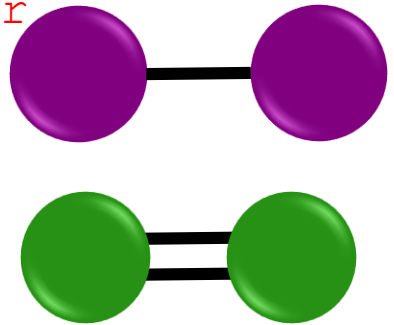

## Bond Angle



## Bond Enthalpy




## Bond Order



# Bond Length

Equilibrium distance between the nuclei of two bonded atoms in a molecule

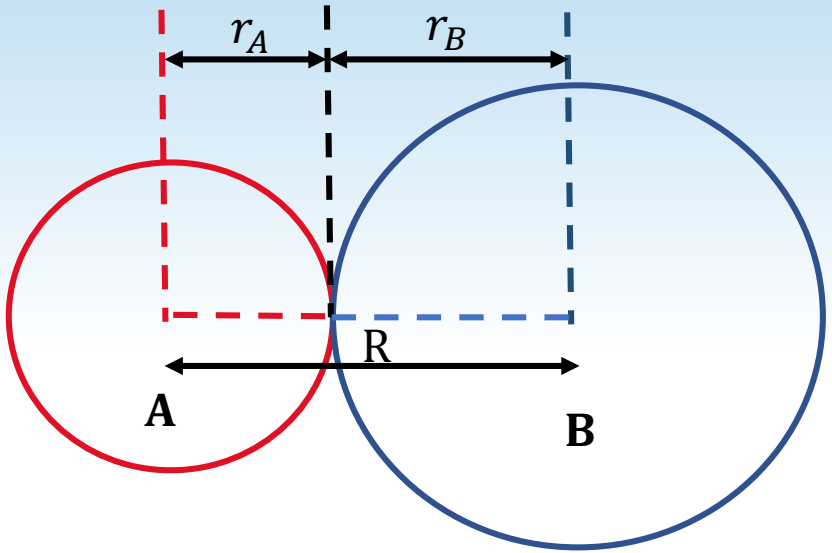
Measured by:

- Spectroscopy
- X-ray diffraction
- Electron-diffraction

**Bond Length**

Covalent radius

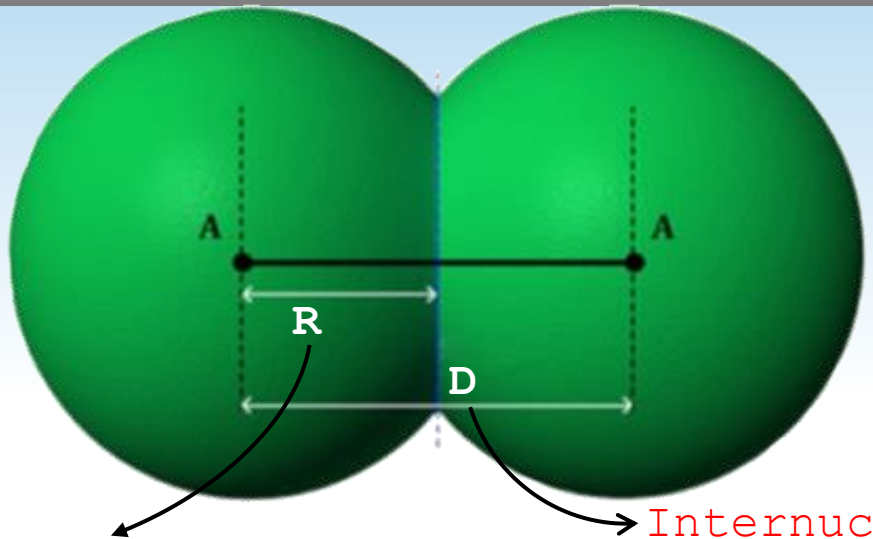
van der waals radius




$$R = r_A + r_B$$

# Covalent radius

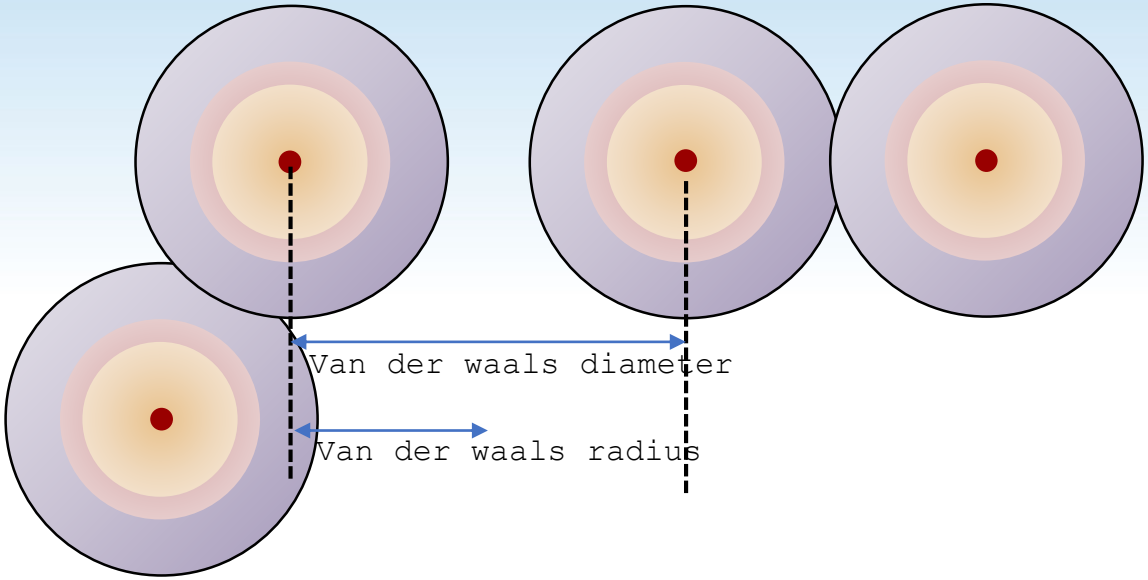
Half of the internuclear distance between the nuclei of two bonded atoms




$$R = \frac{D}{2}$$

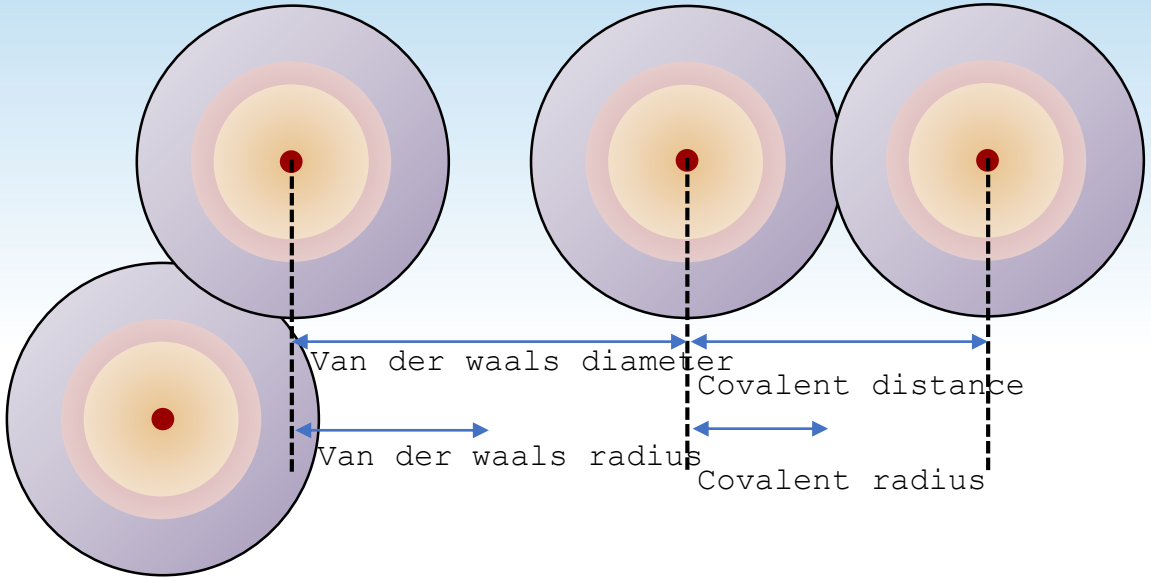
# Van der Waals radius

Half of the distance between two nearest similar atoms in separate molecules in a solid



# Covalent v/s van der Waal radius

Covalent and van der waals radii in a chlorine molecule

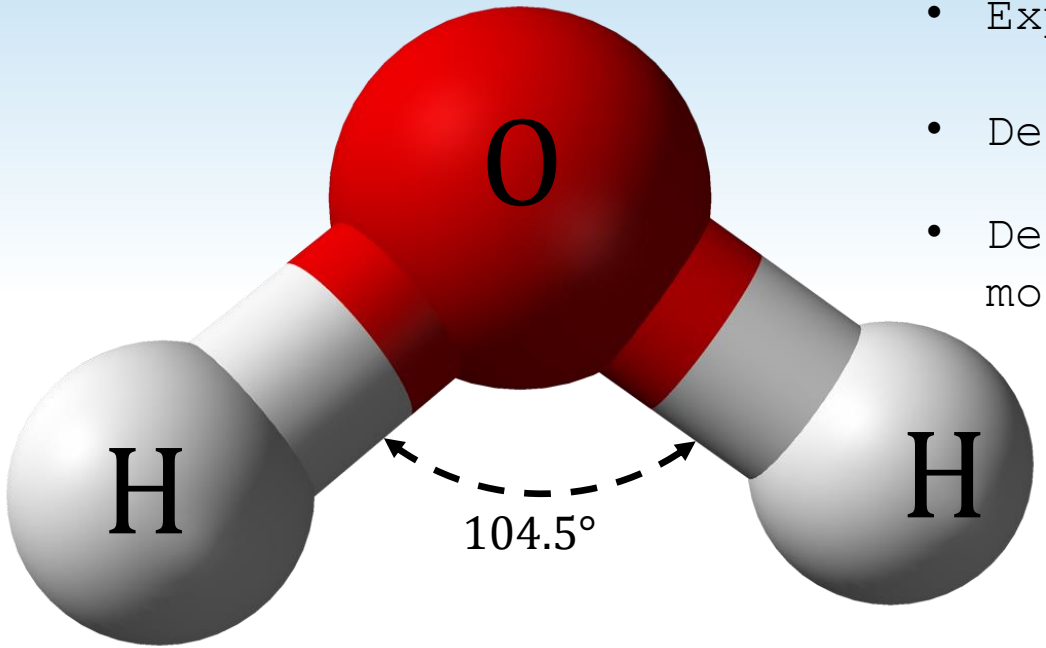


# Bond Length

| Molecule | Bond length(pm) |
|---|---|
| $H_2(H-H)$ | 74 |
| $F_2(F-F)$ | 144 |
| $Cl_2(Cl-Cl)$ | 199 |
| $Br_2(Br-Br)$ | 228 |
| $I_2(I-I)$ | 267 |
| $N_2(N\ N)$ | 109 |
| $O_2(O=O)$ | 121 |
| $HF(H-F)$ | 92 |
| $HCl(H-Cl)$ | 127 |
| $HBr(H-Br)$ | 141 |
| $HI(H-I)$ | 160 |

# Bond Angle

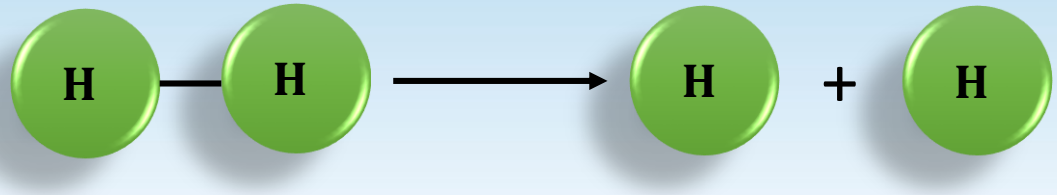
Angle between the orbitals containing bonding electron pairs around the central atom in a molecule/ complex ion




- Expressed in degree
- Determined by spectroscopy
- Determines the shape of a molecule

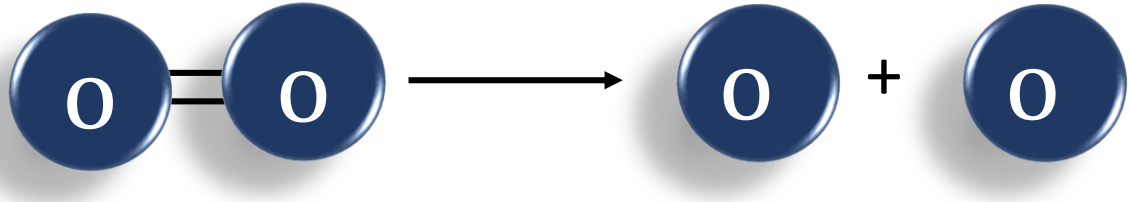
# Bond Enthalpy

Energy required to break one mole of bonds between two atoms in gaseous state

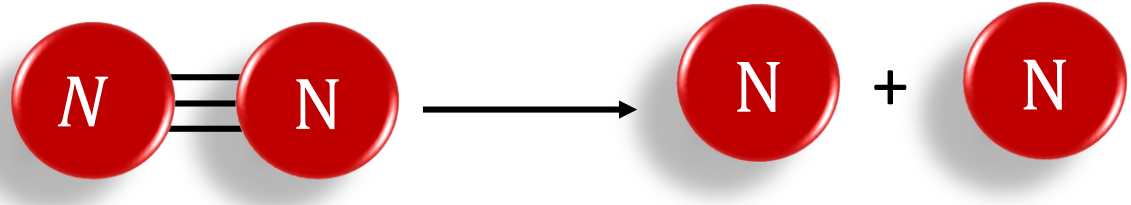



$\Delta_a H^{\boldsymbol{o}} = 435.8\ KJmol^{-1}$

For molecules containing multiple bonds,



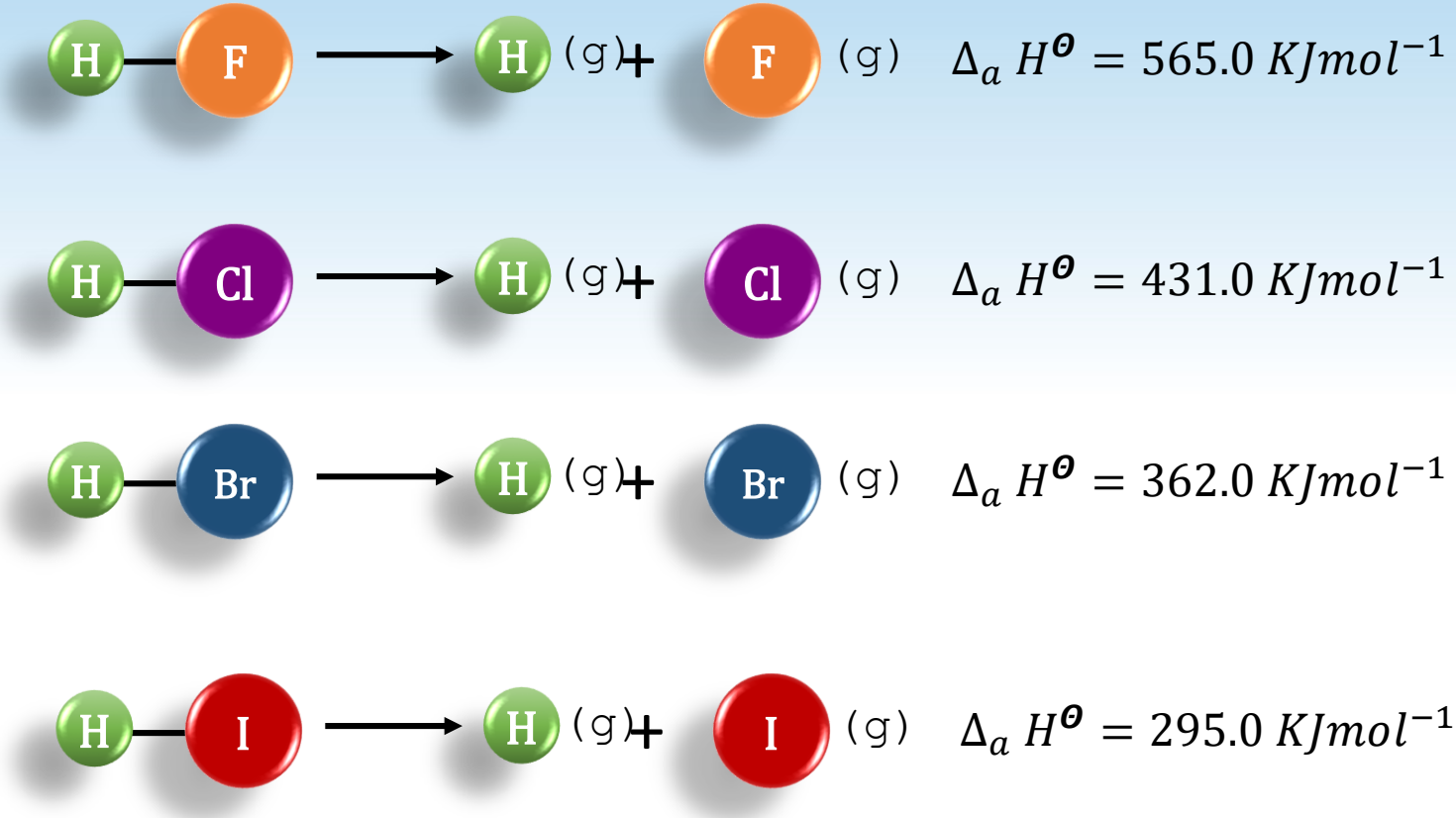

$\Delta_a H^{\boldsymbol{o}} = 498\ KJmol^{-1}$




$\Delta_a H^{\boldsymbol{o}} = 946.0\ KJmol^{-1}$

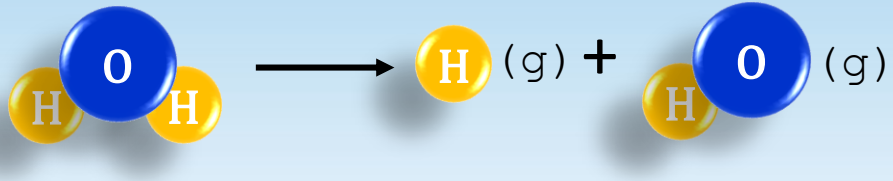
# Bond Enthalpy

For a heteronuclear diatomic molecules

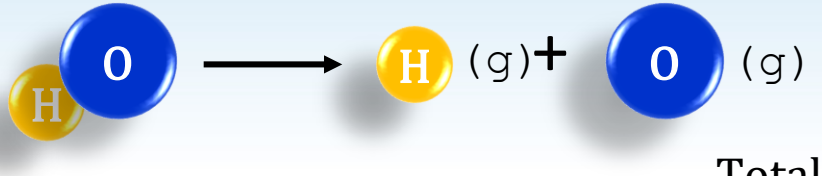


# Bond Enthalpy

For polyatomic molecules, measurement of bond strength is complicated




$$\Delta_a H_1^{\boldsymbol{o}} = 502\ KJmol^{-1}$$




$$\Delta_a H_2^{\boldsymbol{o}} = 427\ KJmol^{-1}$$

$$\text{Mean or average bond enthalpy} = \frac{\text{Total bond dissociation enthalpy}}{\text{Number of bonds broken}}$$

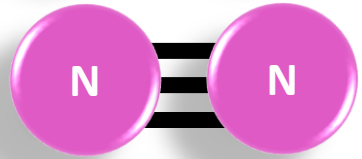
For water molecule,

$$\text{Mean or average bond enthalpy} = \frac{502 + 427}{2}$$

$$= 464.5\ \text{KJ mol}^{-1}$$

# Bond Order

Number of bonds between the two atoms in a molecule

| Molecule | Bond Order |
|---|---|
|   | 1 |
|   | 2 |
|   | 3 |
|   | 3 |

# Bond Order

Isoelectronic molecules and ions have identical bond orders

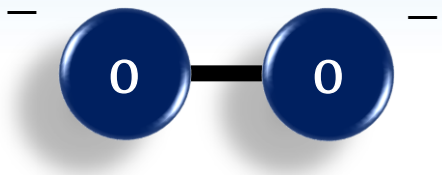
| Molecule/Ion | Total Number of electrons | Bond order |
|---|---|---|
| $F_2$ | 18 | 1 |
| $O_2^{2-}$ | 18 | |
| $N_2$ | 14 | 3 |
| CO | 14 | |

NOTE: Bond order ↑ Bond Enthalpy↑ Bond Length ↓

**Q.** What is the bond order of $O_2^{2-}$?

**Sol.**

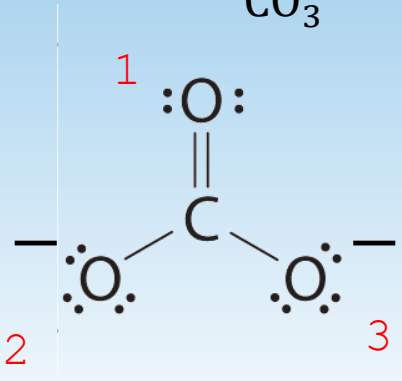


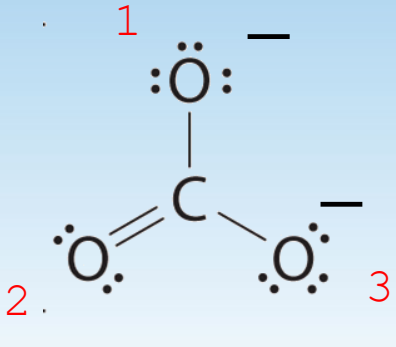

$O_2^{2-}$ has bond order equal to 1

# Resonance

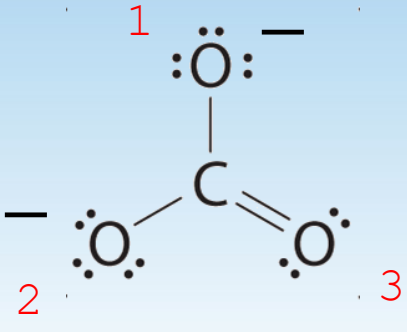
Lewis structure of $CO_3^{2-}$




A




B




C

Problems-

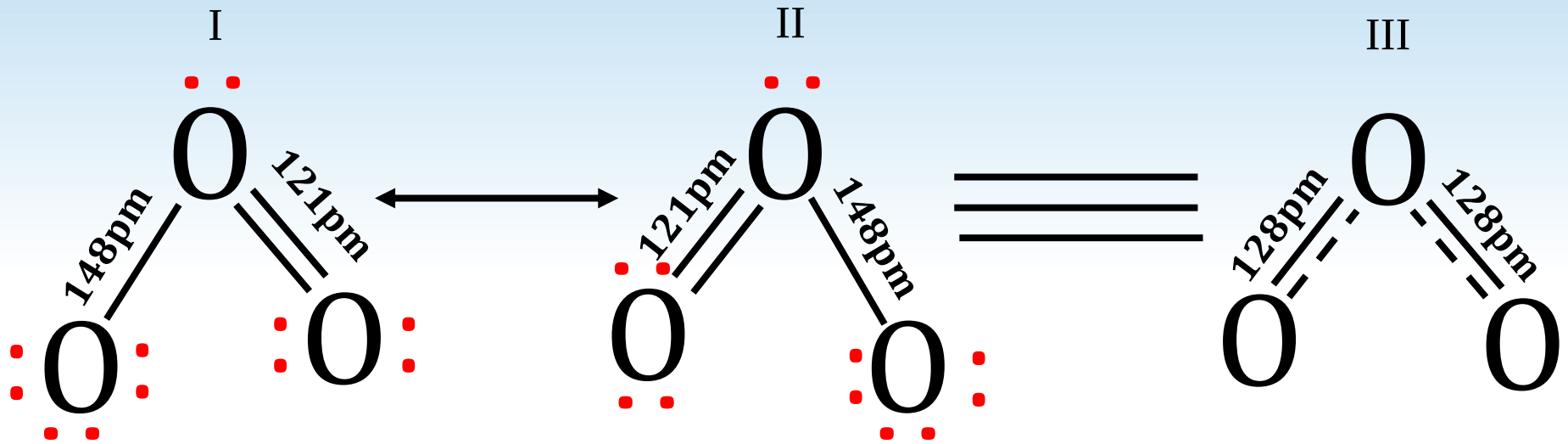
Which structure is correct?

Are all the structure correct?

# Resonance

When a single Lewis structure can not explain a molecule accurately

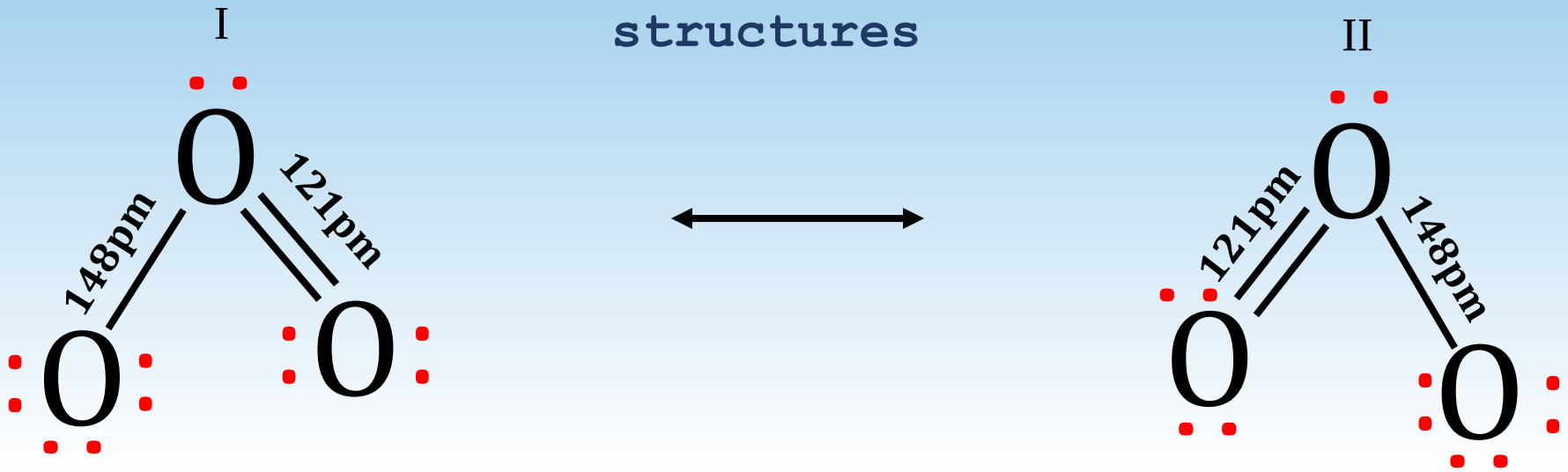
We use canonical and hybrid structure of molecules




Canonical structure or Resonance structure

Resonance Hybrid

# Characteristics of Resonance structures



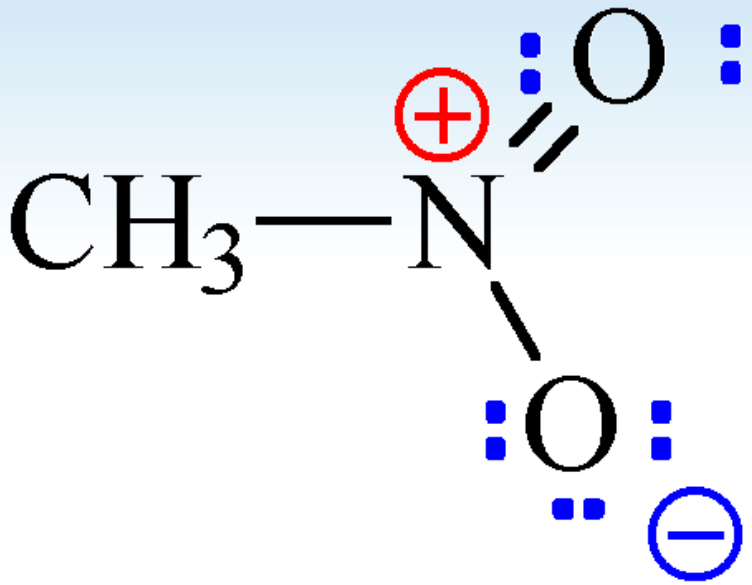

Similar energy

Similar positions of nuclei

Same bonding and non-bonding pairs of electrons

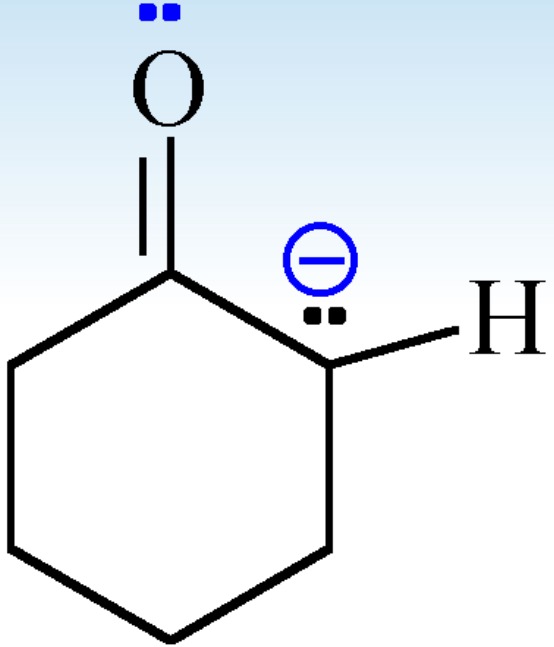
# Misconceptions of Resonance

Cannonical forms have no real existence



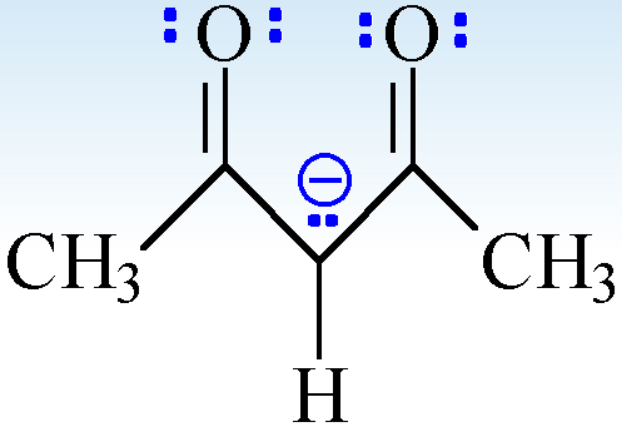
# Misconceptions of Resonance

They do not exist for a certain fraction of time in one canonical form and for other fractions of time in other canonical forms

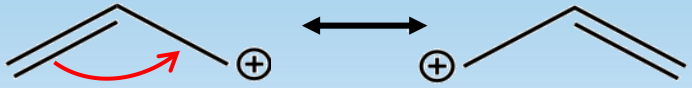


# Misconceptions of Resonance

There is no such equilibrium between the canonical forms as we have between tautomeric forms (keto and enol) in tautomerism



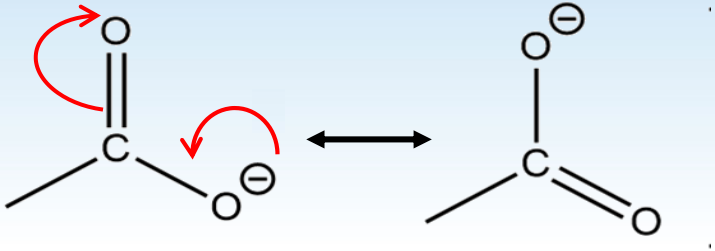
# Misconceptions of Resonance



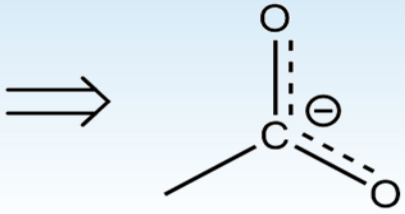
Resonating structures- Allyl carbocation



Resonance hybrid



Resonating structures- Acetate Ion



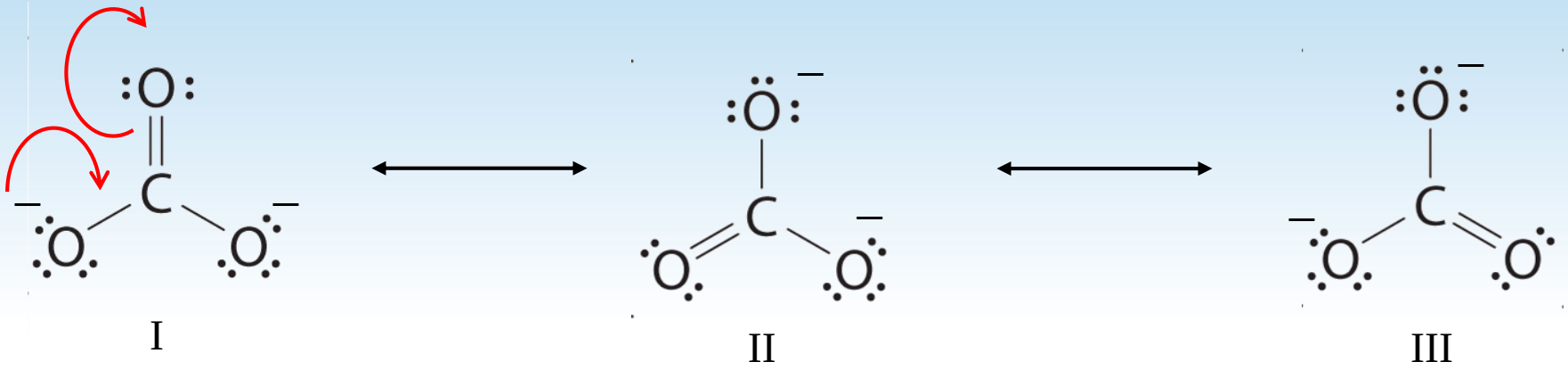
Resonance hybrid

Single hybrid structure

Cannot as such be depicted by a single structure

**Q.** Explain the structure of $CO_3^{2-}$ in terms of resonance?

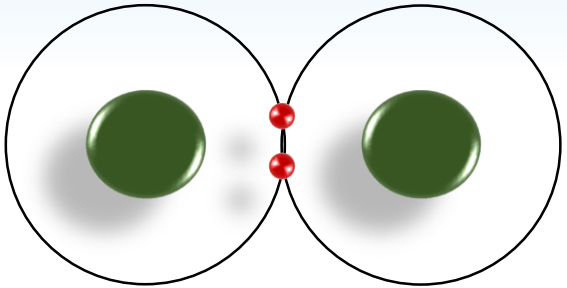
**Sol.**



# Dipole moment

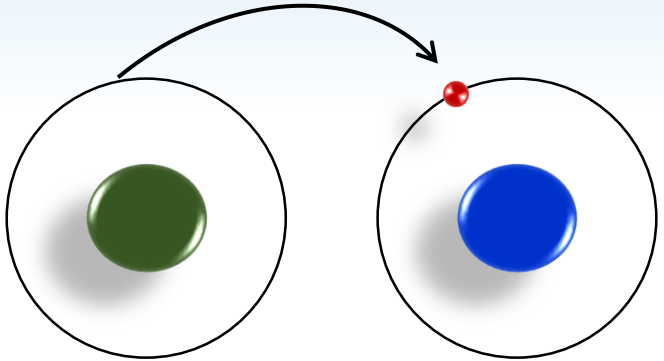
No bond or a compound is either completely covalent or ionic

Electrons shared



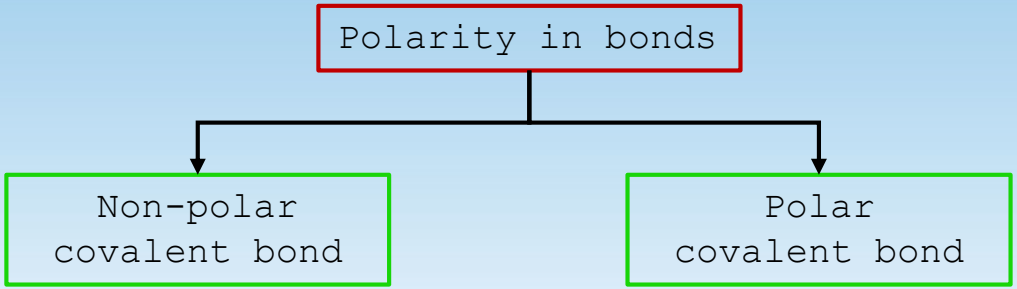
Covalent bond

Electron transferred

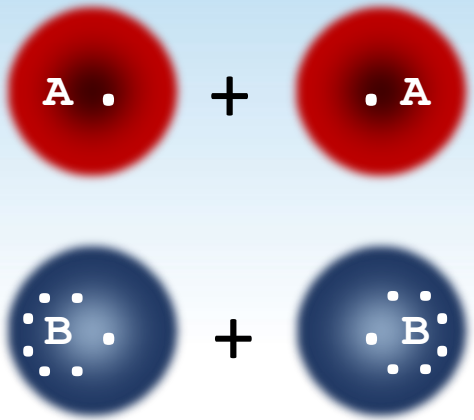


Ionic bonds

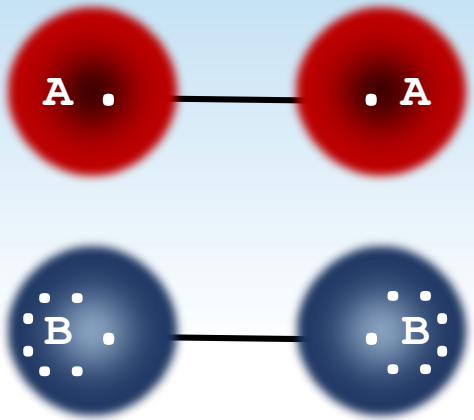
# Dipole moment



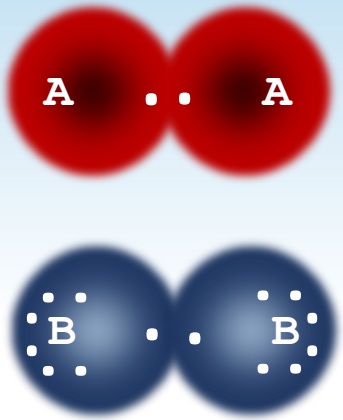
# Non-polar covalent molecule



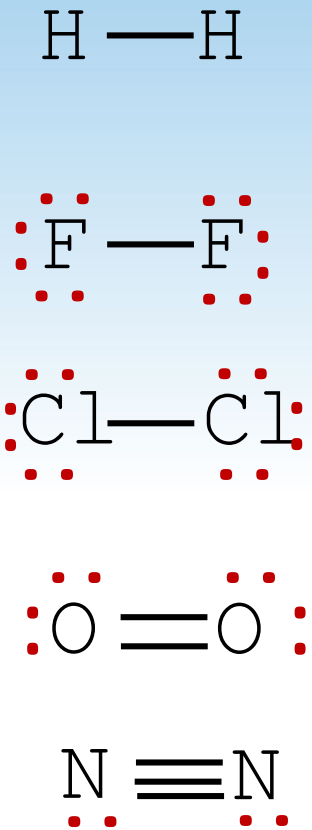
# Non-polar covalent molecule



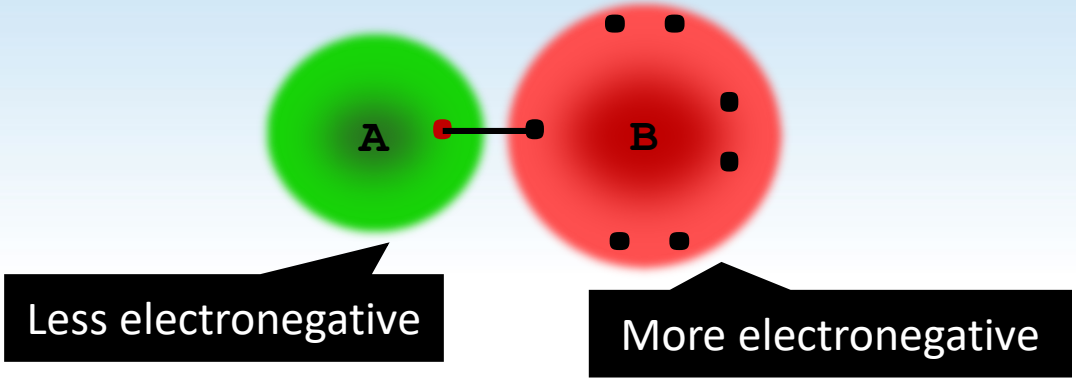
# Non-polar covalent molecule



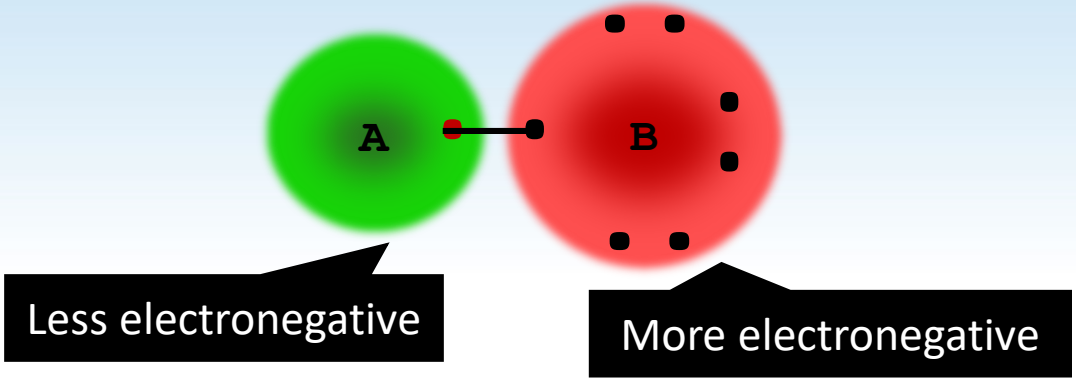

Non-polar
covalent molecule



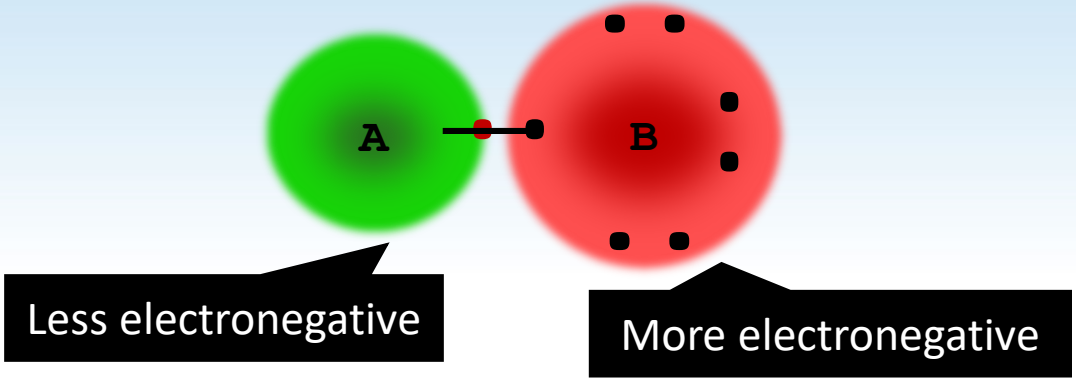
# Polar covalent Molecule



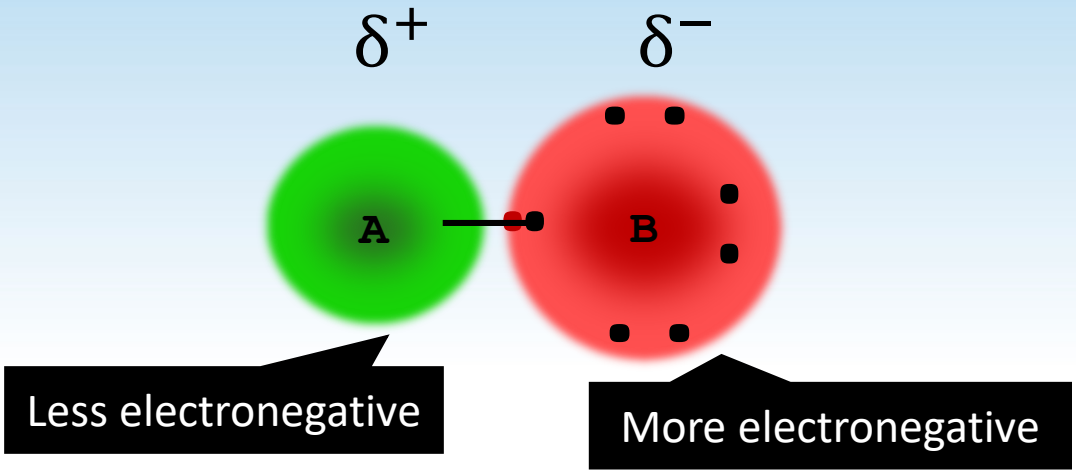
# Polar covalent Molecule



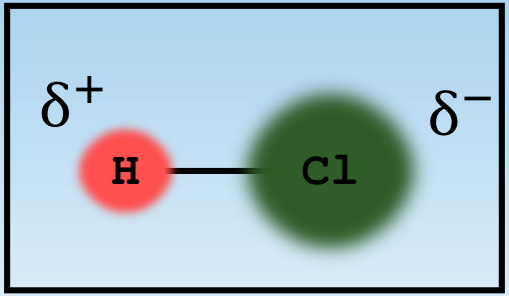
# Polar covalent Molecule

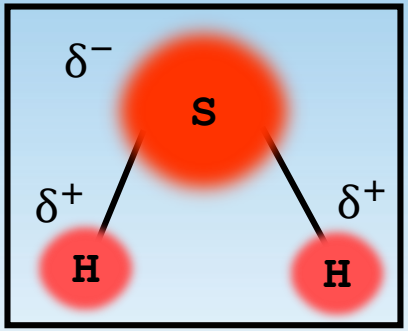


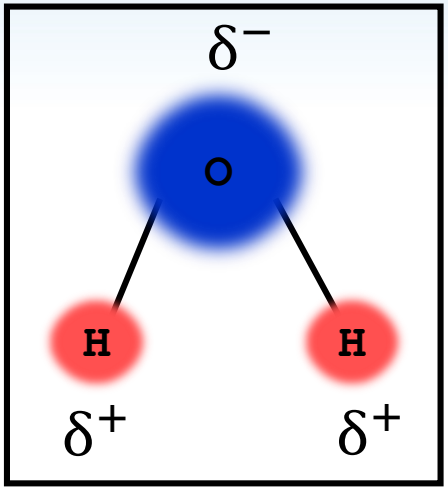
# Polar covalent Molecule

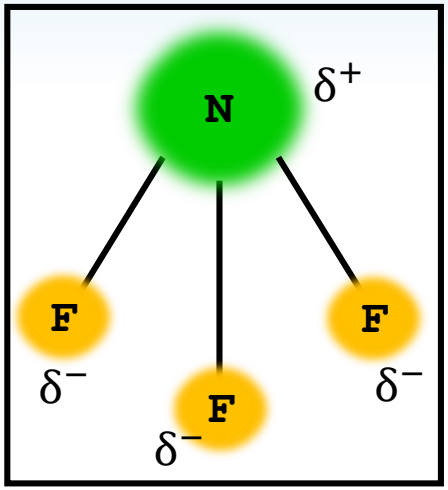


# Polar covalent Molecule

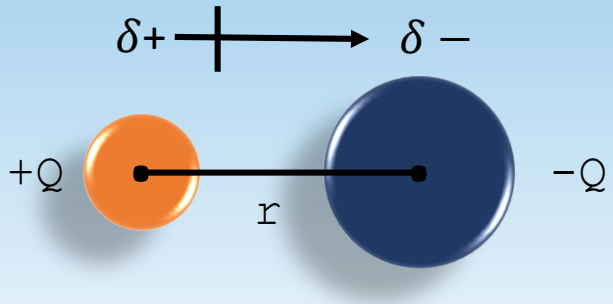


# How to calculate Dipole moment




Dipole moment (μ) = charge(Q) × distance of separation (r)

In coulombs

In meters

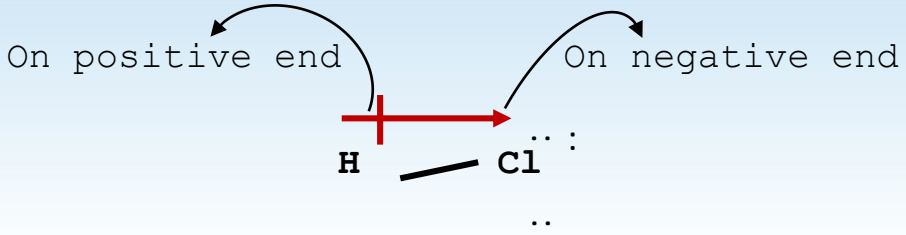
Usually expressed in Debye units

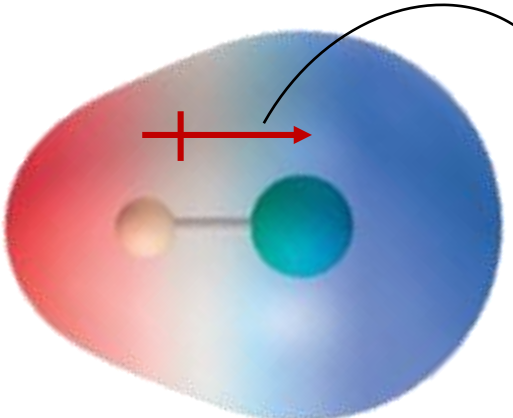
$1D = 3.33564 \times 10^{-30}$ Cm

Where, C = coloumb and m is meter

# Representation of Dipole moment

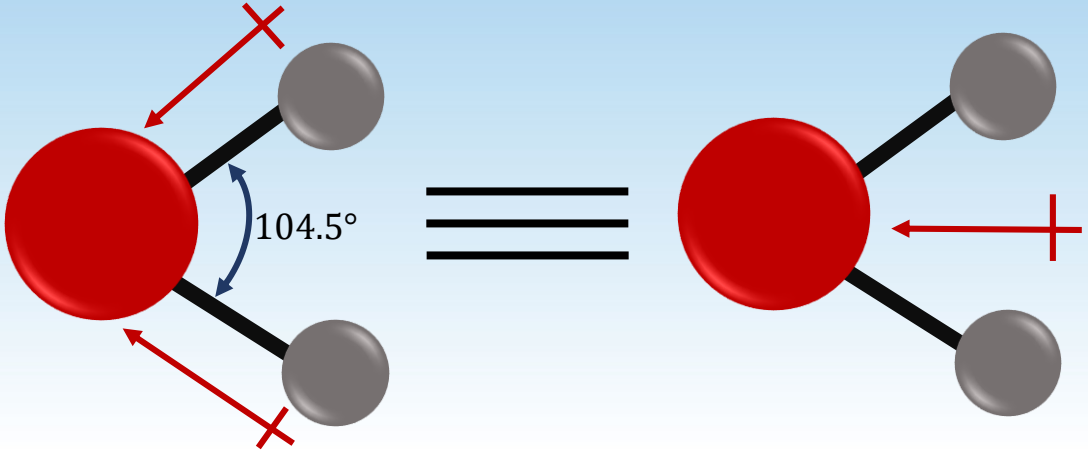
Dipole moment is represented by the crossed arrow ( ) put on lewis structure of the molecule



Direction of the shift of electron density in the molecule

# Resultant Dipole moment




Bond Dipole

Resultant Dipole

Net dipole moment, $\mu = 1.85D$

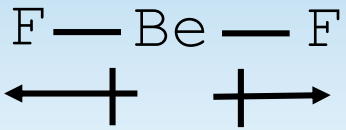
$= 1.85 \times 3.33564 \times 10^{-30}$ Cm

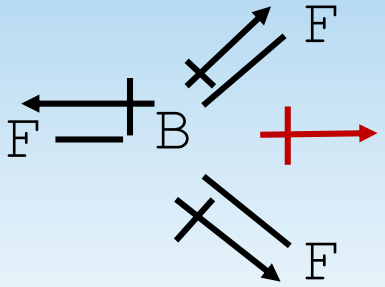
$= 6.17 \times 10^{-30}$C m

$\mu$ depends upon spatial arrangement of various bonds

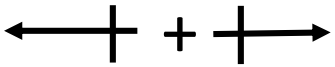
$\mu$ of a molecule is the vector sum of the $\mu$ of various bonds

# Dipole moment




otal dipole moment

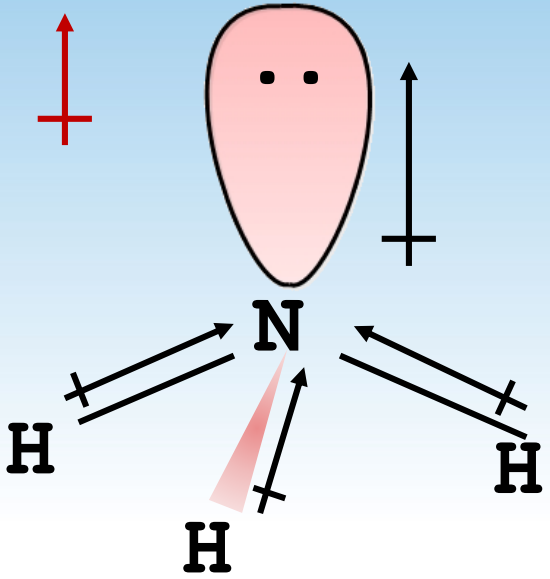




Net dipole moment=0

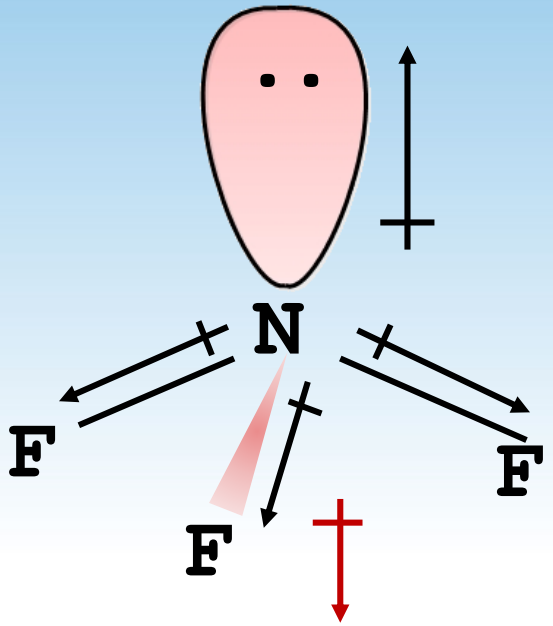
**Q.** Which out of $NH_3$ and $NF_3$ has higher dipole moment and why?

**Sol.**




Resultant dipole moment
$= 4.90 \times 10^{-30}$ Cm

Lone pair is in same direction as resultant dipole moment of $N - H$ bonds

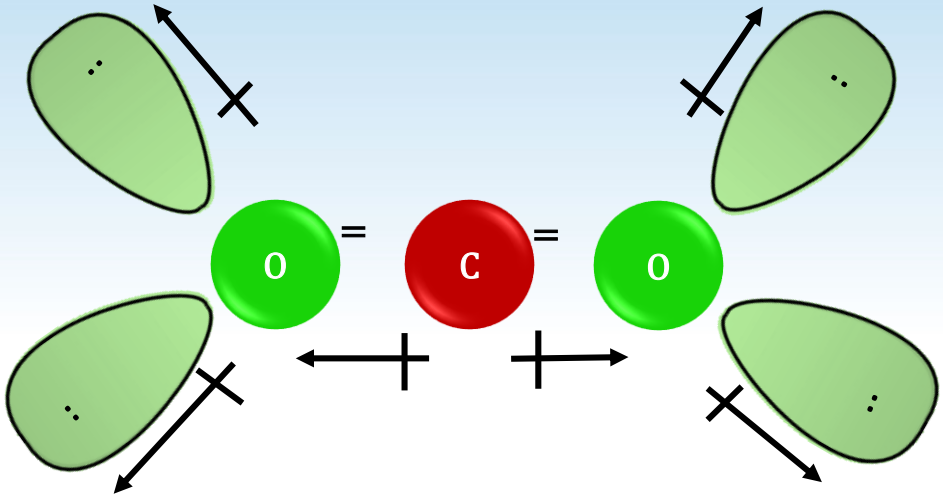



Resultant dipole moment
$= 0.80 \times 10^{-30}$ Cm

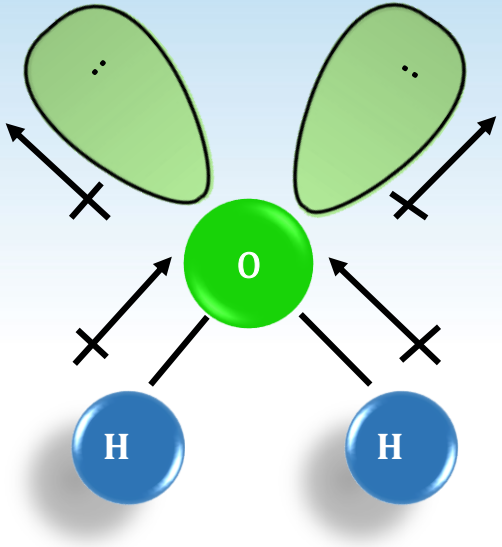
Lone pair is in Opposite direction of resultant dipole moment of $N - F$ bonds

**Q.** Although both $CO_2$ and $H_2O$ are triatomic molecules, the shape of $H_2O$ molecule is bent while that of $CO_2$ is linear. Explain this on the basis of dipole moment.

**Sol.**




Resultant dipole moment $= 0$ D



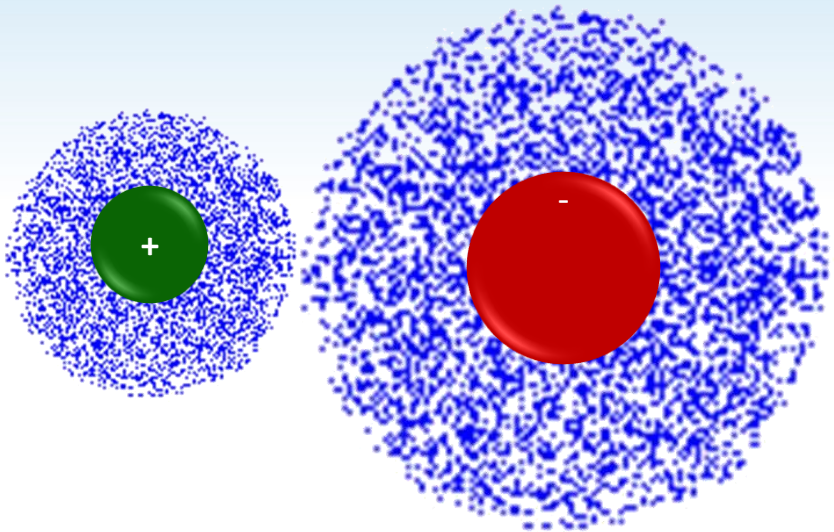

Resultant dipole moment $= 1.84$ D

# Fajans Rule

Just as all the covalent bonds have partial ionic character

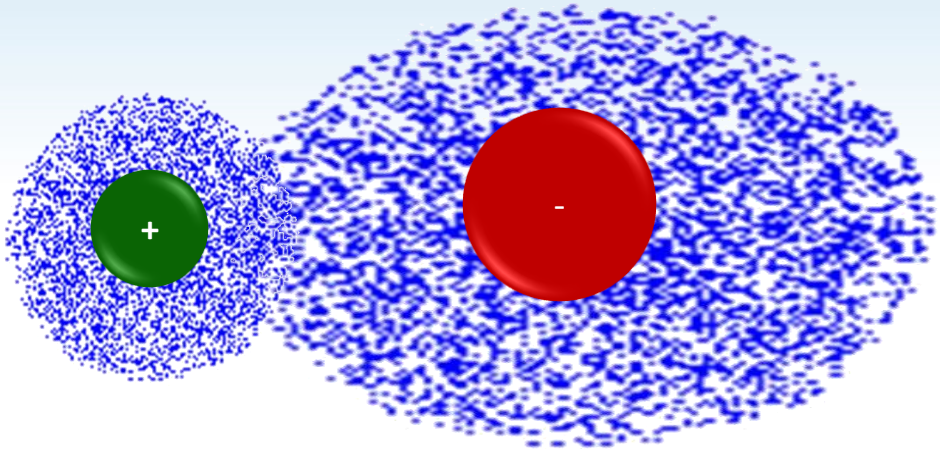
Ionic bonds also have partial covalent character

Polarization



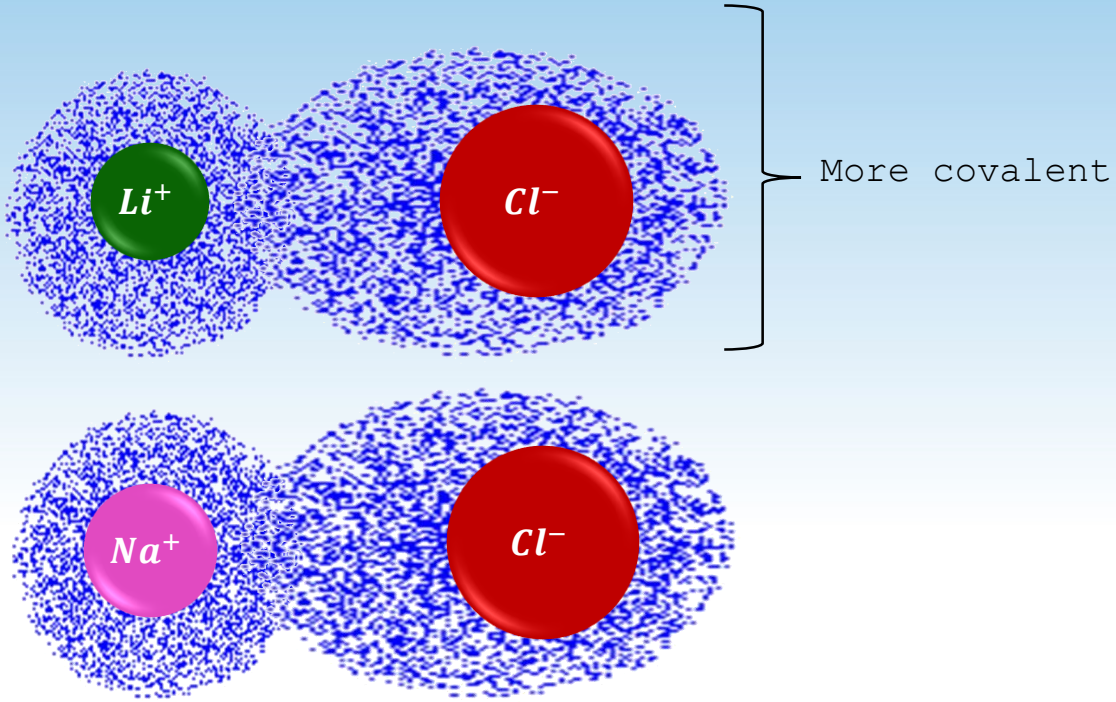
# Fajans Rule

Polarization



# Fajans Rule

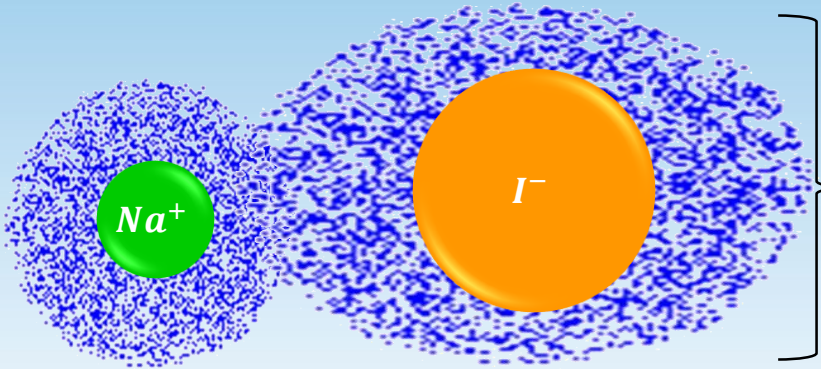
$$\text{Size of cation} \propto \frac{1}{\text{polarisation}}$$




The smaller the size of the cation and larger the size of the anion, the greater the covalent character of an ionic bond

# Fajans Rule

Size of anion ∝ Polarization




More covalent




Charge on the cation or anion ∝ covalent character

# Fajans Rule

The cation with inert gas configuration forms compounds with more ionic character:

$Na^+ = 2, 8$

The cations with pseudo inert gas configuration forms compound with more covalent character:

$Cu^+ = 2, 8, 18$

Pseudo inert gas configuration

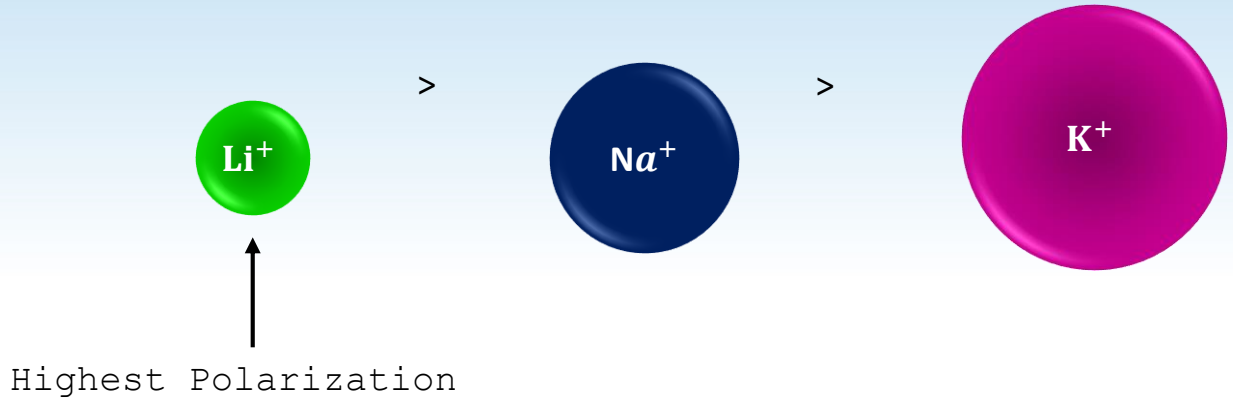
Copper(I)Chloride = CuCl
Sodium chloride = NaCl

Q. Which of the following cation will show highest polarization?

$Li^+, Na^+, K^+$

Sol.

$$\text{Size of cation} \propto \frac{1}{\text{polarisation}}$$



## Summary

- Bond parameters such as bond length, bond angle, bond enthalpy and bond parameter have significant effect on the properties of compounds

- Resonance structure are several forms of a molecule where the chemical connectivity is the same but the electrons are distributed differently around the structure

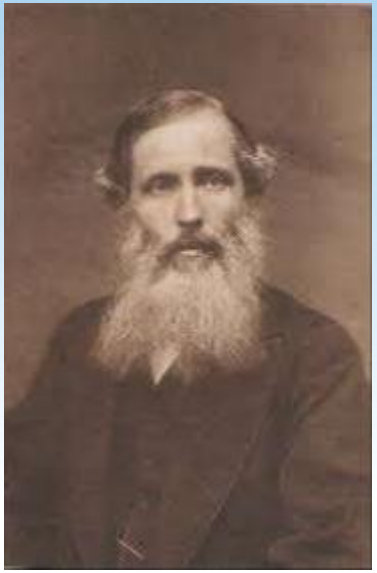
- Dipole moment ($\mu$) = charge(Q) × distance of separation (r)

- The partial covalent character of ionic bonds was discussed by fajans rule

# Valence Shell Electron Pair Repulsion Theory



Powell



Sidgwick

Proposed in 1940

Based on the repulsive interactions of the electron pairs in the valence shell of the atoms

# Valence Shell Electron Pair Repulsion Theory



Ronald Sydney Nyholm

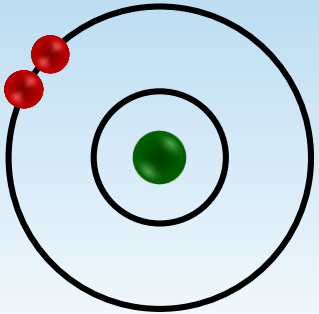


Ronald Gillespie

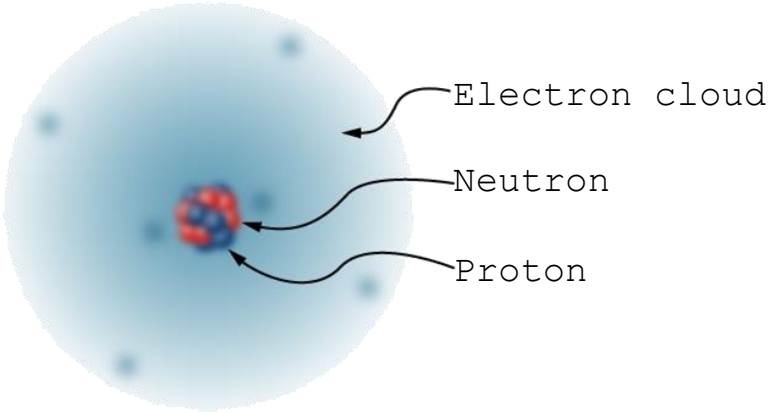
Further developed and refined by Nyholm and Gillespie

# Main postulates of VSEPR theory

1. Shape of the molecule depends upon the number of valence shell electron pairs around the central atom

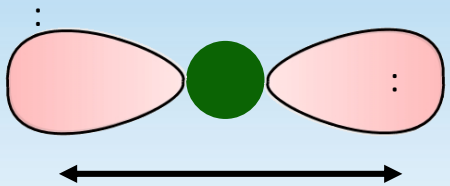


2. Pairs of electrons in the valence shell repel one another since their electron clouds are negatively charged

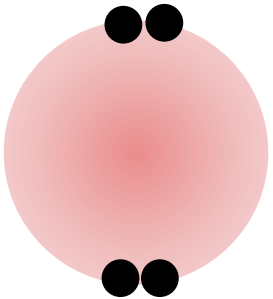


# Main postulates of VSEPR theory

3. These pairs of electrons tend to occupy such positions in space that minimize repulsion and thus maximize distance between them.




4. The valence shell is taken as a sphere with the electron pairs localizing on the spherical surface at maximum distance from one another.

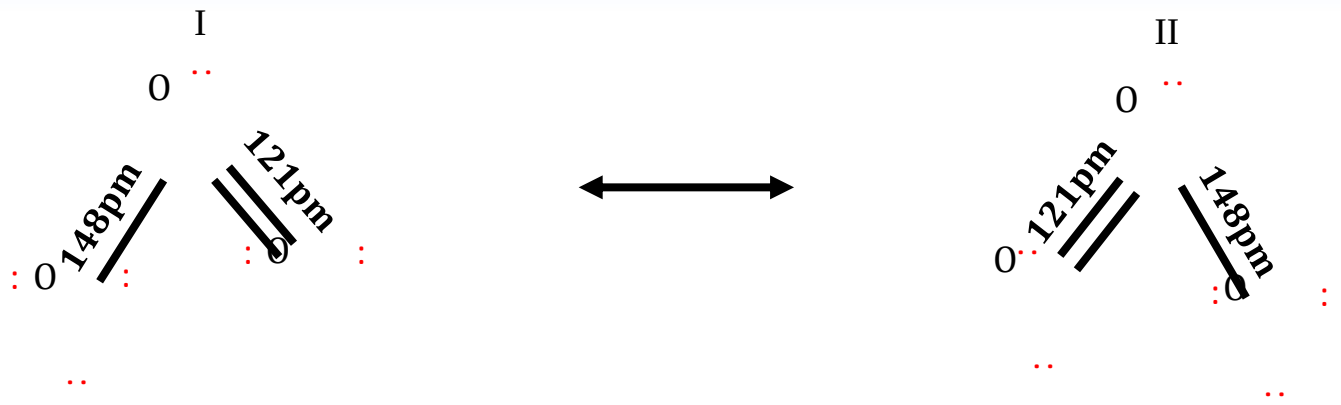


# Main postulates of VSEPR theory

5. Multiple bond → Treated as → Single electron pair

Two or three electron pairs → Treated as → Single super pair

6. Where two or more resonance can represent a molecule, the VSEPR model is applicable to any such structure



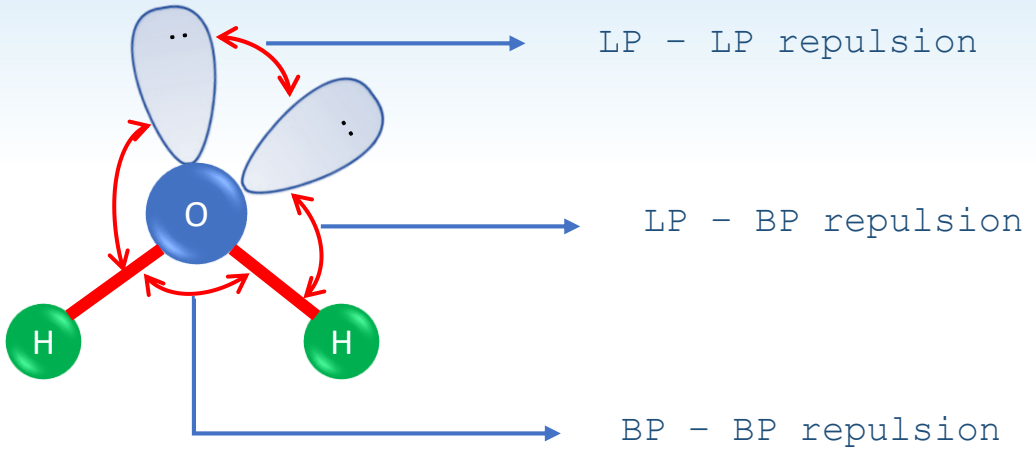
# Order of repulsive interaction

The repulsive interaction of electron pairs decrease in the order:

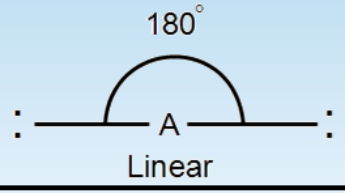
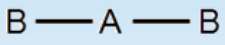
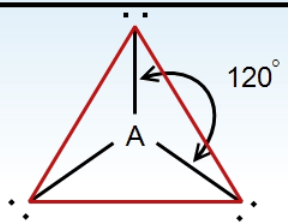
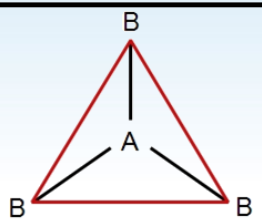
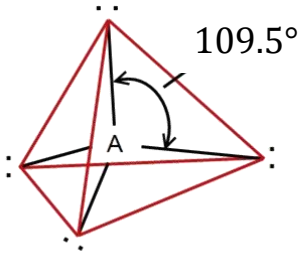
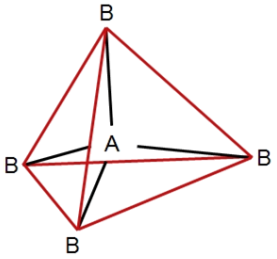
**Lone pair(lp)-Lone pair(lp) > Lone pair(lp)-Bond pair(bp) > Bond pair(bp)-Bond pair(bp)**

Lp are localized on central atom

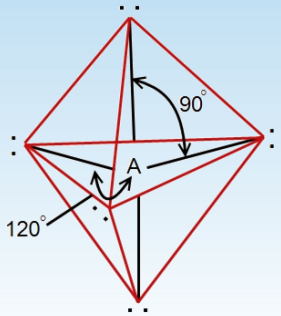
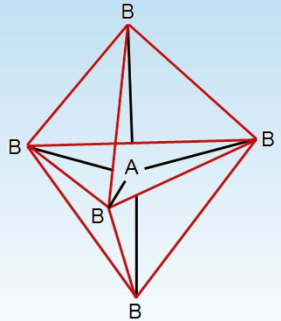
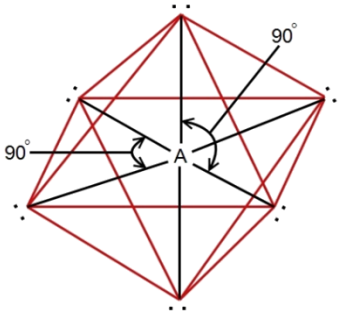
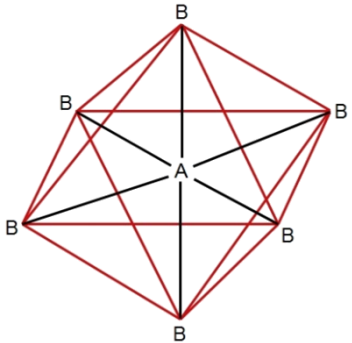
Lp electron occupy more space



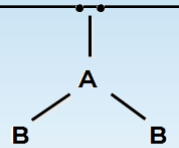
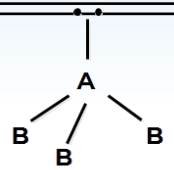
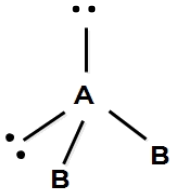
# Geometry of molecules in which the central atom has no lone pair of electrons

| Number of electron pairs | Arrangement of electron pairs | Molecular geometry | Examples |
|---|---|---|---|
| 2 |   Linear |   Linear | $BeCl_2, HgCl_2$ |
| 3 |   Trigonal planar |   Trigonal planar | $BF_3$ |
| 4 |   Tetrahedral |   Tetrahedral | $CH_4, NH_4^+$ |

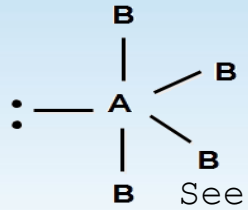
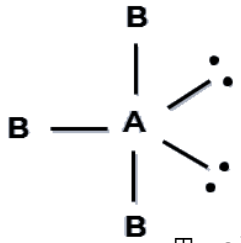
# Geometry of molecules in which the central atom has no lone pair of electrons

| Number of electron pairs | Arrangement of electron pairs | Molecular geometry | Examples |
|---|---|---|---|
| 5 |  Trigonal bipyamidal |  Trigonal bipyamidal | $PCl_5$ |
| 6 |  Octahedral |  Octahedral | $SF_6$ |

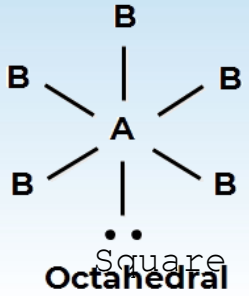
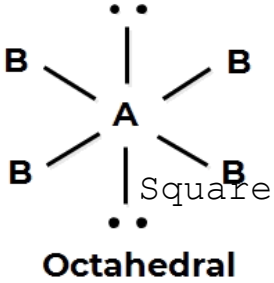
# Molecules/Ions with central Atoms having one or more Lone pairs of Electron

| Molecule type | No. of bonding pairs | No. of lone pairs | Arrangement of electron pairs | Shapes and examples |
|---|---|---|---|---|
| $AB_2E$ | 2 | 1 |   Trigonal planar | Bent, $SO_2, O_3$ |
| $AB_3E$ | 3 | 1 |   Tetrahedral | Trigonal pyramidal, $NH_3$ |
| $AB_2E_2$ | 2 | 2 |   Tetrahedral | Bent, $H_2O$ |

# Molecules/Ions with central Atoms having one or more Lone pairs of Electron

| Molecule type | No. of bonding pairs | No. of lone pairs | Arrangement of electron pairs, its shape and examples |
|---|---|---|---|
| $AB_4E$ | 4 | 1 |   See saw, $SF_4$ Trigonal bi-pyramidal |
| $AB_3E_2$ | 3 | 2 |   T-shape, $ClF_3$ Trigonal bi-pyramidal |

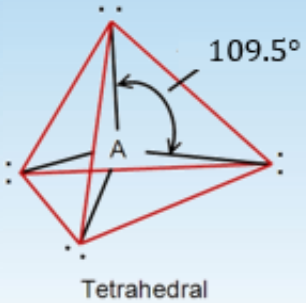
# Molecules/Ions with central Atoms having one or more Lone pairs of Electron

| Molecule type | No. of bonding pairs | No. of lone pairs | Arrangement of electron pairs, its shape and examples |
|---|---|---|---|
| $AB_5E$ | 5 | 1 |   Octahedral Square pyramidal, $BrF_5$ |
| $AB_4E_2$ | 4 | 2 |   Octahedral Square planar, $XeF_4$ |

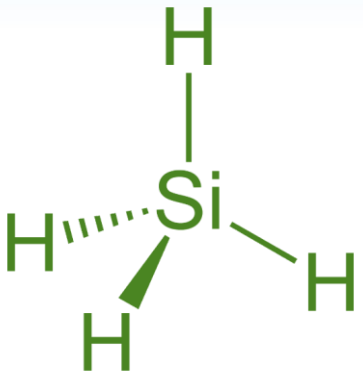
**Q.** Explain the geometry of $SiH_4$ molecule.

**Sol.**

Number of electron pair = 4

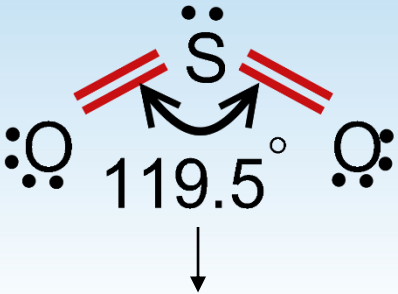



Molecular geometry

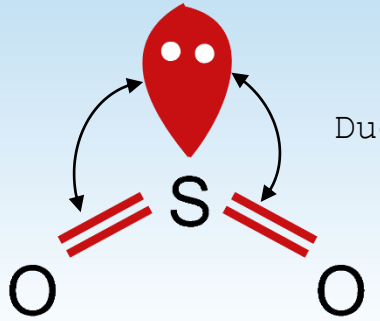


# Shapes of molecules containing Bond pair and lone pair

$AB_2E$ Type: $SO_2$




Angle reduced




Bent

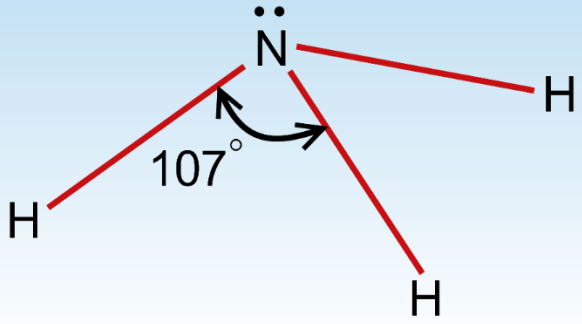
Due to lone pair-bond pair repulsion

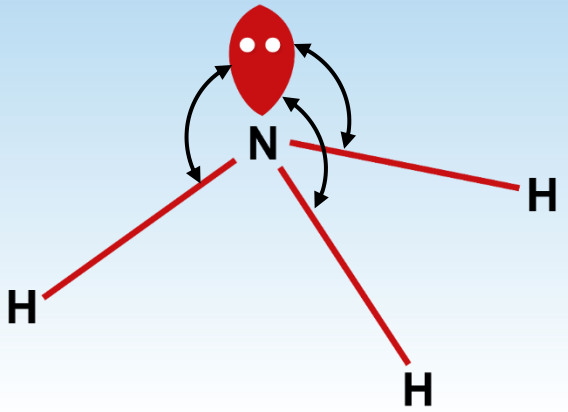
Number of bonding pairs: 4

Number of lone pairs : 1

# Shapes of molecules containing Bond pair and lone pair

$AB_3E$ Type: $NH_3$



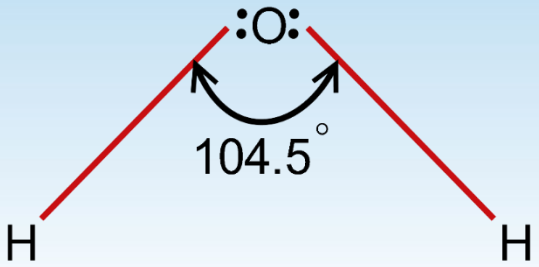

Trigonal Pyramidal
Due to lp - bp repulsion

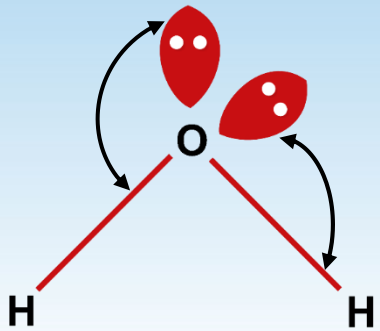
Number of bonding pairs: 3

Number of lone pairs : 1

# Shapes of molecules containing Bond pair and lone pair

$AB_2E_2$ Type: $H_2O$




Bent

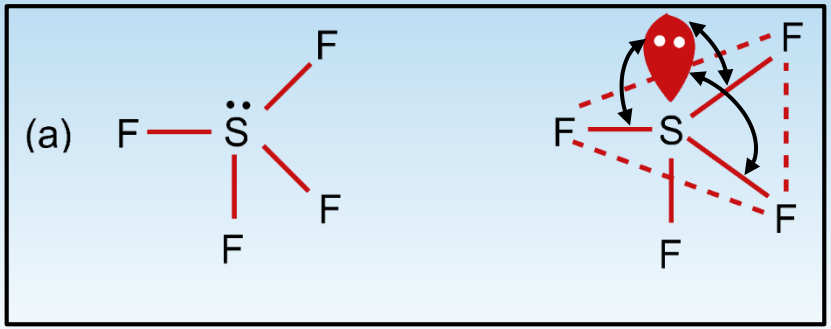
Due to lp - lp > lp – bp > bp – bp repulsion

Number of bonding pairs: 2

Number of lone pairs : 2

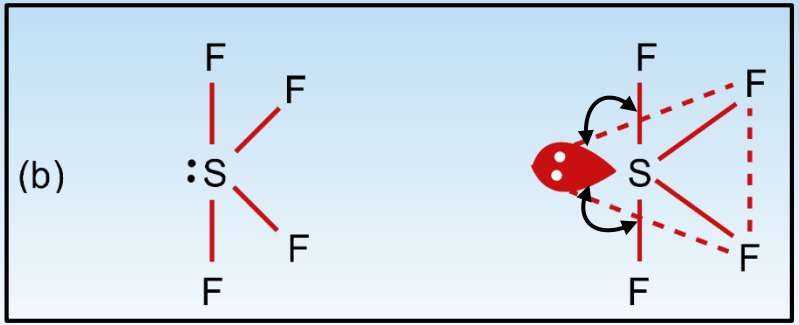
# Shapes of molecules containing Bond pair and lone pair

## $AB_4E$ Type: $SF_4$




Lone pair is present at axial position

3 lp-bp repulsion at 90°




Lone pair is present at equitorial position

2 lp-bp repulsion → More stable

Number of bonding pairs: 4

Number of lone pairs : 1

Distorted tetrahedron

Folded square

See - saw

# Shapes of molecules containing Bond pair and lone pair

$AB_3E_2$ Type: $ClF_3$

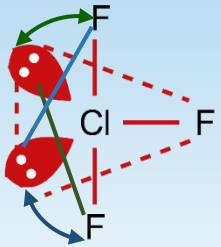
lp are at equatorial position

Less repulsion

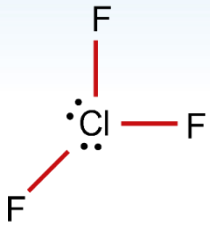
Most stable

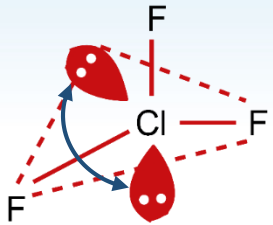
(a)



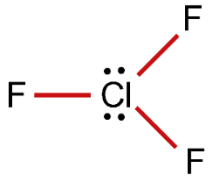



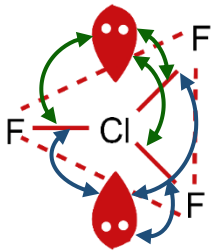

(b)




(c)



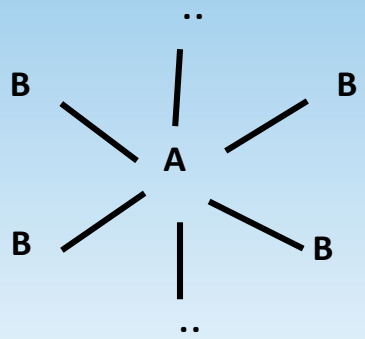




Number of bonding pairs: 3

Number of lone pairs : 2

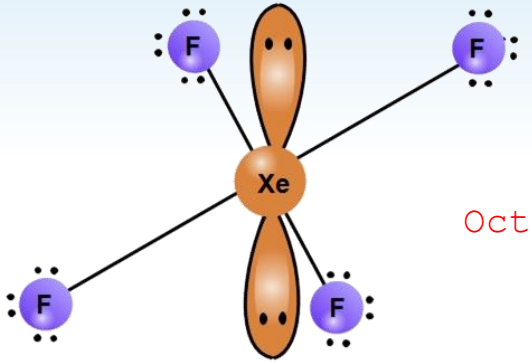
**Q.** According to VSEPR theory, what will be the shape of $XeF_4$?

**Sol.** Molecule type: $AB_4E_2$




Number of bonding pairs: 4

Number of lone pairs: 2




Octahedral

Arrangement of electron pairs:

Square planar shape

# Valence Bond Theory



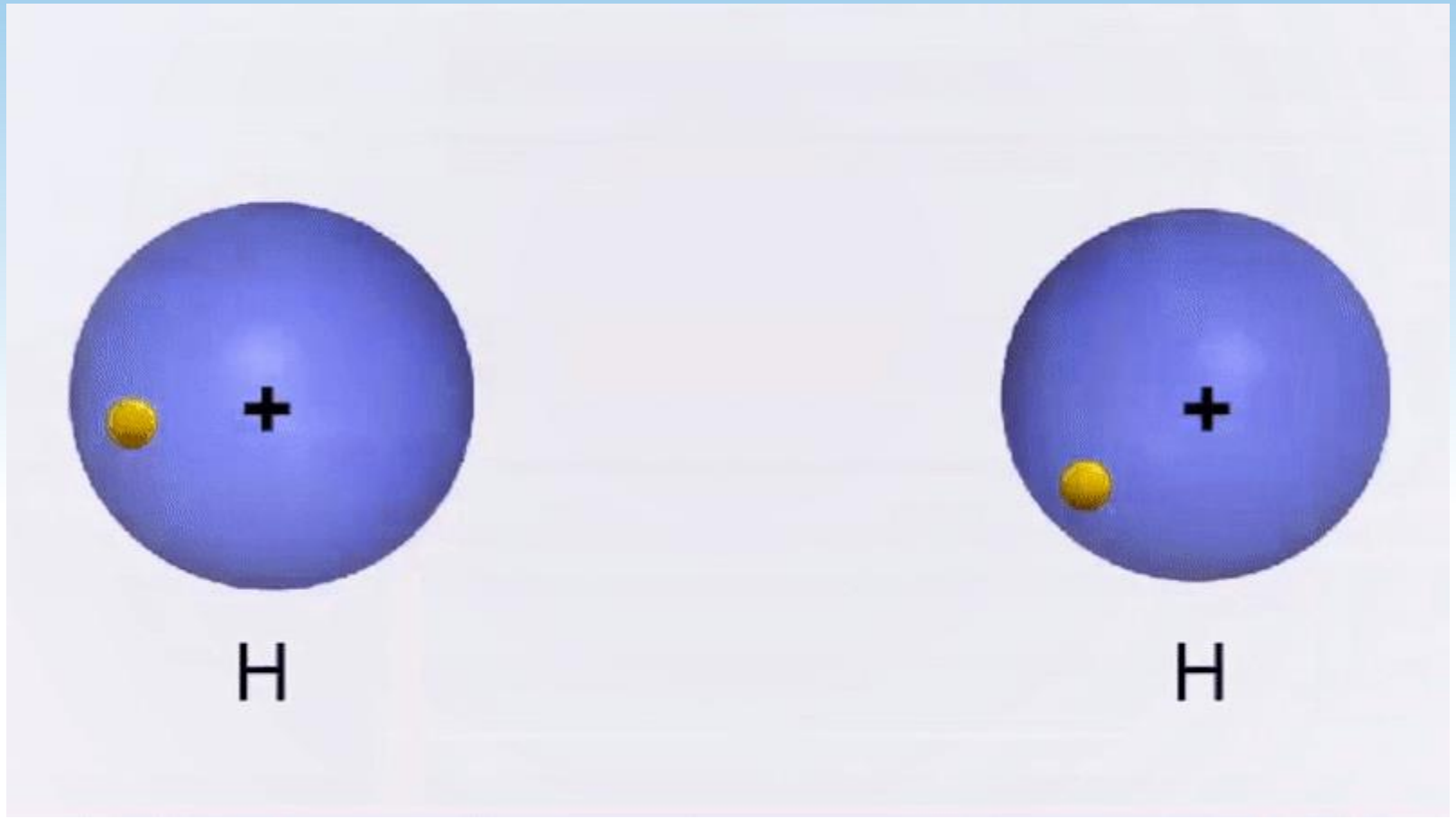
Walter Heitler



Fritz W. London

Valence Bond Theory was introduced by Heitler and London in 1927 and developed further by Pauling and others

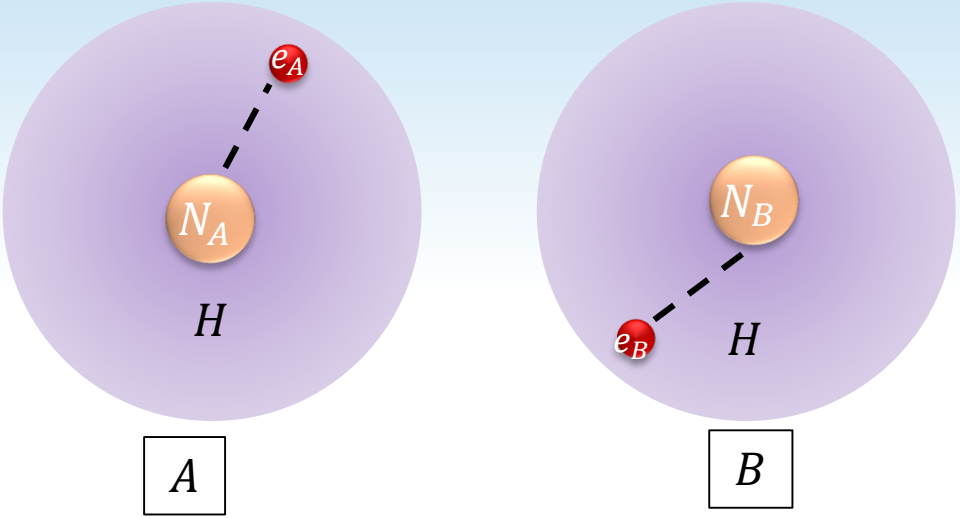
# Valence Bond Theory



# Valence Bond Theory

Attractive forces arise between:

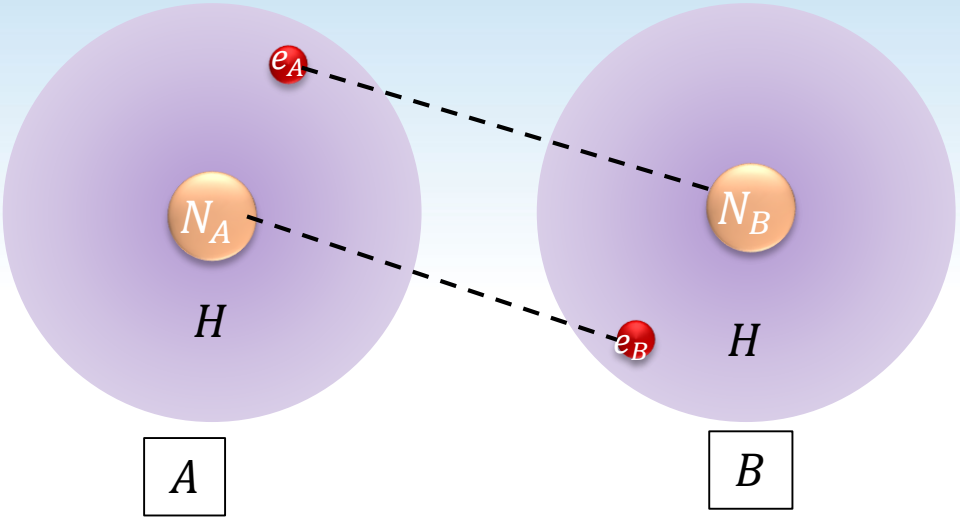
Nucleus of one atom and its own electron that is $N_A - e_A$ and $N_B - e_B$



# Valence Bond Theory

Attractive forces arise between:

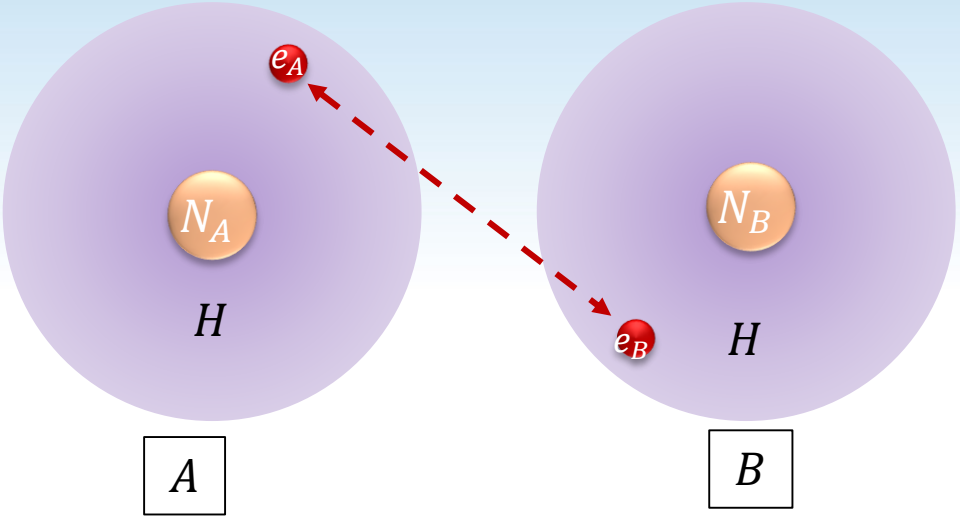
Nucleus of one atom and electron of other atom $N_A - e_B$ and $N_B - e_A$



# Valence Bond Theory

Repulsive forces arise between:

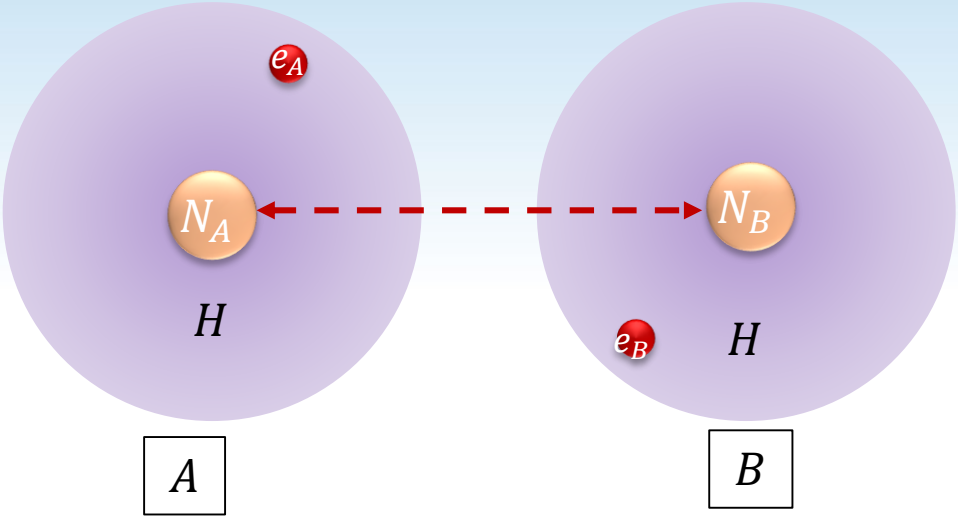
electrons of two atoms like $e_A - e_B$



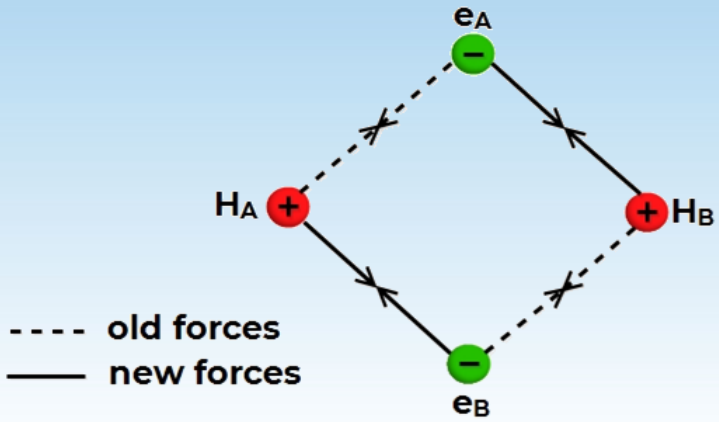
# Valence Bond Theory

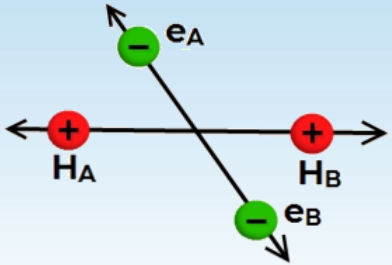
Repulsive forces arise between:

Nucleus of two atoms like $N_A - N_B$



# Valence Bond Theory




Attractive Forces

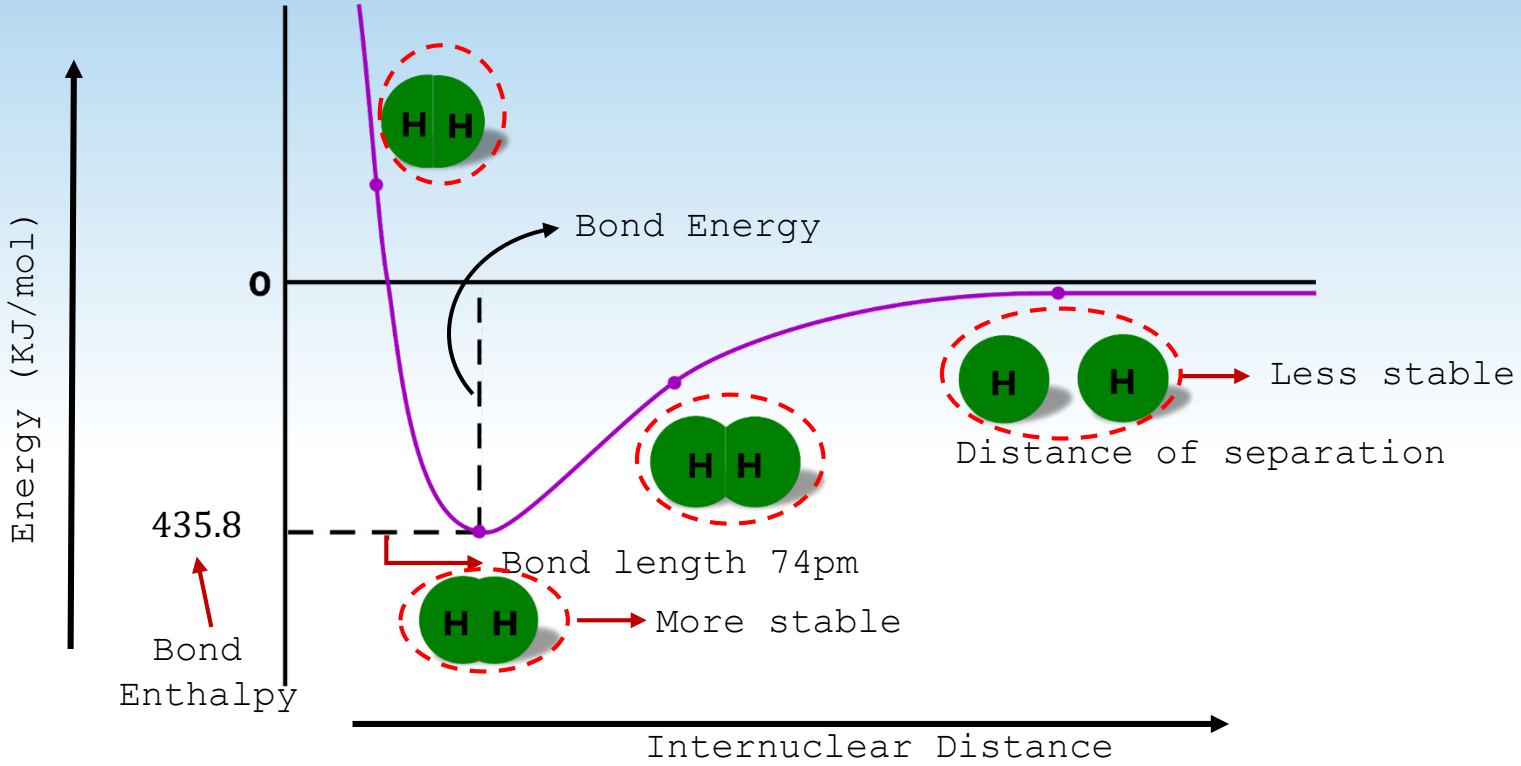
Bring the two atoms close to each other

Repulsive Forces

Push them apart

# Valence Bond Theory

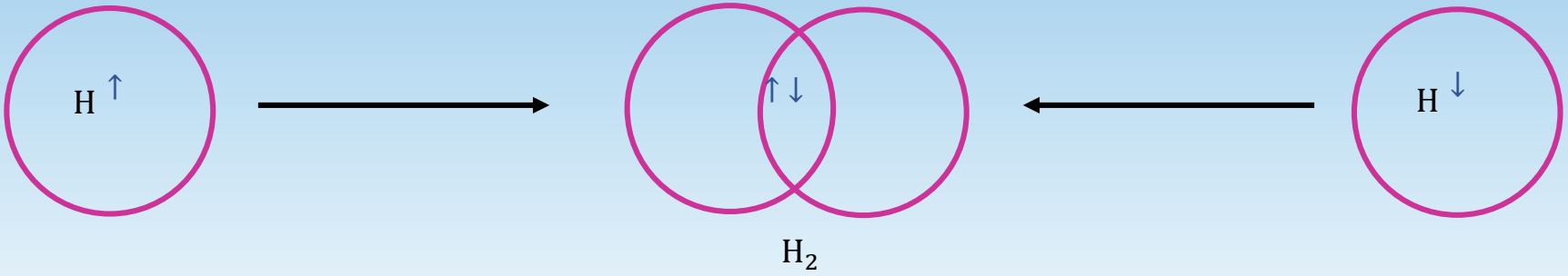
Magnitude of new attractive force is more than the new repulsive forces




Conversly, $H_2(g) + 435.8\ KJ\ mol^{-1} \longrightarrow H(g) + H(g)$

# Orbital overlap Concept

At high energy state



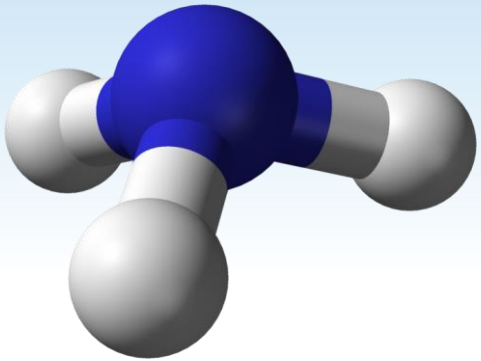

At low energy state

Highly stable

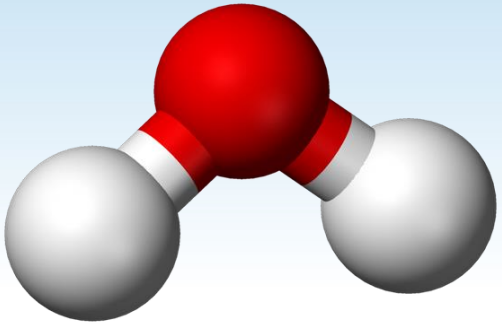
Extent of Overlapping ↑ Bond Strength ↑

# Directional Properties of Bonds

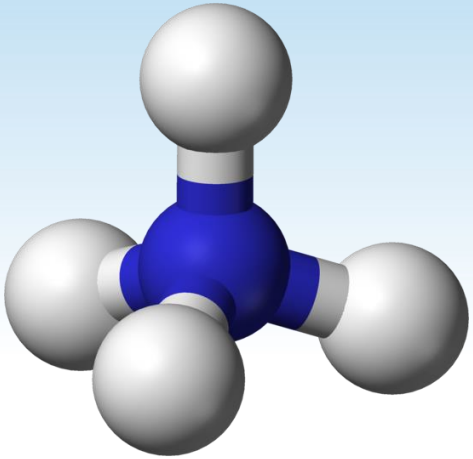
The valence bond theory explains the formation and directional properties of bonds in polyatomic molecules like-



$NH_3$



$H_2O$



$CH_4$

Overlapping Hybridisation

**Q.** Explain the attractive and repulsive forces of $H_2$ molecule on the basis of VBT.

**Sol.** Attractive forces arise between:

i. Nucleus of one atom and its own electron that is:

$N_A - e_A$ and $N_B - e_B$

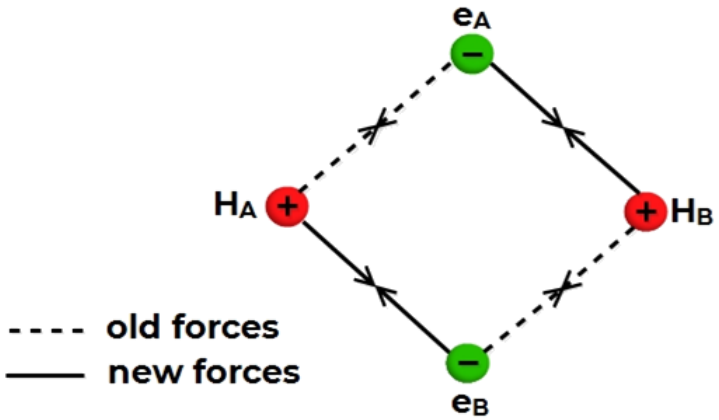
ii. Nucleus of one atom and electron of other atom :

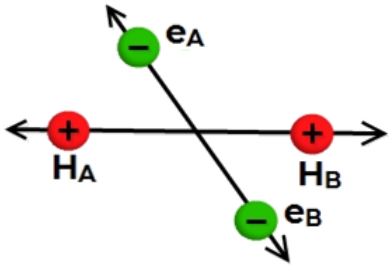
$N_A - e_B$ and $N_B - e_A$

Repulsive forces arise between:

i. electrons of two atoms like $e_A - e_B$

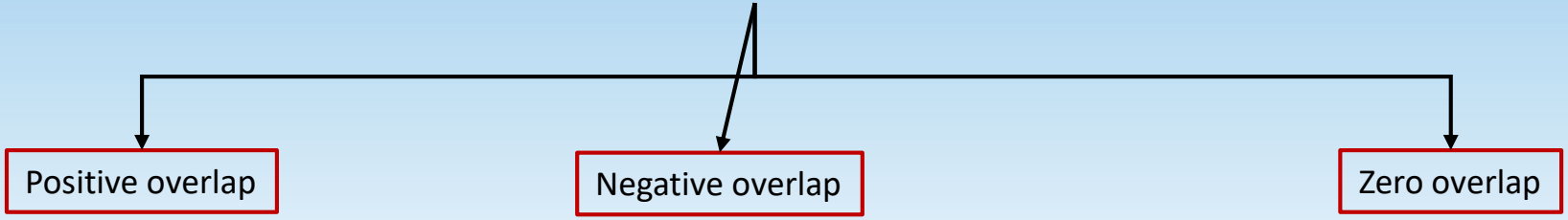
ii. Nucleus of two atoms like $N_A - N_B$



# Overlapping of atomic orbitals

When two atoms come close to each other, there is overlapping of atomic orbitals

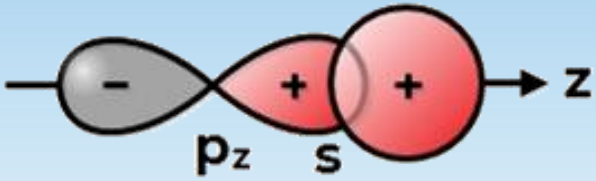


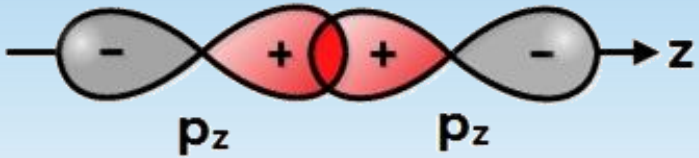

Depend upon the sign(phase) and direction of orientation of amplitude of Orbital wave function in space

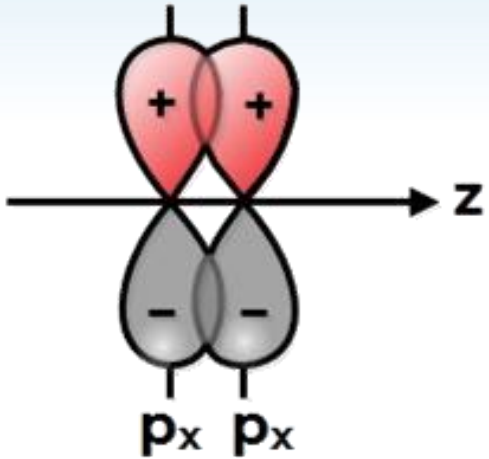
Orbitals forming bond should have same sign (phase) and orientation in space

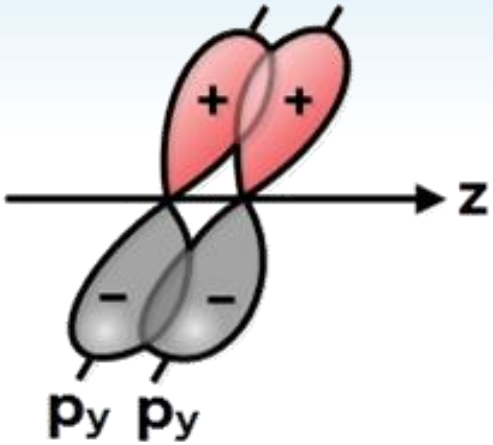
## Positive or in phase overlap

When the phase of two interacting orbital is same



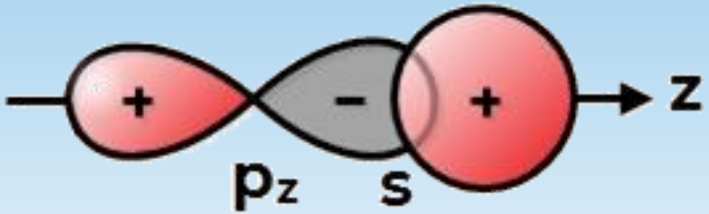



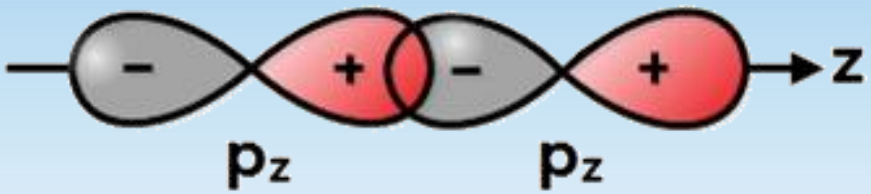



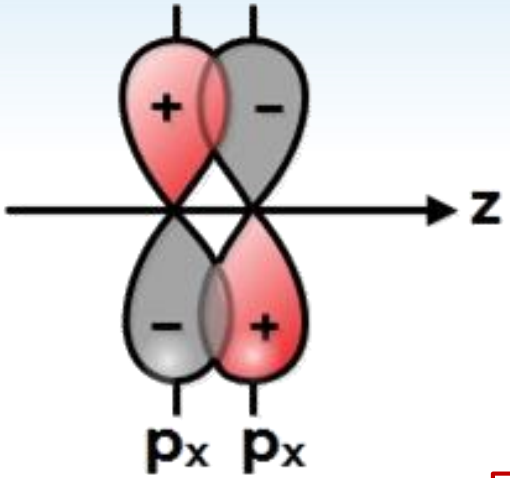

Bond is formed.

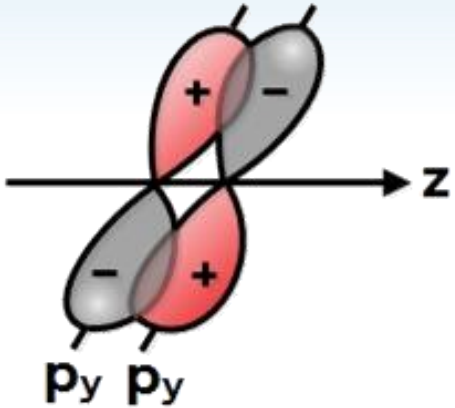
## Negative or out phase overlap

When the phase of two interacting atomic orbital is opposite



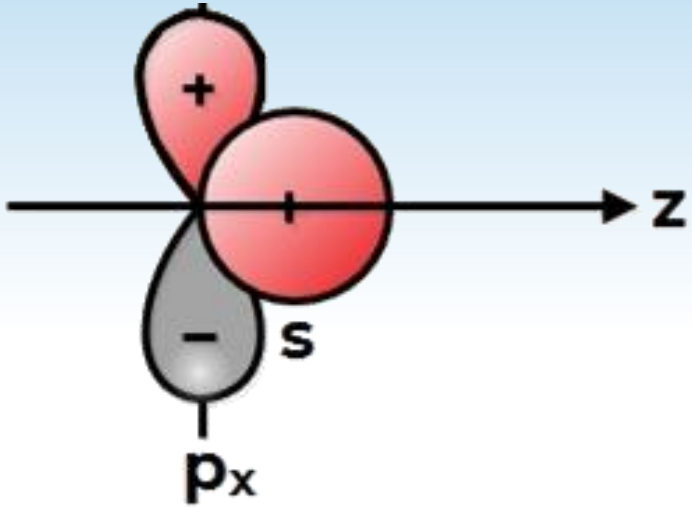



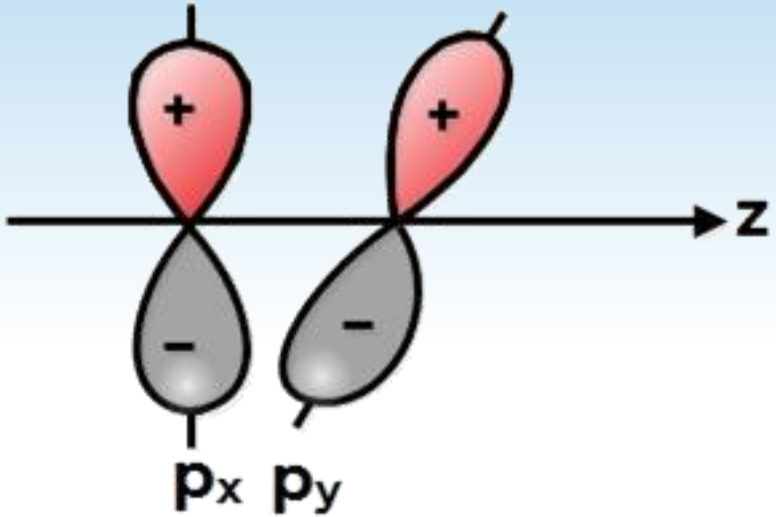




Bond is not formed

# Zero overlap

When the orientation of two interacting atomic orbital is such that there is no overlapping of the orbital

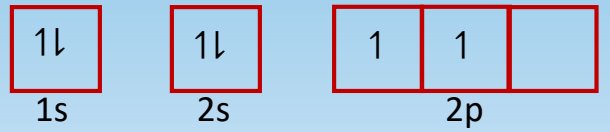


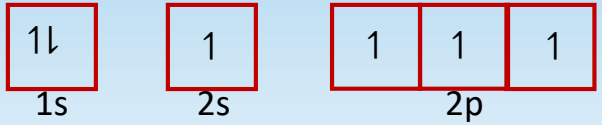
# $CH_4$ Molecule

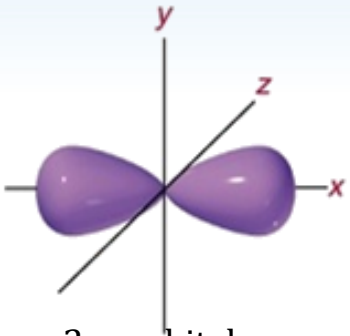
Carbon :

Ground state electronic configuration –

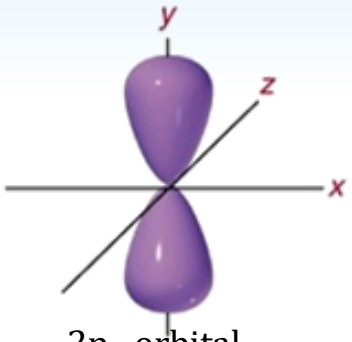



Excited state electronic configuration –

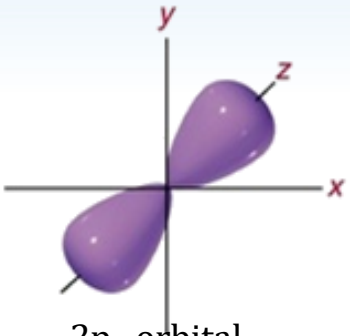



$2p_x$ orbital



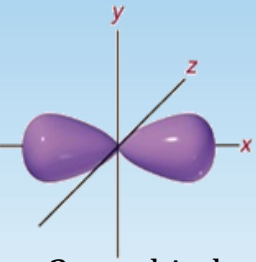

$2p_y$ orbital



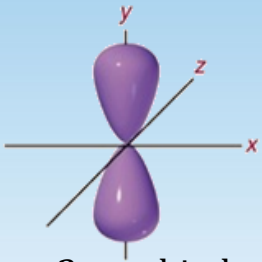

$2p_z$ orbital

3 p-orbital are at 90° to each other

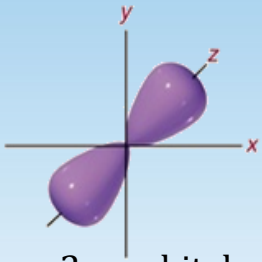
# $CH_4$ Molecule




$2p_x$ orbital

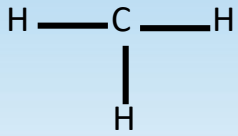



$2p_y$ orbital



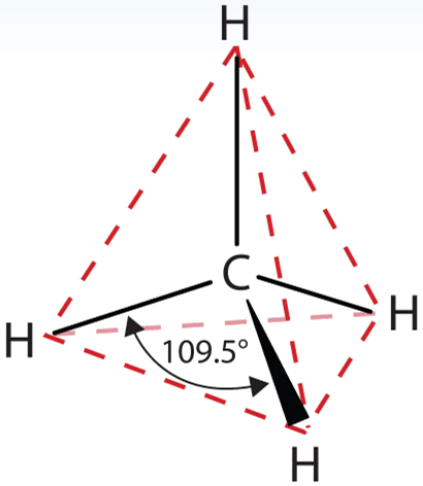

$2p_z$ orbital

3 p-orbital are at 90° to each other




The three C-H bond will be oriented at 90° to one another

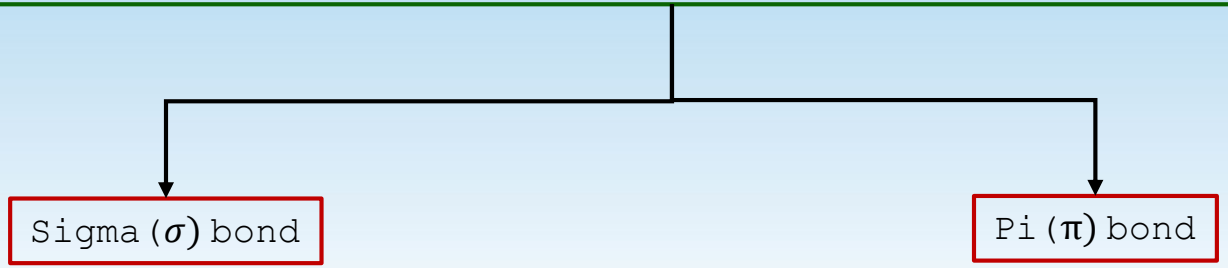
Direction of fourth C-H bond cannot be ascertained




Atomic orbital overlap does not account for the directional characteristics of bond in $CH_4$

# Types of overlapping and nature of covalent bonds

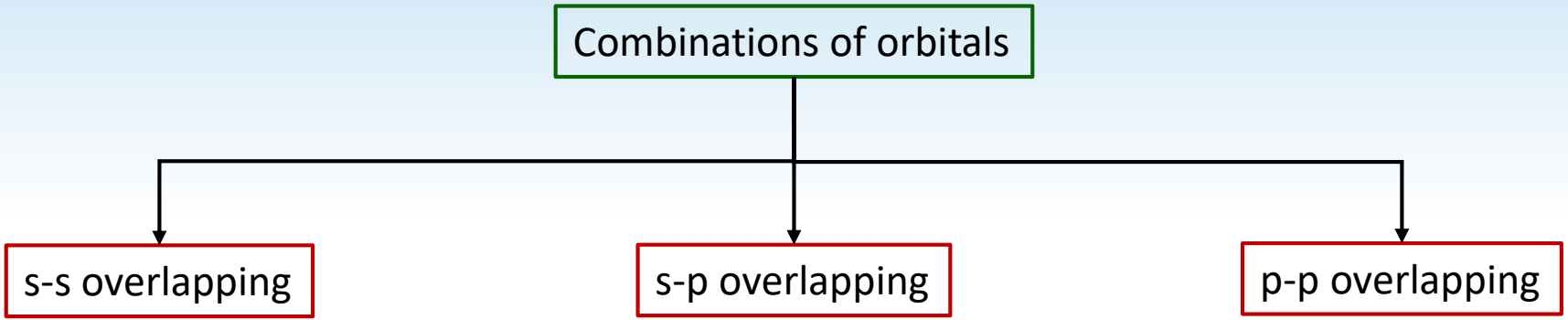
The covalent bond may be classified into two types depending upon the types of overlapping



# Sigma($\sigma$) bond

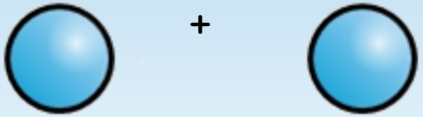
Formed by end to end overlap (head-on) of bonding orbitals along the Internuclear axis

Also called head on overlap or axial overlap

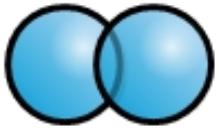


# Sigma($\sigma$) bond

## s-s overlapping



s - orbital  s - orbital

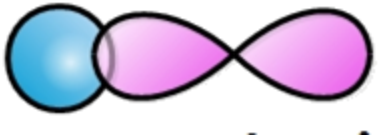


s – s  overlapping

## s-p overlapping

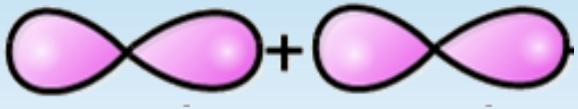


s - orbital  p - orbital



s – p  overlapping

## p-p overlapping

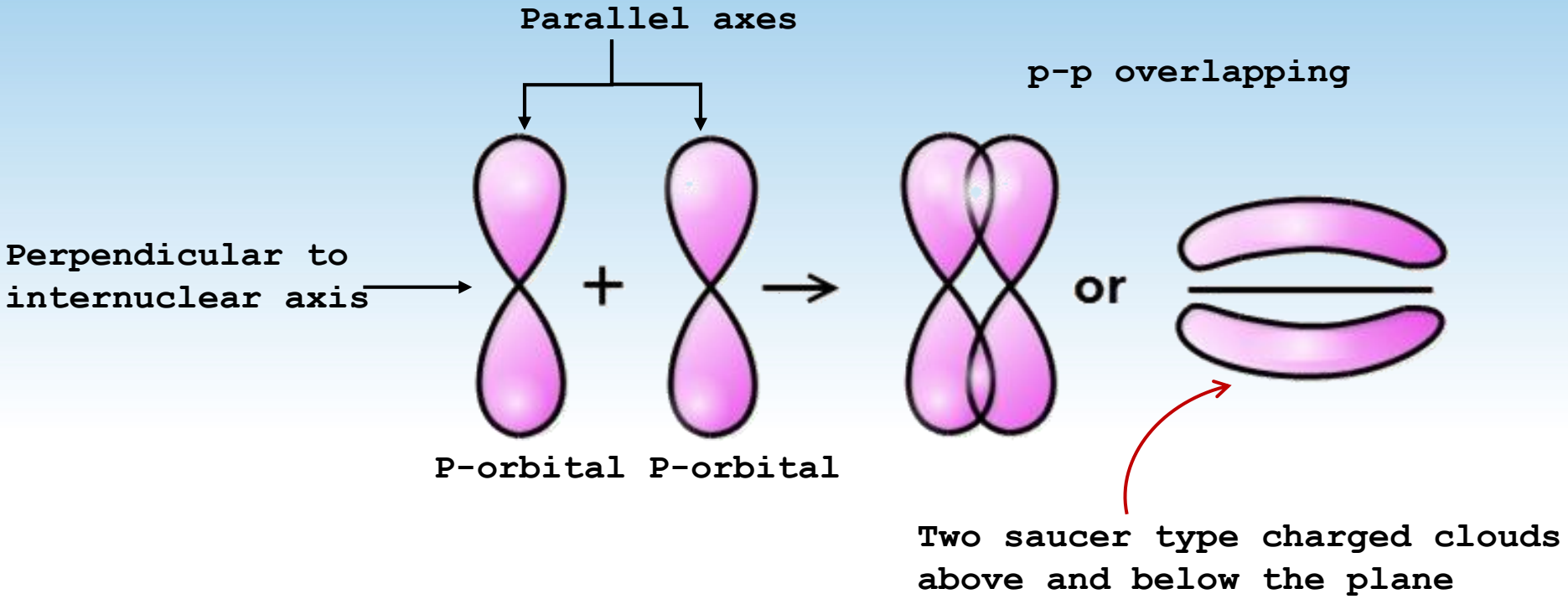


p - orbital  p - orbital



p – p  overlapping

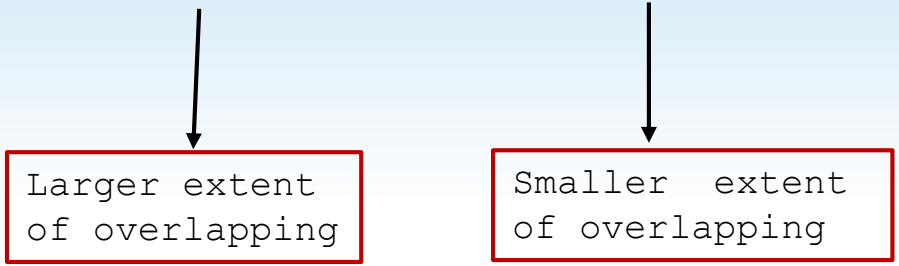
# Pi ($\pi$) bond


Parallel axes
p-p overlapping
Perpendicular to internuclear axis
+
or
P-orbital
P-orbital
Two saucer type charged clouds above and below the plane

# Strength of sigma and pi bond

Strength of bond depends upon extent of overlapping

Sigma bond($\sigma$) >> Pi bond($\pi$)



**Q.** Distinguish between sigma and pi bond.

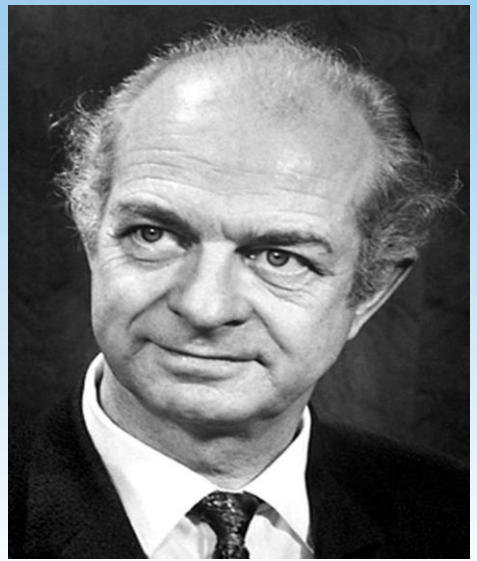
**Sol.**

| Sigma ( $\sigma$ ) bond | Pi ($\pi$) bond |
|---|---|
| • Formed by end to end overlap of orbitals | • Formed by lateral overlap of orbitals |
| • Strong Bond | • Weak Bond |
| • The orbitals involved are: s-s, s-p and p-p | • Formed by p-p overlap only |
| • Free rotation about $\sigma$ bonds is possible | • Rotation is restricted |

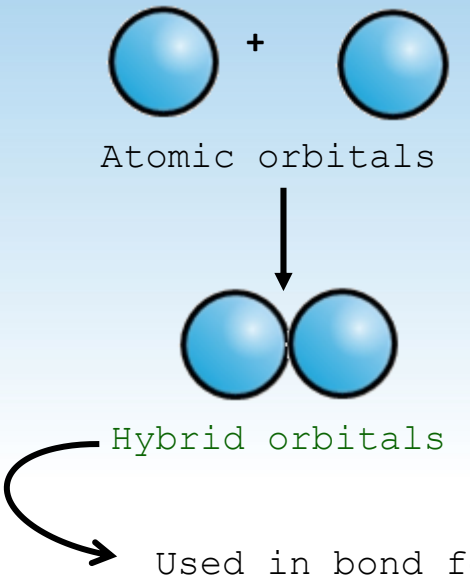
## Summary

- According to VSEPR model, molecular geometry is determined by repulsions between lp - lp; lp - bp and bp - bp
- The order of these repulsions being: lp – lp > lp – bp > bp - bp
- Valence bond Theory (VBT) discusses bond formation in terms of overlapping of atomic orbitals
- Formation of $H_2$ molecule from two hydrogen atoms involves the overlap of the 1s orbitals of the two hydrogen atoms which are singly occupied.
- Types of overlapping – positive, negative and zero
- Covalent bond – sigma and pi

# Hybridisation



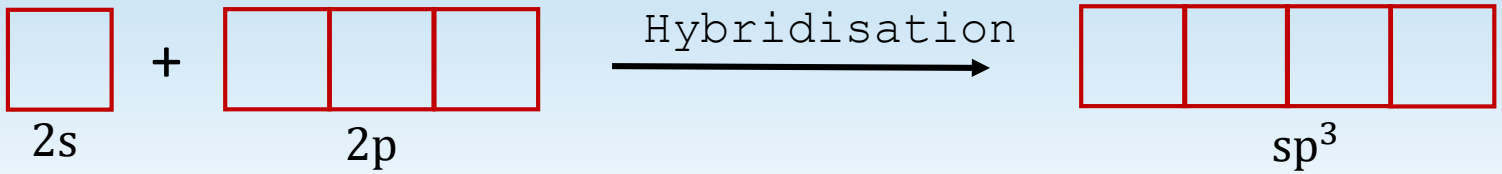
Linus Carl Pauling



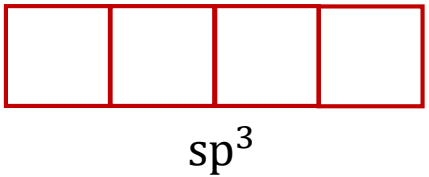

Intermixing of the orbitals of slightly different energies, so as to redistribute their energies resulting in the formation of new set of orbitals of equivalent energies and shape.

# Salient features of Hybridisation

1. The number of hybrid orbitals is equal to the number of the atomic orbitals that get hybridised




2. The hybridised orbital are always equal in energy and shape

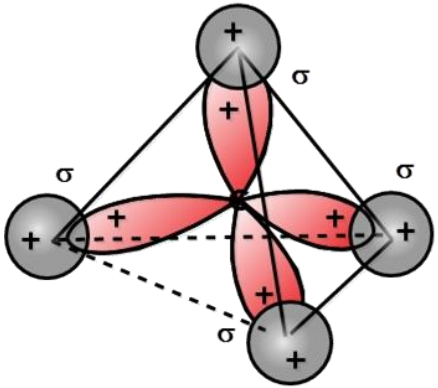



Equal energy and shape

# Salient features of Hybridisation

3. The hybrid orbitals are more effective in forming stable bonds
   pure atomic orbitals

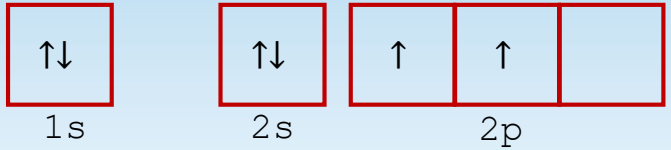
4. Hybrid orbitals are directed in space in some preferred direct
   have minimum repulsion between electron pairs and thus stabl
   arrangement



# Important conditions for Hybridisation

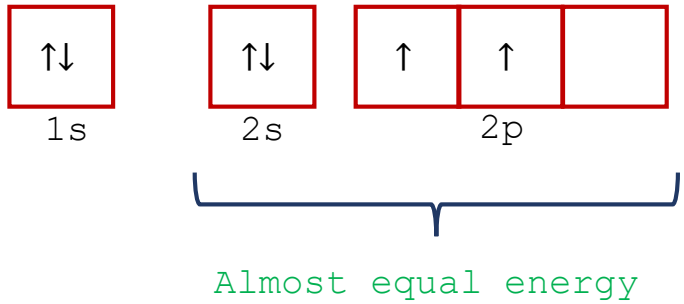
1. Orbitals present in the valence shell of the atoms are hybridized

For carbon :




2. Orbitals undergoing hybridization should have almost equal energy

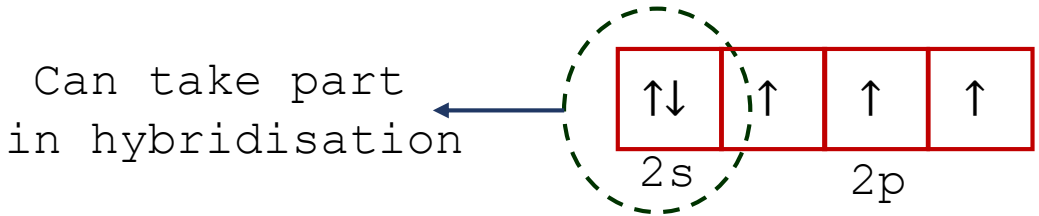
For carbon :



# Important conditions for Hybridisation

3. Promotion of electron is not essential condition prior to hybr

4. It is not necessary that only half filled orbitals participate
   In some cases, even filled orbitals of valence shell take par



# Types of Hybridisation




sp

$sp^2$

$sp^3$

$dsp^2$

$sp^3d$

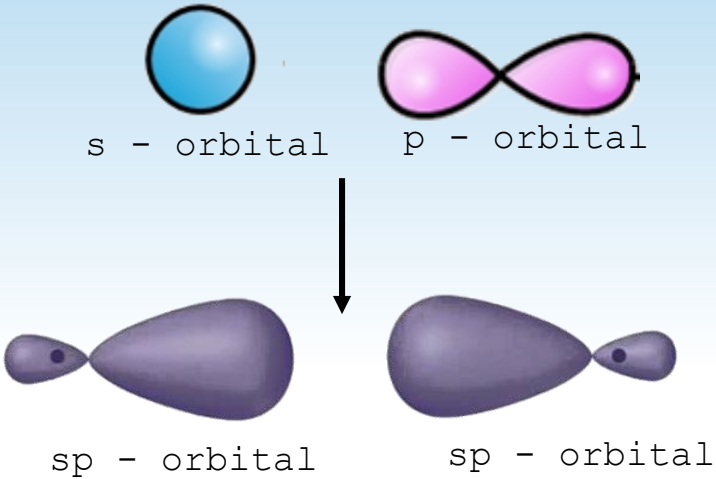
$dsp^3$

$sp^3d^2$

$d^2sp^3$

# sp Hybridisation

Also known as diagonal hybridisation



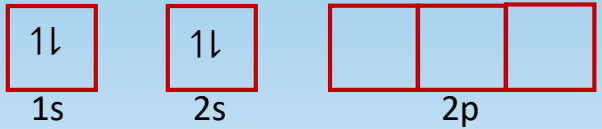

Each sp hybrid orbitals has 50% s-character and 50% p-character

Such molecules possesses linear geometry

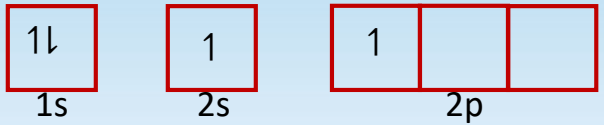
# Hybridisation in $BeCl_2$

Beryllium :

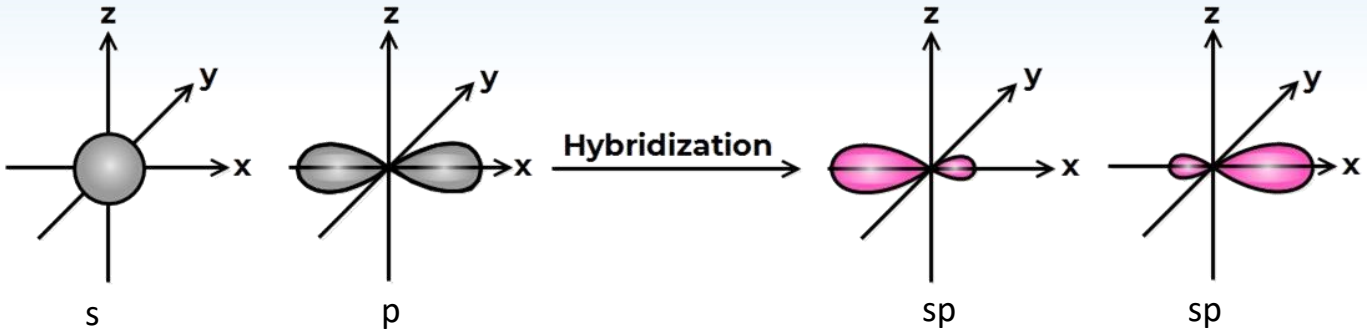
Ground state electronic configuration –




Excited state electronic configuration –




## Formation of sp Hybrid Orbitals

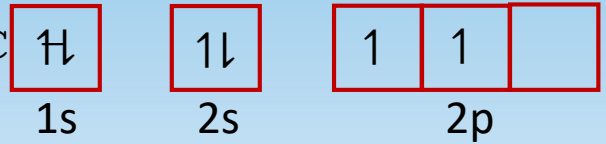



Formation of $BeCl_2$



# Hybridisation in $C_2H_2$

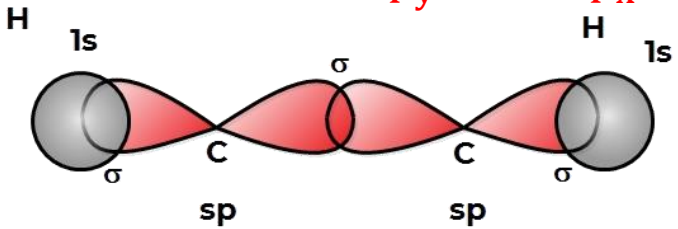
G.S electronic configuration of C



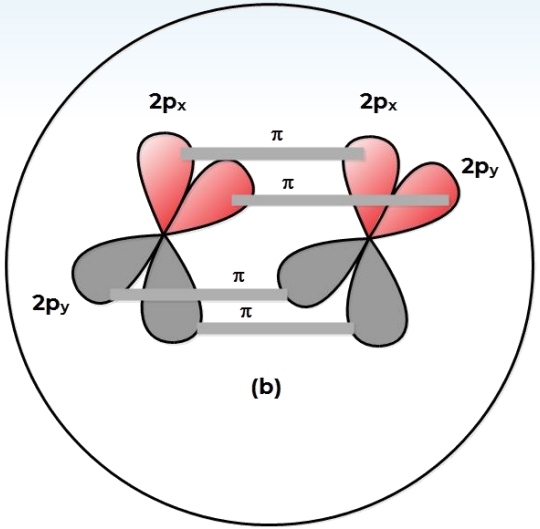

E.S electronic configuration of C




Both carbon atoms undergo sp-hybridisation having two unhybridised orbital i.e. $2p_y$ and $2p_x$

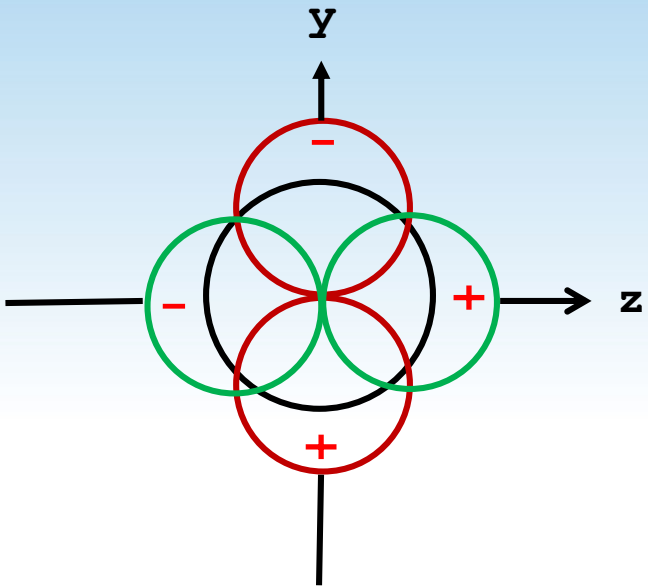



(a)

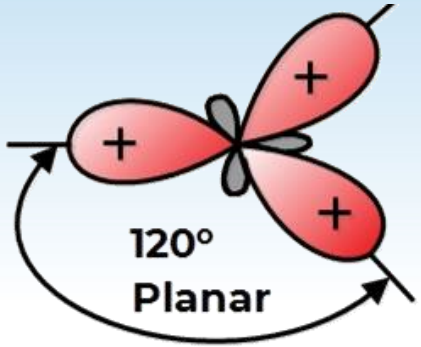



(b)

# $sp^2$ Hybridisation




$s + p_y + p_z$

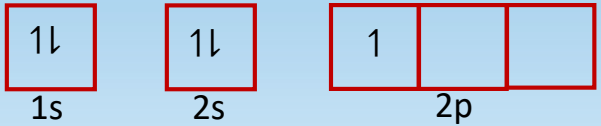



$sp^2$ hybrids

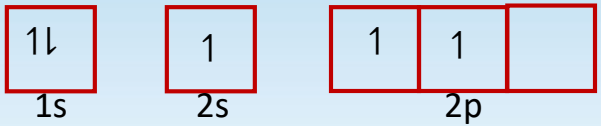
# $sp^2$ Hybridisation in $BCl_3$

Boron :

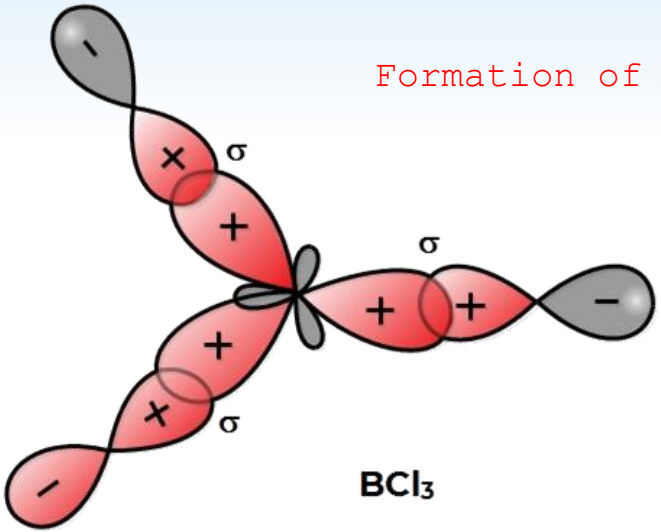
Ground state electronic configuration –




Excited state electronic configuration –

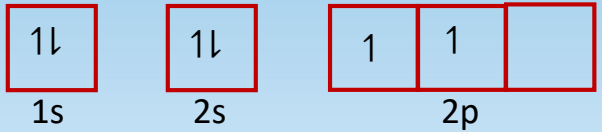



Formation of $BCl_3$

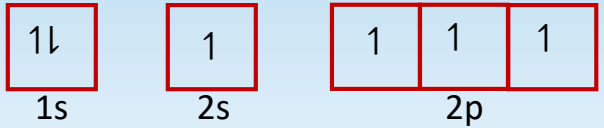


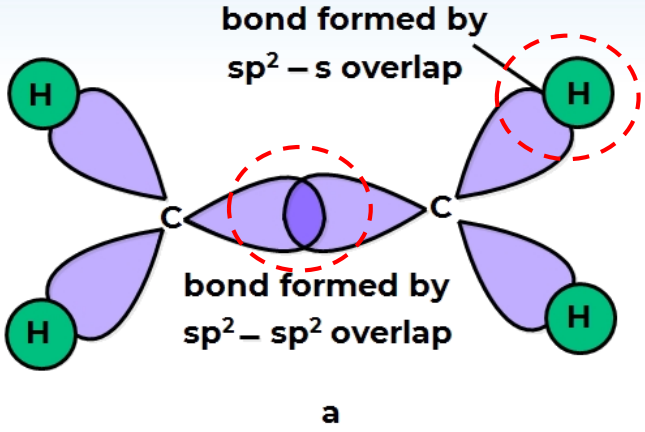
# Hybridisation in $C_2H_4$

G.S electronic configuration of C –

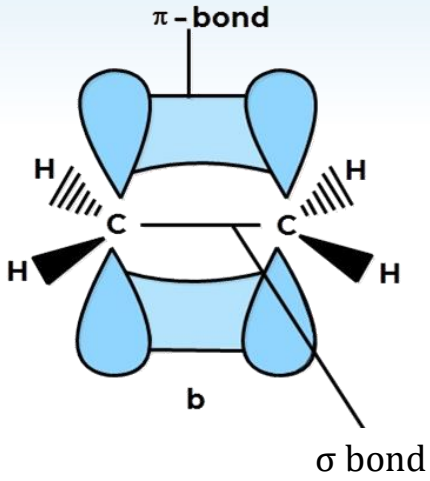



E.S electronic configuration of C –




a




C – C Bond length = 134 pm

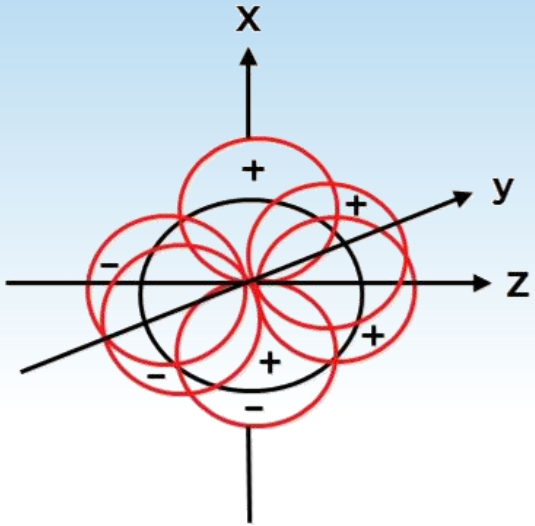
C – H bond length = 108 pm

H – C – H bond angle is 117.6°

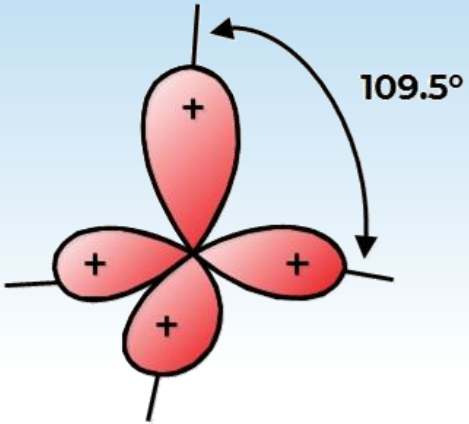
H – C – C angle is 121°

# $sp^3$ Hybridisation

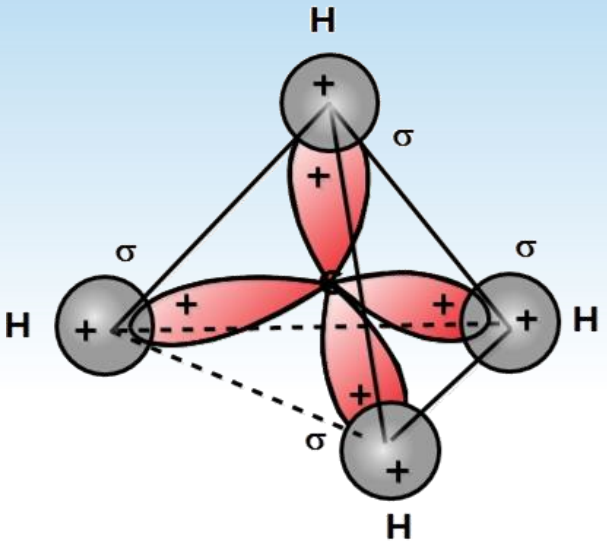
There is 25% s-character and 75% p-character in each $sp^3$ hybrid orbital




$s + p_x + p_y + p_z$

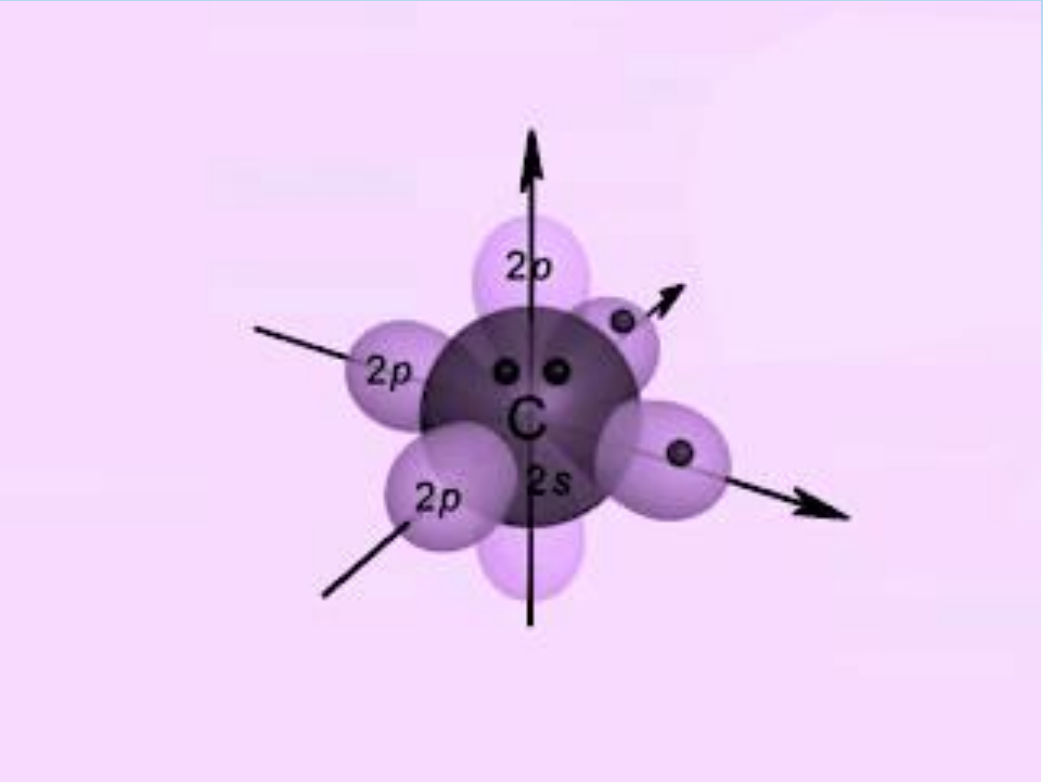



$sp^3$ hybrids



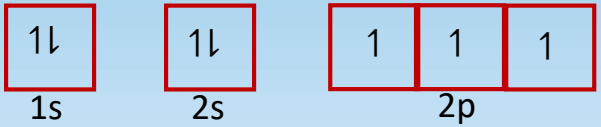

$CH_4$

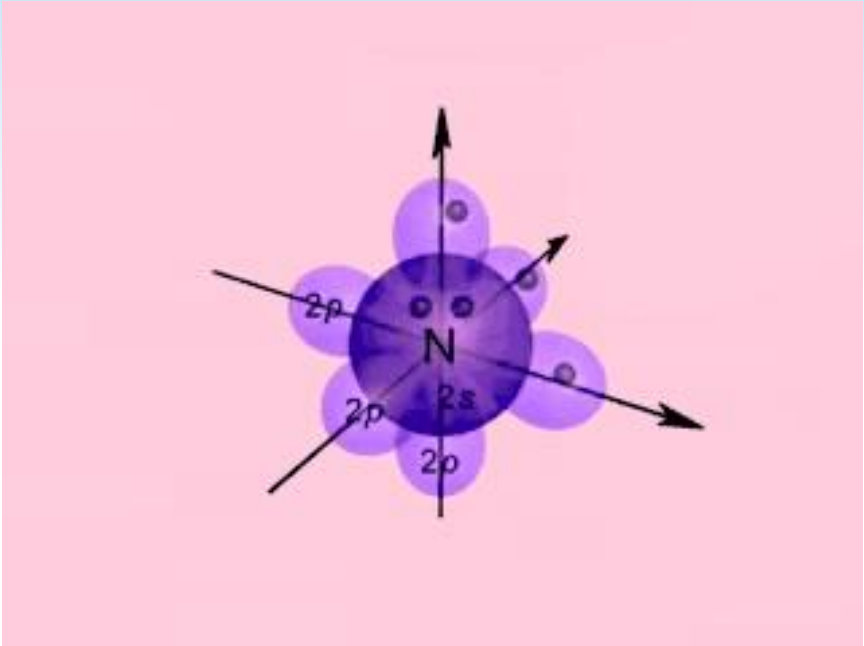
# $sp^3$ Hybridisation in $CH_4$



# $sp^3$ Hybridisation in $NH_3$

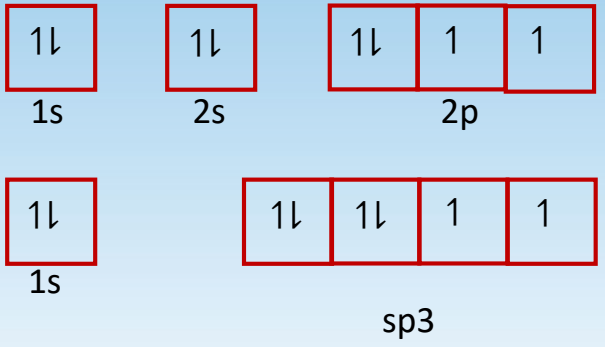
Nitrogen :
Ground state electronic configuration –



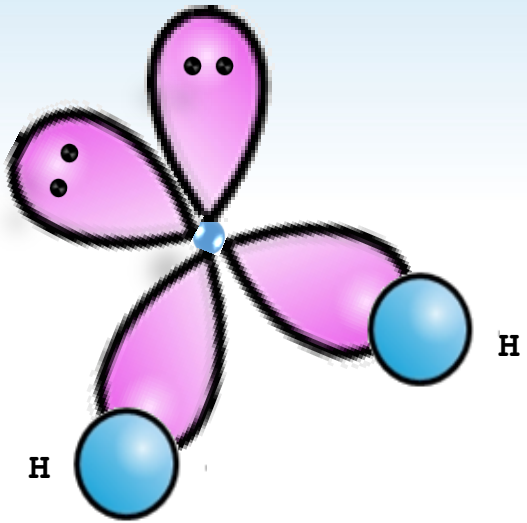



**Q.** Explain the hybridisation in $H_2O$ molecule?

**Sol.** G.S electronic configuration of O –




Hybrid orbitals –




$H_2O$

$sp^3$ hybrid orbitals acquire tetrahedral geometry

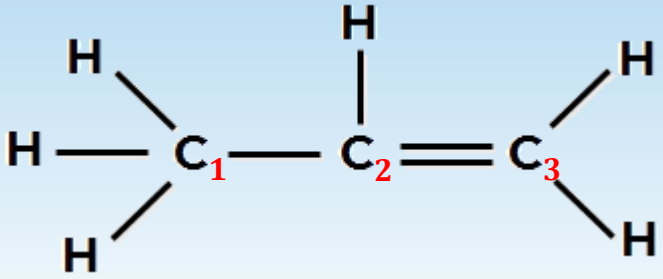
Bond angle reduced to 104.5°

V – shape or angular geometry

**Q.** Which hybrid orbitals are used carbon atoms in the following mo

a) $CH_3 - CH = CH_2$ b) $CH_3COOH$

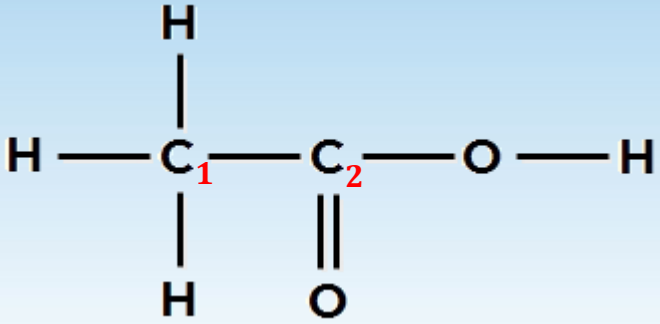
**Sol.**




$C_1$ is $sp^3$ hybridised

$C_2$ is $sp^2$ hybridised

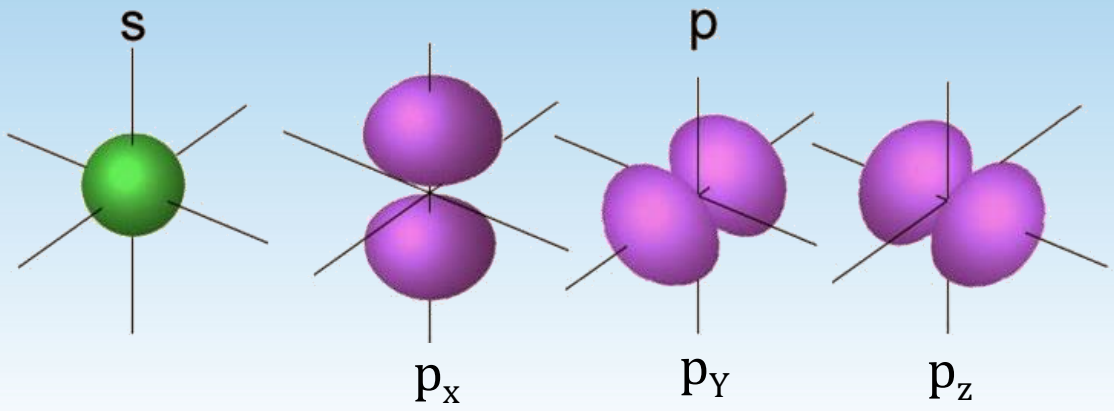
$C_3$ is $sp^2$ hybridised

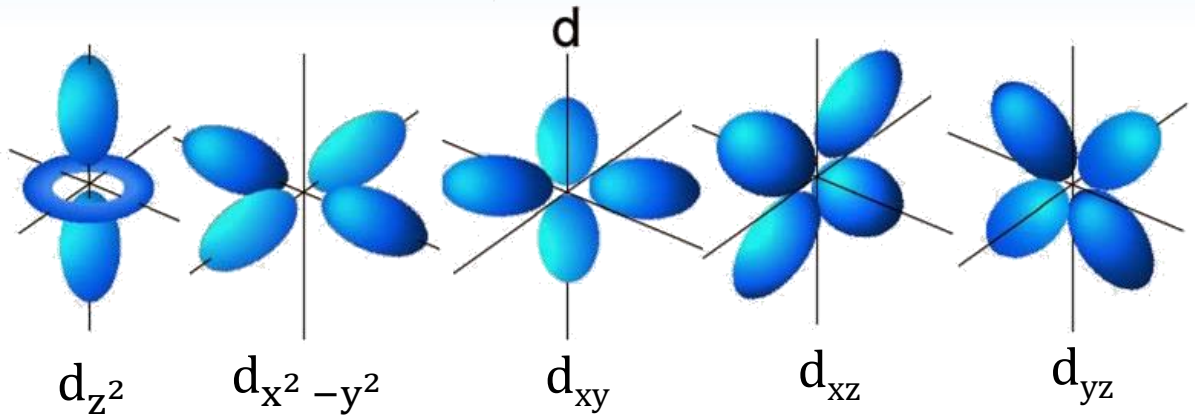



$C_1$ is $sp^3$ hybridised

$C_2$ is $sp^2$ hybridised

# Hybridisation of Elements involving d Orbitals

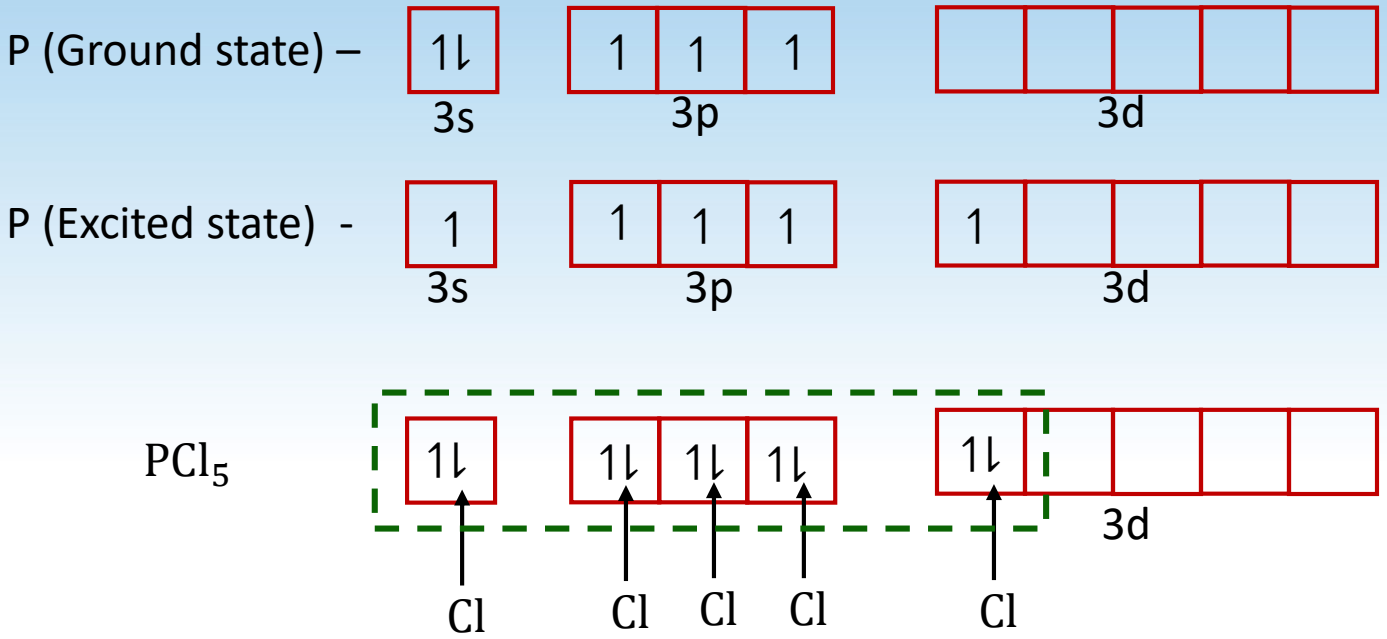


# Hybridisation of Elements involving d Orbitals

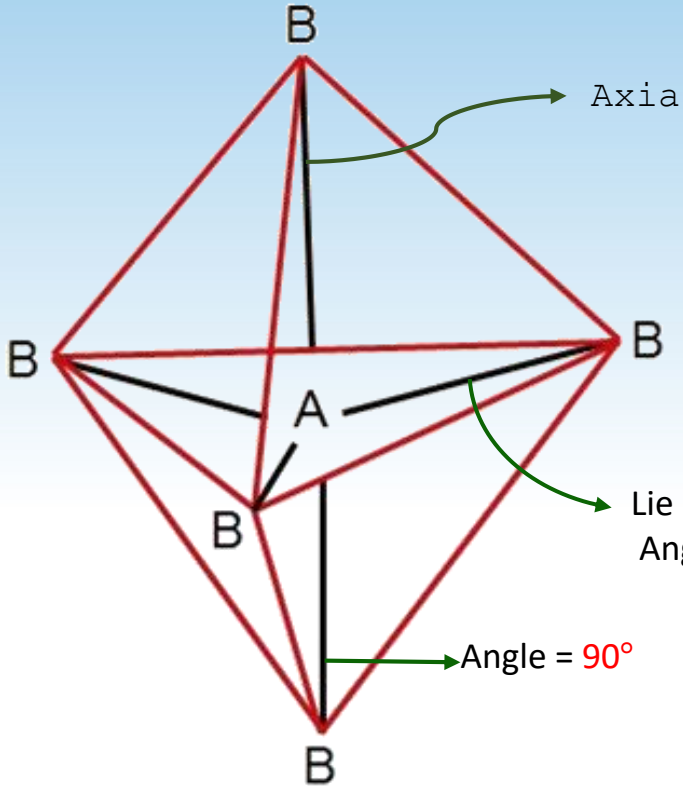
| Atomic orbitals | Hybridisation | Shapes of molecules or ions | Examples |
|---|---|---|---|
| $d + s + p(2)$ | $dsp^2$ | **Square planar** | $[Ni(CN)_4]^{2-}$, $[Pt(Cl)_4]^{2-}$ |
| $s + p(3) + d$ | $sp^3d$ | **Trigonal bipyramidal** | $PF_5, PCl_5$ |
| $d + s + p(3)$ | $dsp^3$ | **Square pyramidal** | $BrF_5, XeOF_4$ |
| $s + p(3) + d(2)$<br>$d(2) + s + p(3)$ | $sp^3d^2$<br>$d^2sp^3$ | **Octahedral** | $SF_6, [CrF_6]^{3-}$ |

# Formation of $PCl_5$ ($sp^3d$ Hybridisation )




$sp^3d$ hybridisation

# Formation of $PCl_5$ ($sp^3d$ Hybridisation )




Axial Bond pairs suffer more repulsive interaction

Axial bonds are slightly longer and weaker than the equatorial bonds

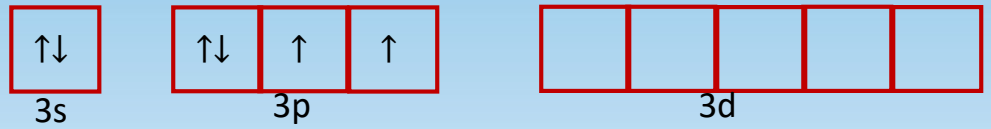
Lie in one plane
Angle = 120°

Angle = 90°

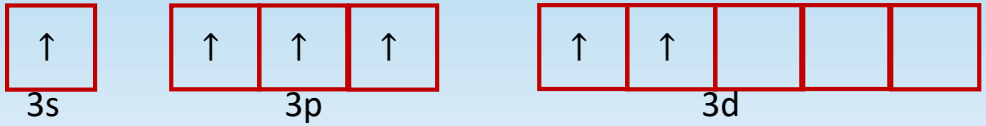
Trigonal bipyramidal

## Formation of $SF_6$ ($sp^3d^2$ Hybridisation )

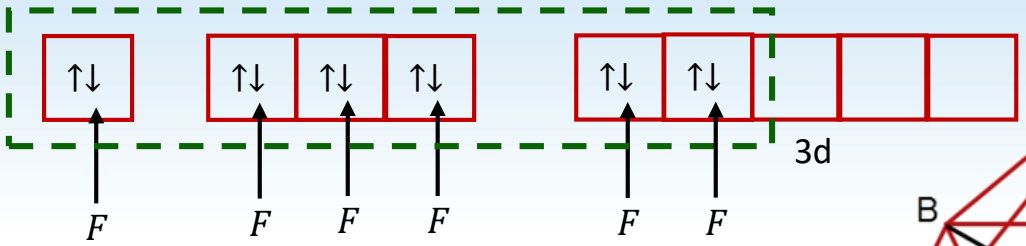
S (Ground state) –



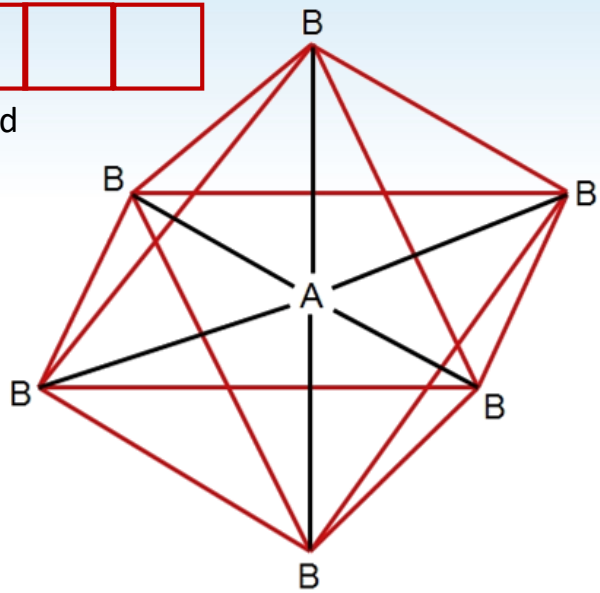

S (Excited state) –




$SF_6$



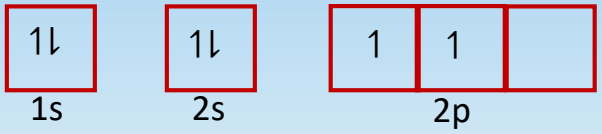

$sp^3d^2$ hybridisation



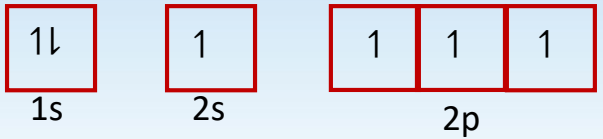

Octahedral geometry of $SF_6$ molecule

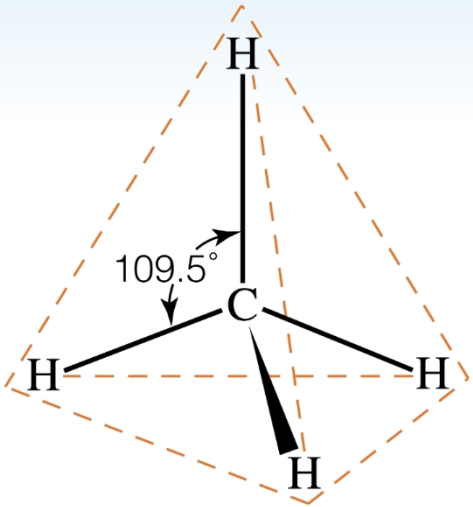
**Q.** Apart from tetrahedral geometry, another possible geometry for $CH_4$ is Square planar. Explain why $CH_4$ is not square planar ?

**Sol.** G.S electronic configuration of C




E.S electronic configuration of C



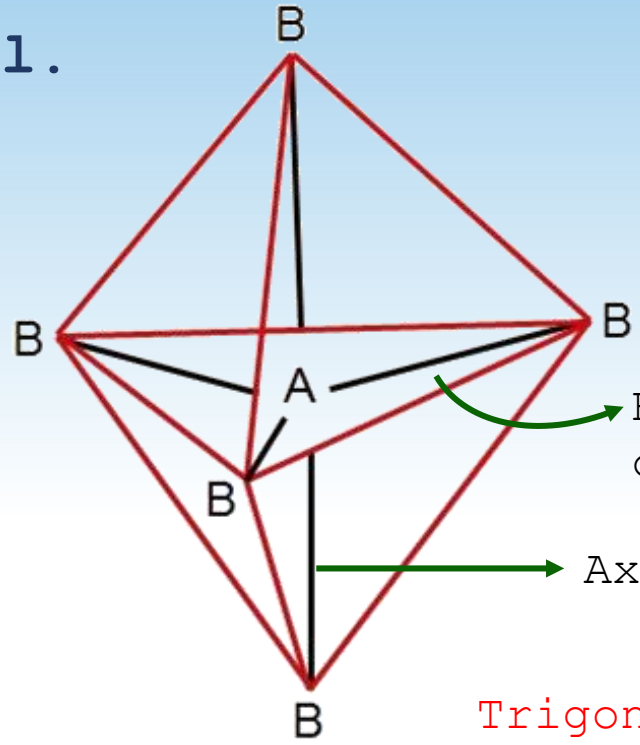




For a square planar shape, the hybridisation of the central atom has to be $dsp^2$

However, an atom of carbon does not have d-orbitals to undergo $dsp^2$ hybridisation

**Q.** Why are the axial bonds longer as compared to equatorial bonds?

**Sol.**




Five $P-Cl$ $(\sigma)$ sigma bonds in $PCl_5$

Equatorial bond making $120°$ with each other

Axial bond making $90°$ with the plane

Trigonal bipyramidal

Axial bond pair suffer more repulsion than equatorial bond pair. Hence, axial bonds are slightly longer than equatorial bond

## Summary

- Pauling introduced the concept of hybridisation of atomic orbitals.
- $sp, sp^2, sp^3$ hybridisation of atomic orbitals of $Be, B, C, N$ and $O$ are used to explain the formation and geometrical shapes of molecules like $BeCl_2, BCl_3, CH_4, NH_3$ and $H_2O$
- They also explain the formation of multiple bonds in molecules like $C_2H_2$ and $C_2H_4$

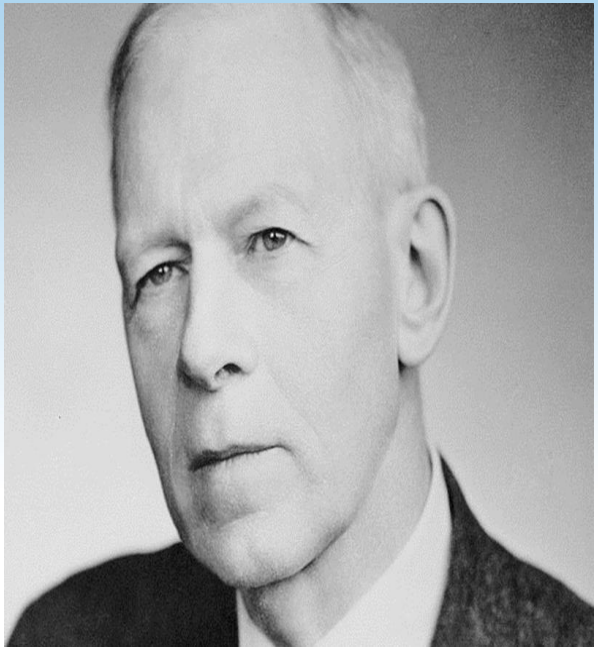
### Types of Hybridisation

| | | | | | |
|---|---|---|---|---|---|
| s + p | sp | $sp^2$ | $sp^3$ | | |
| s + p + d | $dsp^2$ | $sp^3d$ | $dsp^3$ | $sp^3d^2$ | $d^2sp^3$ |

# Molecular Orbital Theory



Friedrich Hund

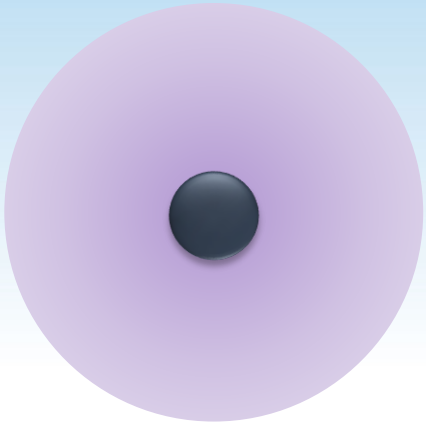


Robert S. Mulliken

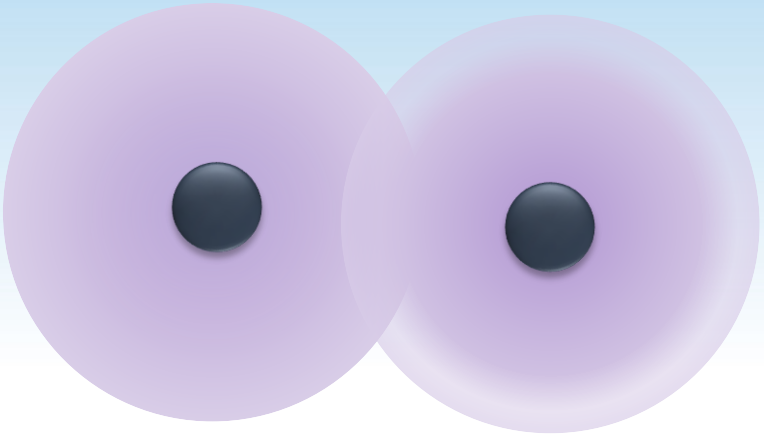
Molecular Orbital Theory was developed in 1932

# Salient features of Molecular Orbital Theory

1. The electrons in a molecule are present in the various molecular orbitals as the electrons of atoms are present in the various atomic orbitals



Electrons in atomic orbitals



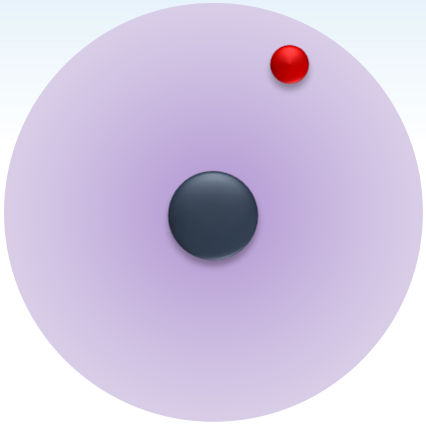
Electron in molecular orbitals

2. The atomic orbitals of comparable energies and proper symmetry combine to form molecular orbitals

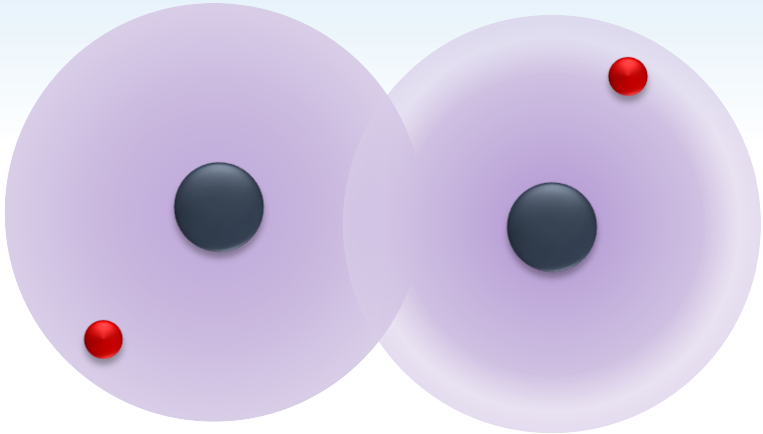
# Salient features of Molecular Orbital Theory

3. Electron in an atomic orbital is influenced by one nucleus

Electron in a molecular orbital is influenced by two or more nuclei depending upon the number of atoms in the molecule



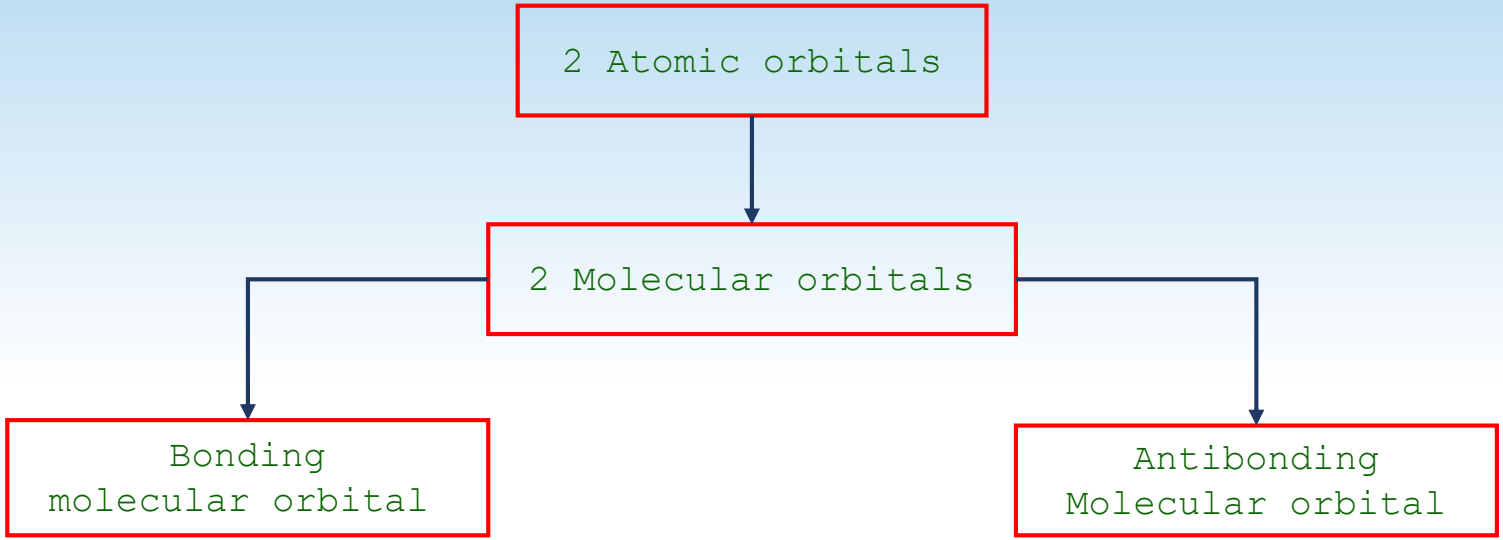
Monocentric atom



Polycentric molecule

## Salient features of Molecular Orbital Theory

4. Molecular orbitals formed = Number of combining atomic orbitals




Lower energy and greater stability

Higher energy and lesser stability

## Salient features of Molecular Orbital Theory

5. The electron probability distribution around a group of nuclei in a molecule is given by molecular orbital.

6. Like atomic orbitals, molecular orbitals are filled in accordance with –

Aufbau principle

Obeying-

Pauli's exclusion principle

Hund's rule

# Linear Combination of Atomic Orbitals (LCAO)

According to wave mechanics –

Atomic orbitals → $\psi$ (wave function) → Amplitude of electron wave

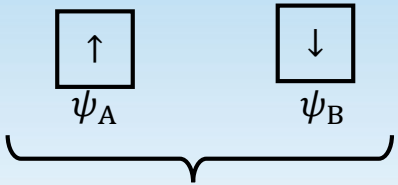
Limitations – Difficult to explain for multielectron systems

Solution – LCAO

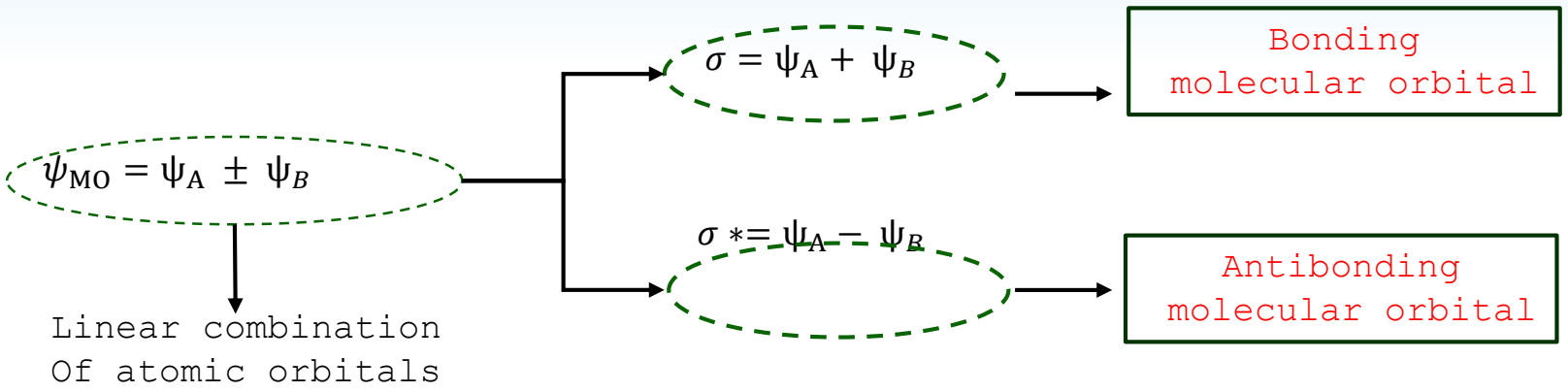
# Linear Combination of Atomic Orbitals (LCAO)

Homonuclear diatomic hydrogen molecule, consisting of A and B atoms

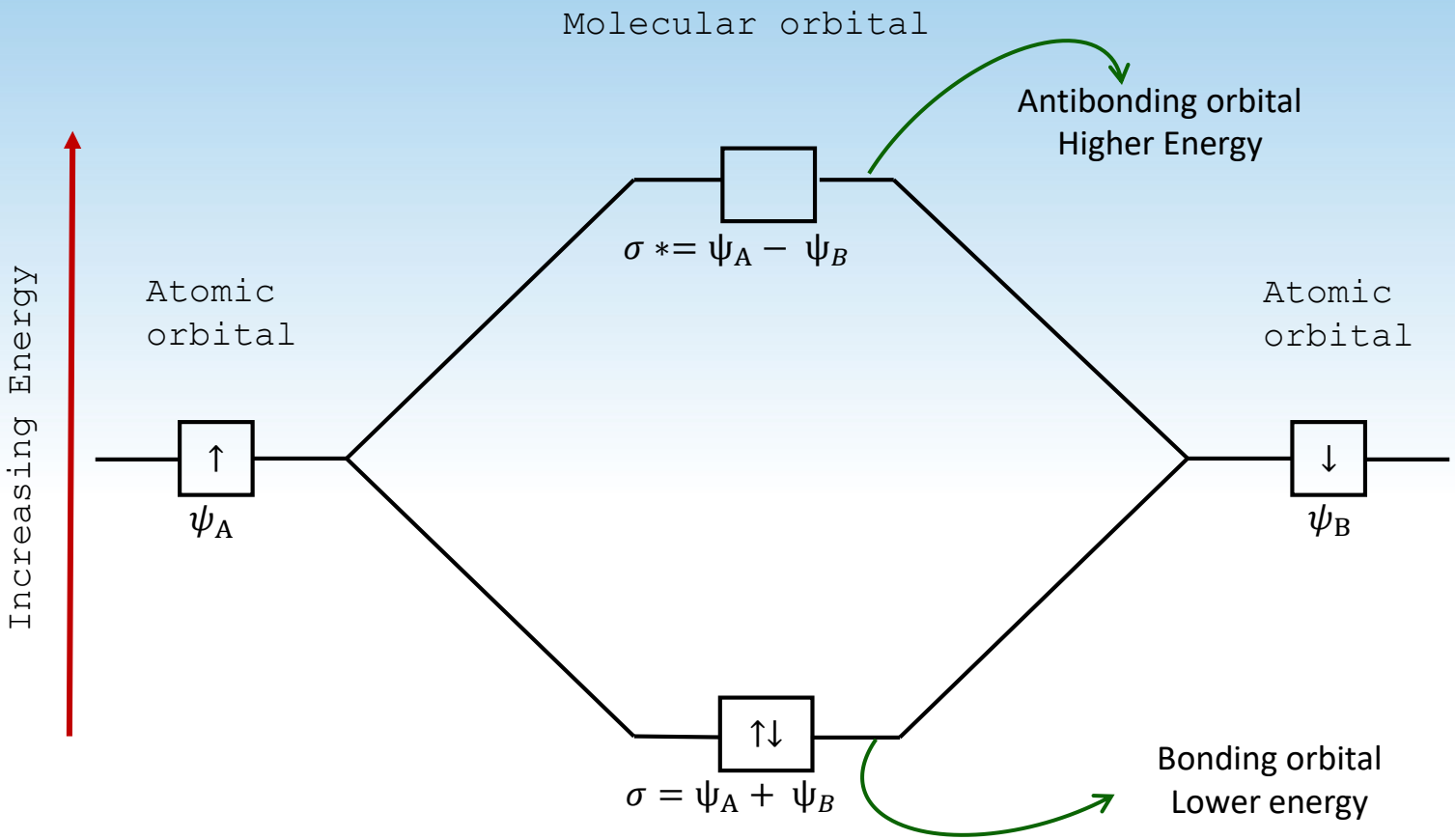
Each hydrogen atom in ground state has one electron in 1s orbital



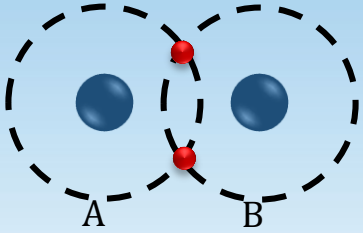

Atomic orbitals

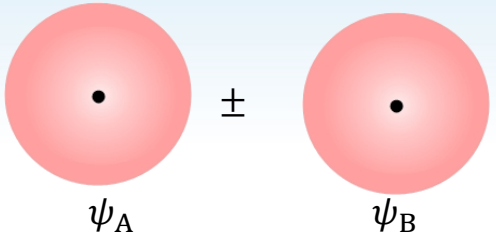


# Linear Combination of Atomic Orbitals (LCAO)



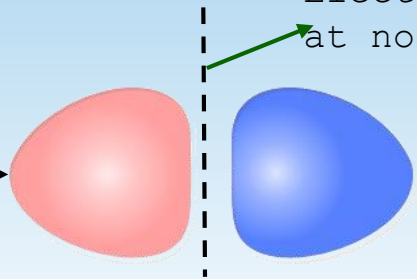
# Linear Combination of Atomic Orbitals (LCAO)




$\sigma * = \psi_A - \psi_B$

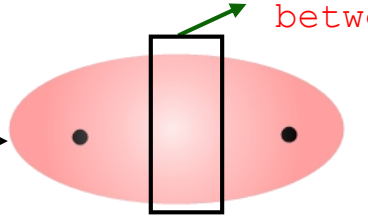
Destructive interference



Electron density is zero at nodal plane

Constructive interference

$\sigma = \psi_A + \psi_B$

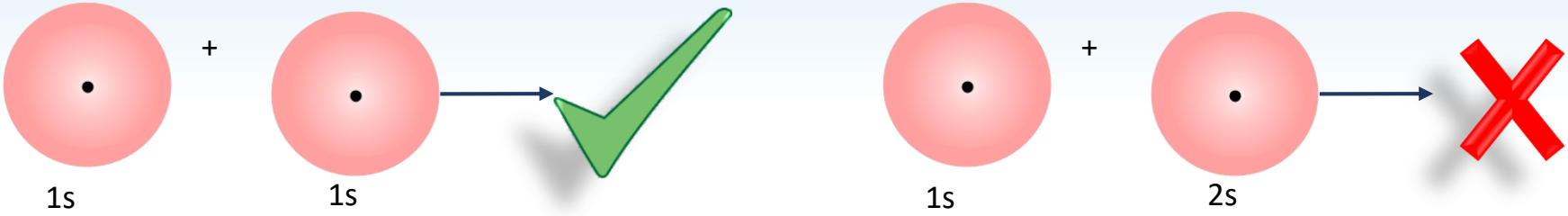


Electron density is between the two nuclei

The total energy of two molecular orbitals remains the same as that of two original atomic orbitals

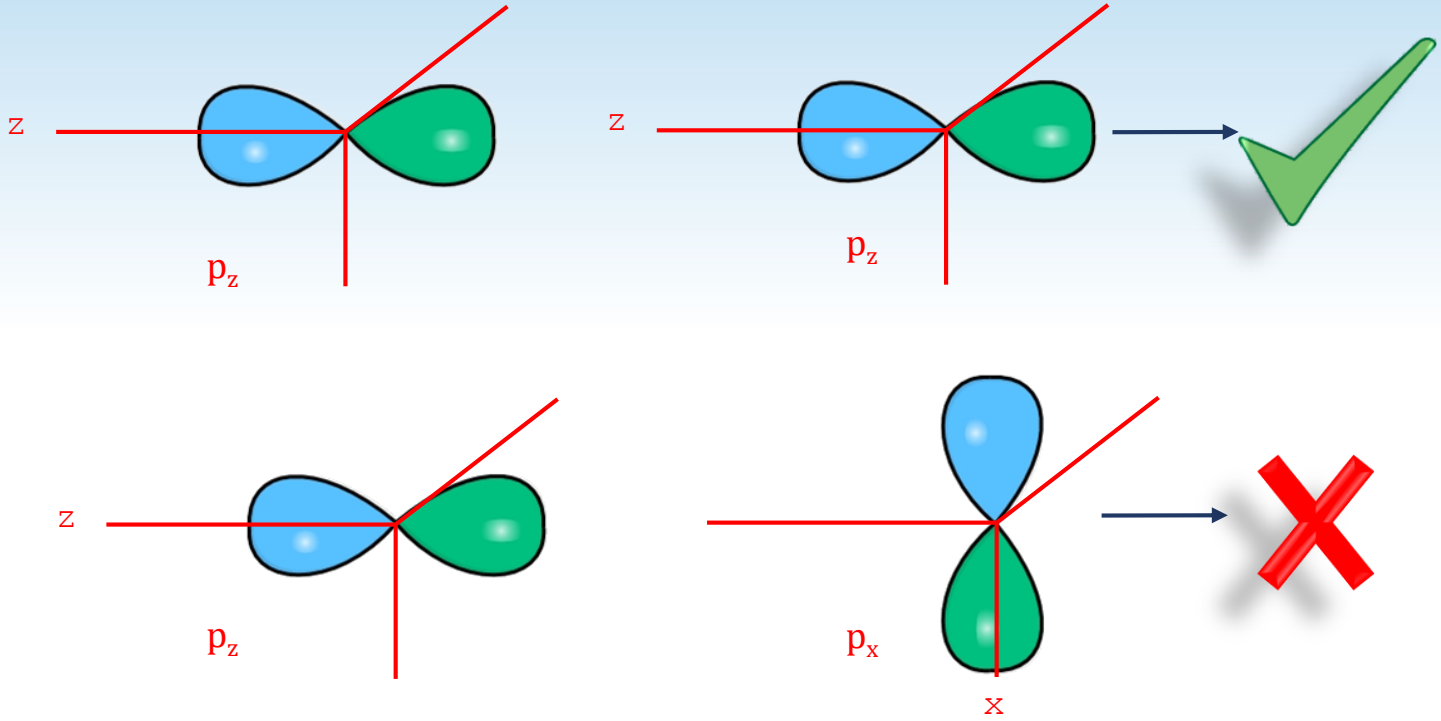
# Conditions for the Combination of Atomic Orbitals

1. The combining atomic orbitals must have the same or nearly the same energy



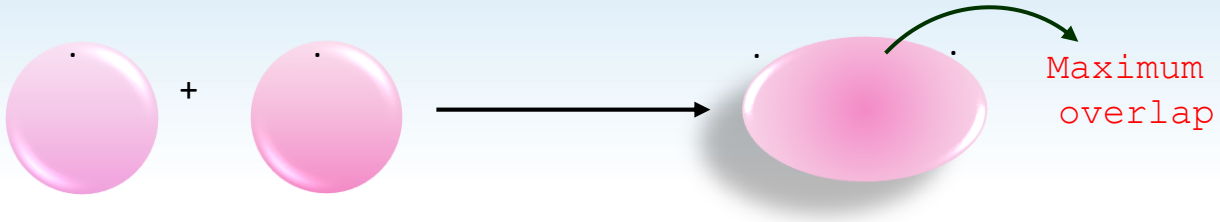
# Conditions for the Combination of Atomic Orbitals

2. The combining atomic orbitals must have the same symmetry about the molecular axis (Z by convention)



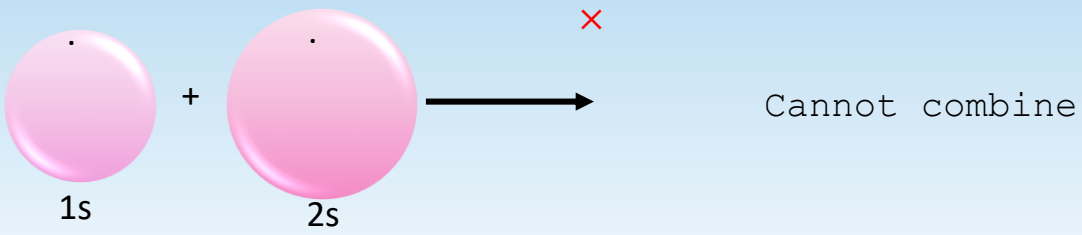
# Conditions for the Combination of Atomic Orbitals

3. The combining atomic orbitals must overlap to the maximum extent.



**Q.** Can 1s orbital combine with another 2s orbital?

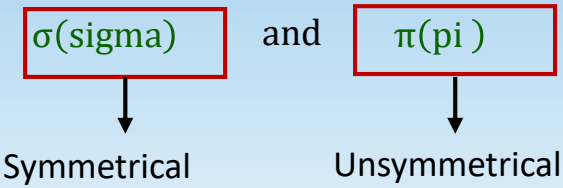
**Sol.** 1 s orbital cannot combine with 2s orbital.



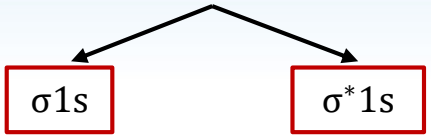

1 s orbital can combine with another 1s orbital but not with 2s orbital because
the energy of 2s orbital is appreciably higher than that of 1 s orbital

# Types of Molecular Orbitals

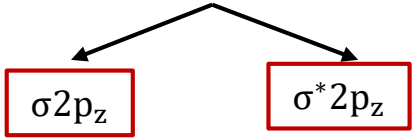
Molecular orbitals of diatomic molecules are designated as :




Linear combination of 1s orbitals centred on two nuclei produces two molecular orbitals which are symmetrical around the bond axis




If internuclear axis is taken to be in the z-direction, it can be seen that a linear Combination of $2p_z$ orbitals of two atoms also produces two sigma MO

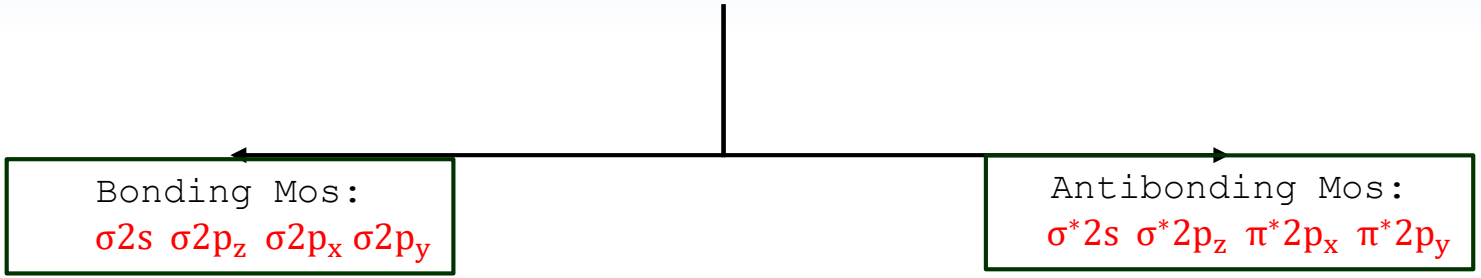


## Energy level Diagram for Molecular orbitals

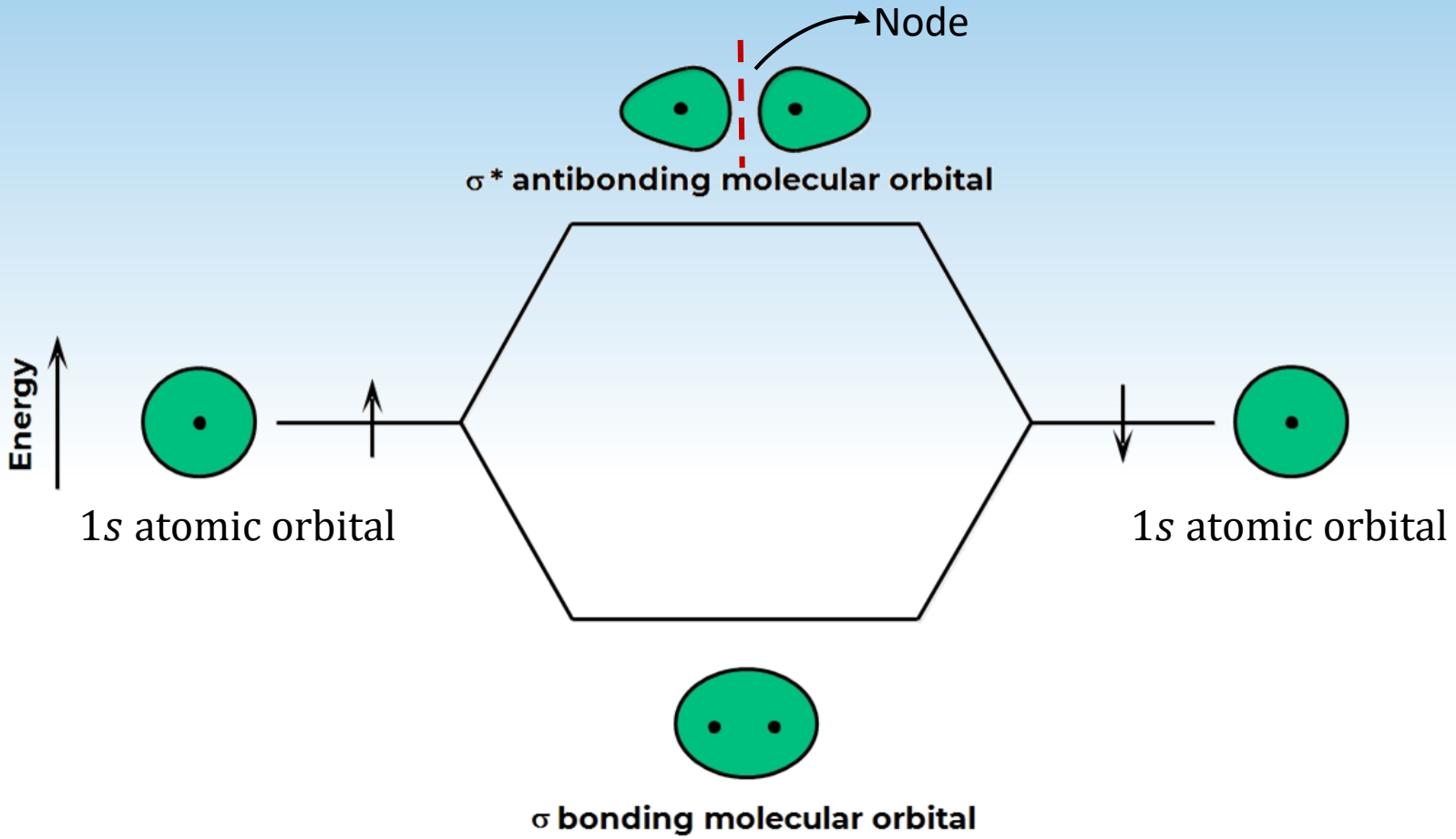
Two $1s$ atomic orbitals combine to form two molecular orbitals :



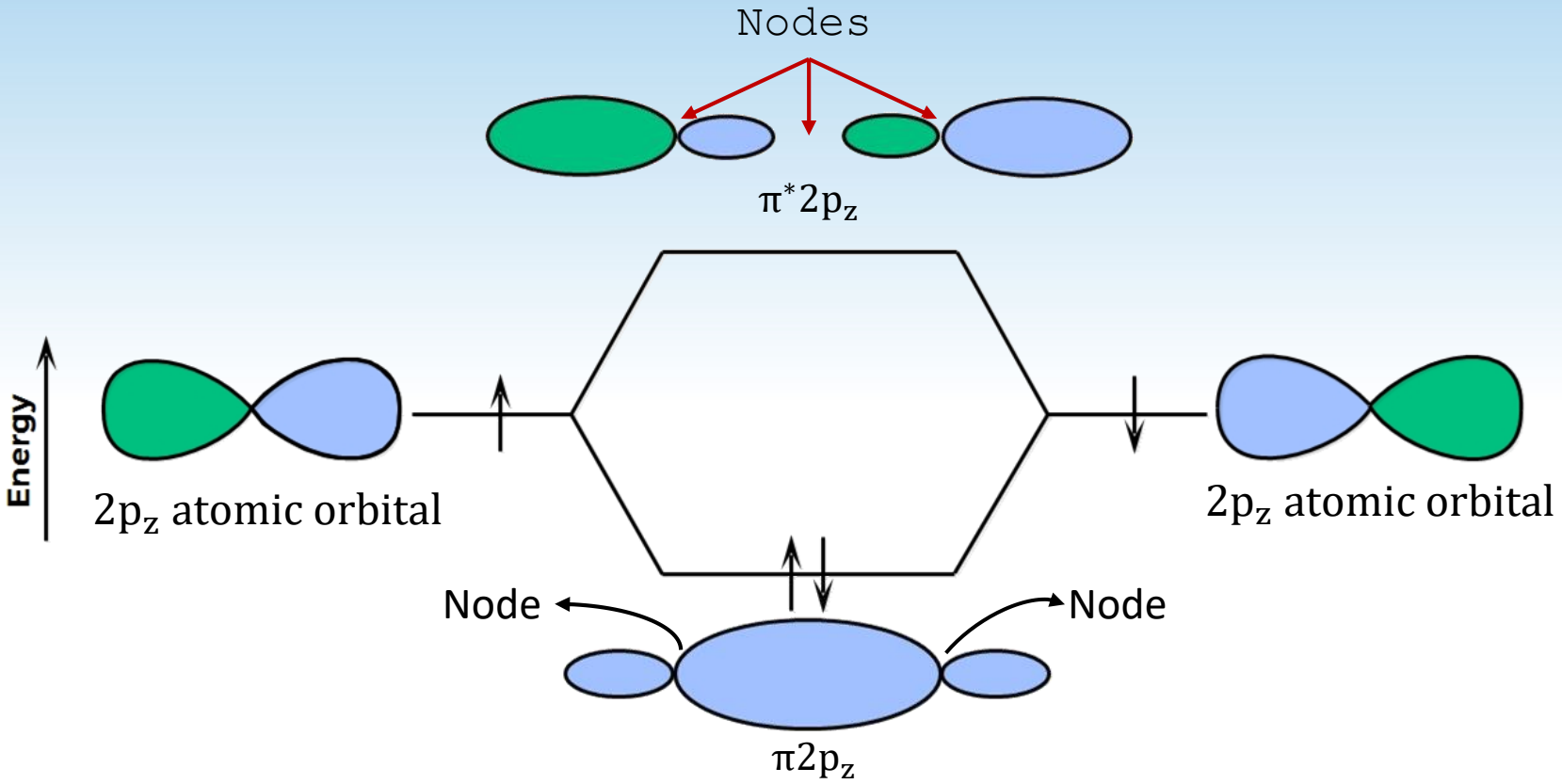

$2s$ and $2p$ atomic orbitals give rise to the following eight molecular orbitals:



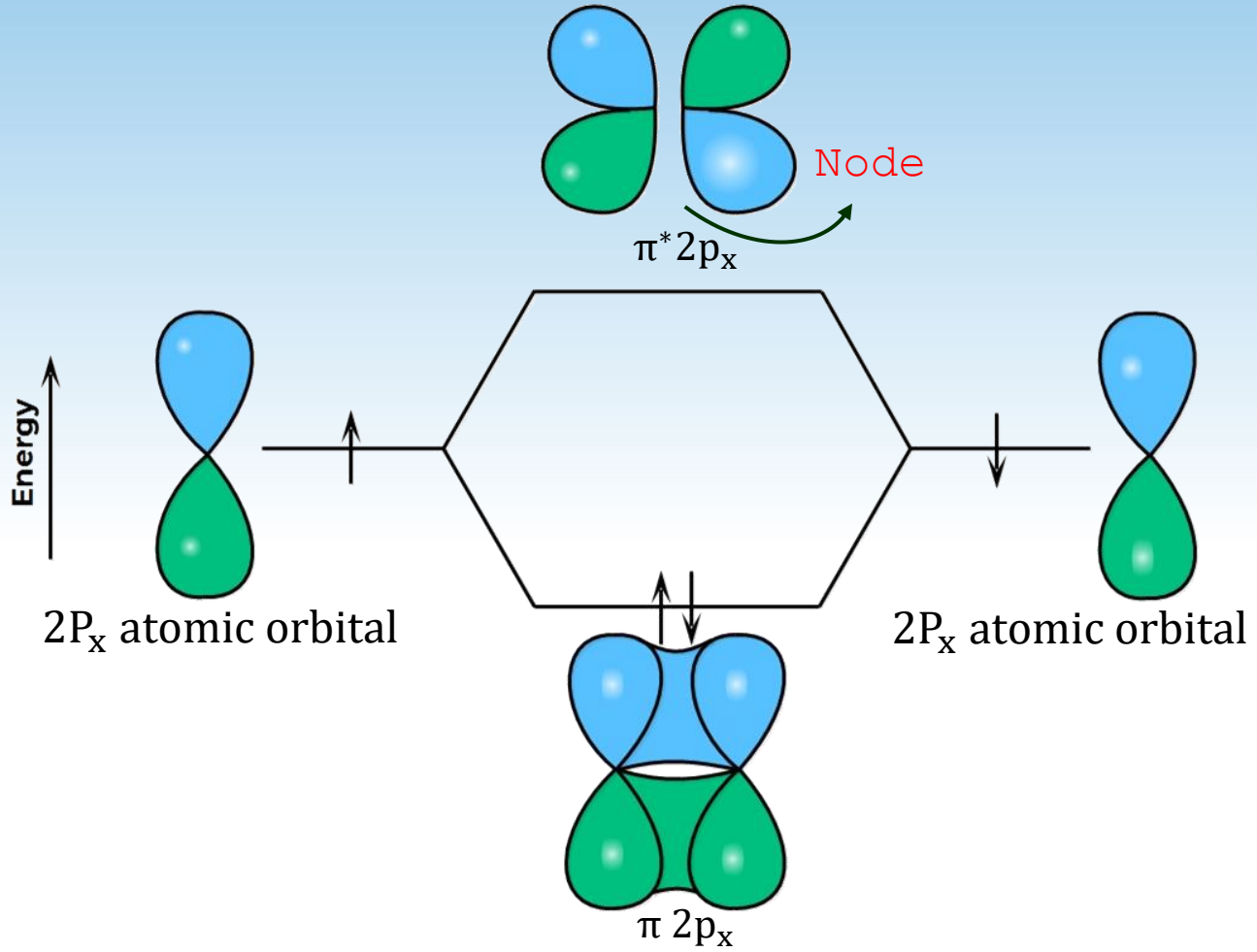
# Energy level Diagram for 1s Atomic Orbitals



# Energy level Diagram for $2p_z$ Atomic Orbitals



# Energy level Diagram for $2p_x$ Atomic Orbitals



# Energy level Diagram for Molecular orbitals

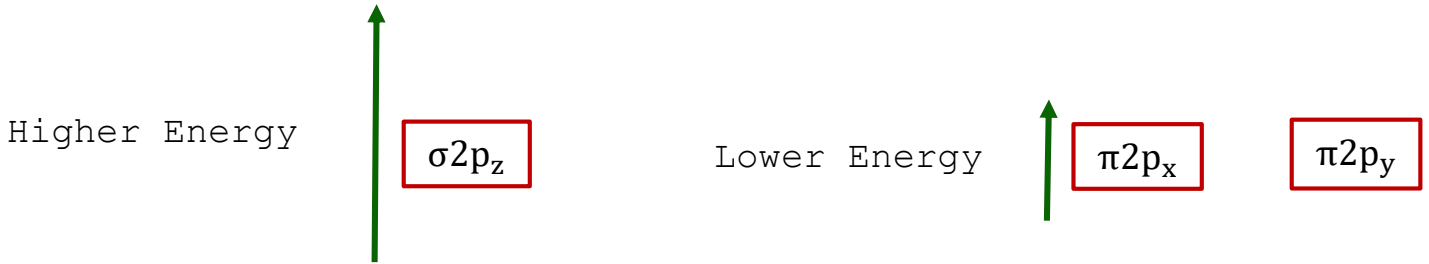
Experimentally determined from spectroscopic data for homonuclear diatomic molecules of second row elements of the periodic table

The increasing order of energies of various molecular orbitals for $O_2$ and $F_2$ is :

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$$

For $B_2, C_2, N_2$ the increasing order of energies of various Molecular orbital is:

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$$



**Q.** Arrange the following molecular orbital in increasing order of their energy for carbon molecule.

$$\sigma^*2s,\quad \sigma^*2p_z,\quad (\pi^*2p_x = \pi^*2p_y),\quad \sigma 2s,\quad \sigma 1s,\quad \sigma^*1s,\quad (\pi 2p_x = \pi 2p_y),\quad \sigma 2p_z$$

**Sol.** Increasing order of energy :

$$\sigma 1s < \sigma^*1s < \sigma 2s < \sigma^*2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^*2p_x = \pi^*2p_y) < \sigma^*2p_z$$

# Electronic Configuration and Molecular Behaviour

Distribution of electrons in various molecular orbitals is called the electronic configuration of the molecule

**Stability of molecules**

If,

$N_b$ → Number of electrons occupying bonding orbitals

$N_a$ → Number of electrons occupying antibonding orbitals

Then,

i. The molecule is stable if $N_b$ is greater than $N_a$

ii. The molecule is unstable if $N_b$ is less than $N_a$

# Electronic Configuration and Molecular Behaviour

## Bond Order

$$\text{Bond Order (BO)} = \frac{1}{2}(N_b - N_a)$$

$N_a$: Number of electrons in the antibonding orbitals

$N_b$: Number of electrons in the bonding orbitals

Positive bond order (i.e., $N_b > N_a$) ⟶ Stable molecule

Negative (i.e., $N_b < N_a$) or zero (i.e., $N_b = N_a$) ⟶ Unstable molecule

# Electronic Configuration and Molecular Behaviour

## Nature of bond

Integral bond order values

1 → Single Bond

2 → Double Bond

3 → Triple Bond

**Bond-Length :** bond order ↑ Bond length ↓

## Magnetic Nature

- Diamagnetic (repelled by magnetic field)
- Paramagnetic (Attracted by magnetic field)

**Q.** If a molecule has 4 electrons in antibonding orbital and 10 electrons in bonding orbital calculate its Bond order.

**Sol.**

$$\text{Bond Order (BO)} = \frac{1}{2}(N_b - N_a)$$

$$= \frac{1}{2}(10 - 4)$$

$$= \frac{1}{2}(6)$$

$$= 3$$

Therefore, its bond order is 3

## Summary

- MOT describes bonding in terms of the combination and arrangement of atomic orbitals to form molecular orbitals that are associated with the molecule as a whole

- The number of molecular orbitals are always equal to the number of atomic orbitals from which they are formed

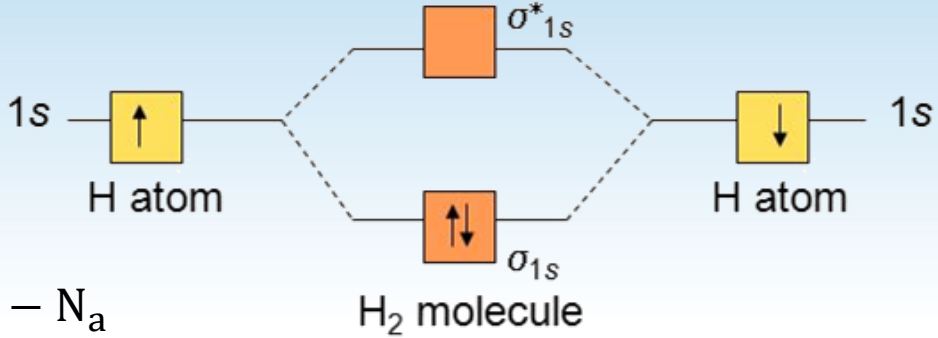
- Bonding MO increase electron density between the nuclei and are lower in energy than the individual atomic orbitals

- Antibonding MO have a region of zero electron density between the nuclei and have more energy than the individual atomic orbitals

# Hydrogen Molecule ($H_2$)

Each hydrogen atom has one electron in 1s orbital

Electronic configuration of Hydrogen molecule is: $(\sigma 1s)^2$




$$\text{Bond Order} = \frac{N_b - N_a}{2}$$

$$= \frac{2-0}{2} = 1 \rightarrow \text{Single covalent bond}$$

B. D. E = 438 kJ $mol^{-1}$

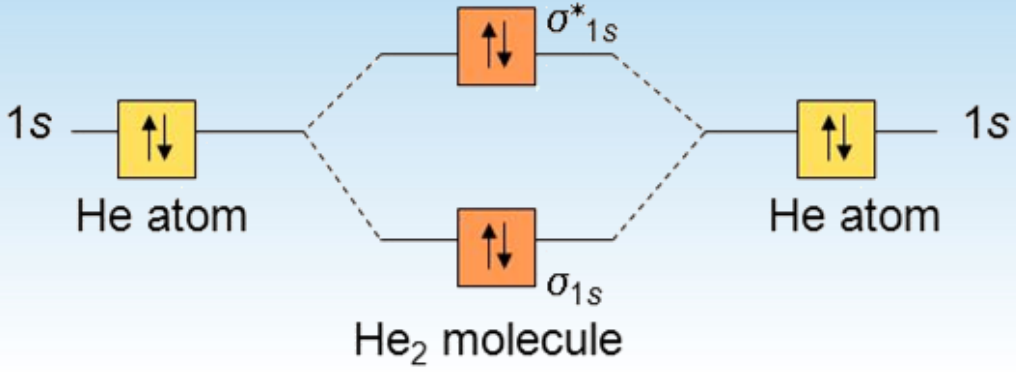
Bond Length = 74 pm

**H — H**

Diamagnetic

# Helium Molecule ($He_2$)

Electronic configuration of Helium molecule is: $(\sigma 1s)^2 \ (\sigma^* 1s)^2$

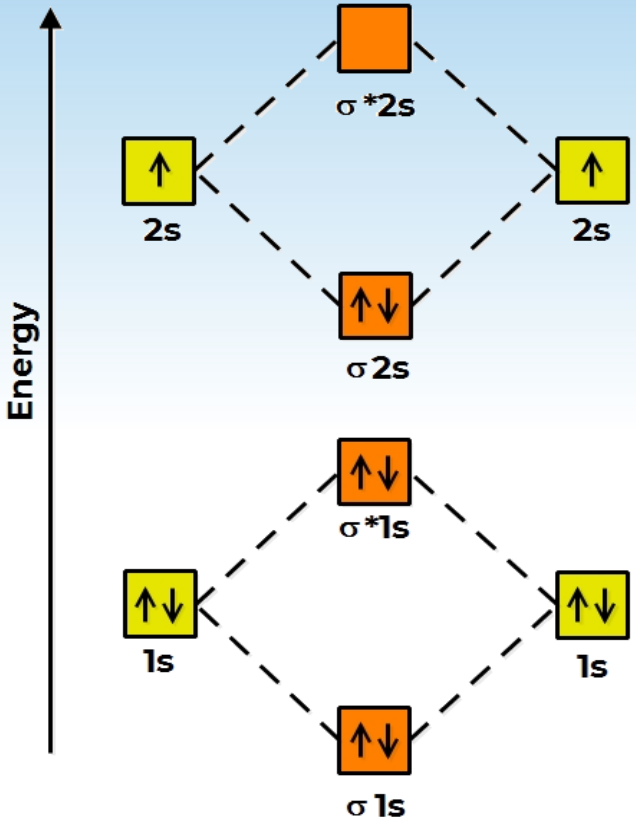



$$\text{Bond Order} = \frac{N_b - N_a}{2} \qquad = \frac{2-2}{2} = 0$$

$He_2$ molecule is unstable and does not exist

**Q.** Explain the bonding in Lithium molecule.

**Sol.** Electronic configuration of Lithium molecule is: $(\sigma 1s)^2 \ \ (\sigma^* 1s)^2 \, (\sigma 2s)^2$




$$\text{Bond Order} = \frac{N_b - N_a}{2} \qquad = \frac{4-2}{2} = 1$$

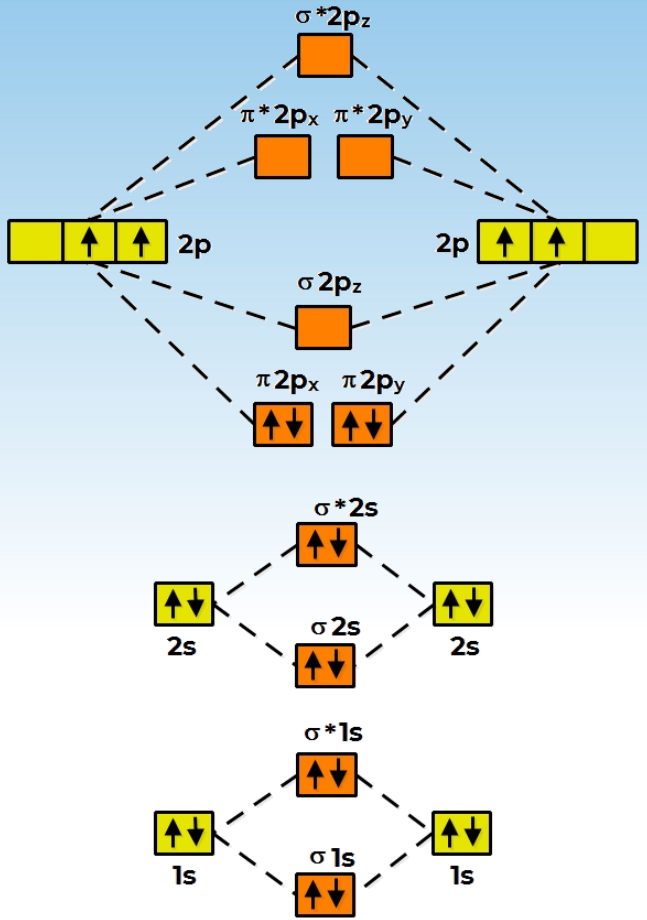
Li—Li

Li molecule is stable

Diamagnetic

In vapour phase

# Carbon Molecule ($C_2$)




Electronic configuration $C_2$:

$$(\sigma 1s)^2 \quad (\sigma^* 1s)^2 \, (\sigma 2s)^2 \quad (\sigma^* 2s)^2 \quad \left(\pi 2p_x^2 = \pi 2p_y^2\right)$$

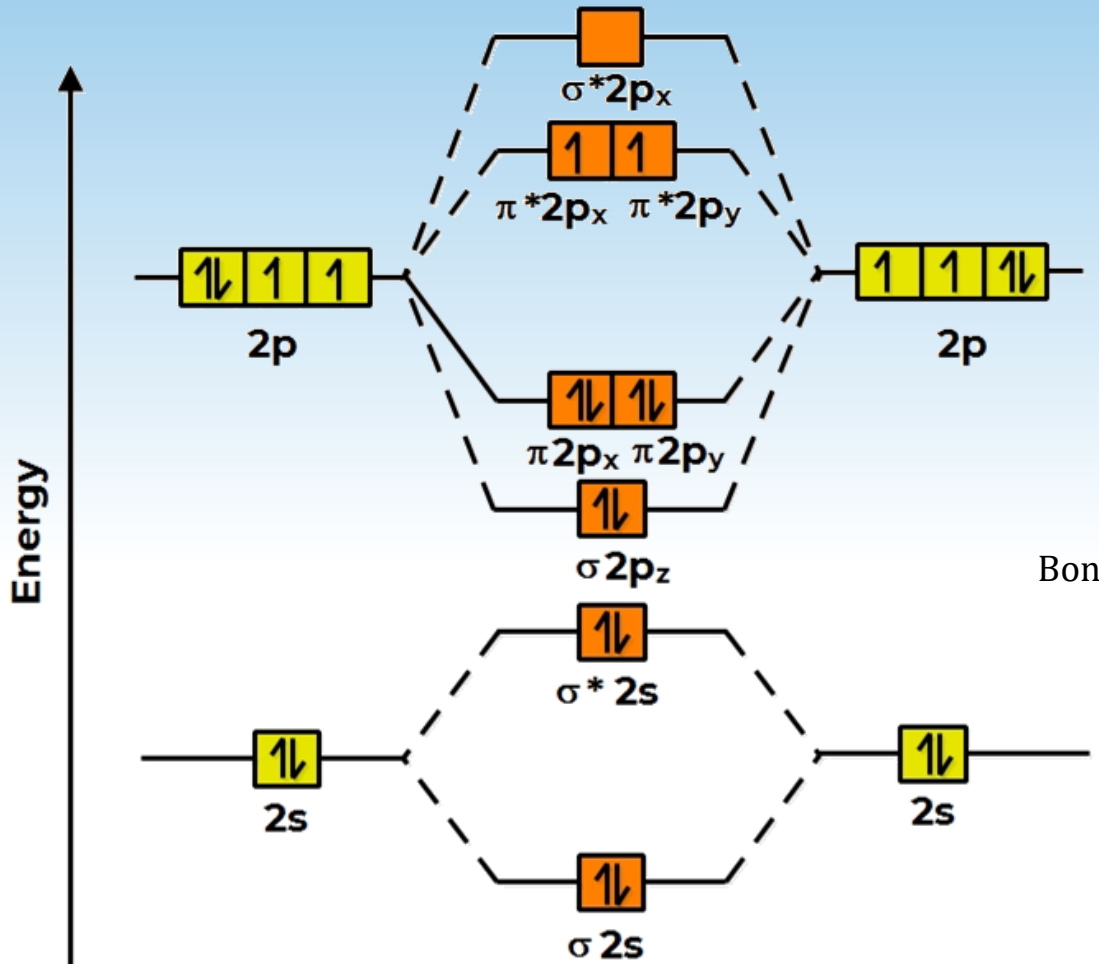
$$\text{Bond Order} = \frac{N_b - N_a}{2} \qquad = \frac{8-4}{2} = 2$$

**C = C**

C molecule is stable and it is diamagnetic

Note: Double bond in $C_2$ Consists of both $\pi$ bonds because of the presence of four electrons in two $\pi$ molecular orbitals

# Oxygen Molecule ($O_2$)




Electronic configuration of $O_2$:

$$(\sigma 1s)^2\,(\sigma^* 1s)^2\,(\sigma 2s)^2\,(\sigma^* 2s)^2\,(\sigma 2p_z)^2$$

$$\left(\pi 2p_x^2 = \pi 2p_y^2\right)\left(\pi^* 2p_x^1 = \pi^* 2p_y^1\right)$$

$$\text{Bond Order} = \frac{N_b - N_a}{2} \qquad = \frac{10-6}{2} = 2$$

**O = O**

Paramagnetic in nature

**Q.** Calculate the bond order of $N_2$ and $O_2$?

**Sol.** Electronic configuration of $N_2$ :

$$(\sigma 1s)^2 \ \ (\sigma^* 1s)^2 \ (\sigma 2s)^2 \ \ (\sigma^* 2s)^2$$
$$\left(\pi 2p_x^2 = \pi 2p_y^2\right)(\sigma 2p_z)^2$$

No. of bonding electrons, $N_b$ = 10

No. of anti-bonding electrons, $N_a$ = 4

$$\text{Bond Order} = \frac{N_b - N_a}{2} \qquad = \frac{10-4}{2} = 3$$

Electronic configuration of $O_2$ :

$$(\sigma 1s)^2 \ \ (\sigma^* 1s)^2 \ (\sigma 2s)^2 \ \ (\sigma^* 2s)^2 \ (\sigma 2p_z)^2$$
$$\left(\pi 2p_x^2 = \pi 2p_y^2\right)\left(\pi^* 2p_x^1 = \pi^* 2p_y^1\right)$$

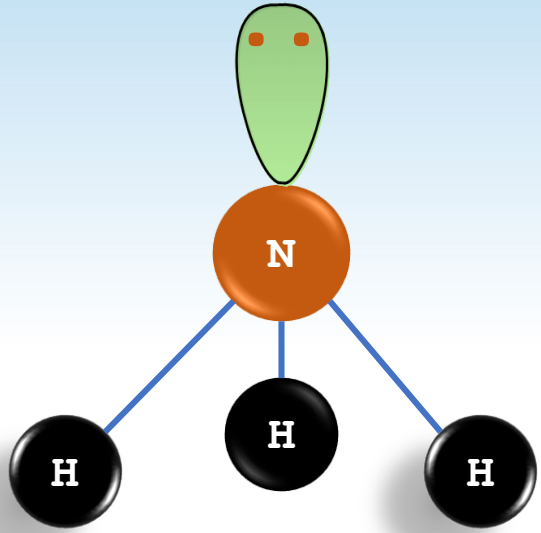
No. of bonding electrons, $N_b$ = 10

No. of anti-bonding electrons, $N_a$ = 6

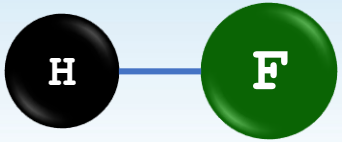
$$\text{Bond Order} = \frac{N_b - N_a}{2} \qquad = \frac{10-6}{2} = 2$$

# Hydrogen Bonding

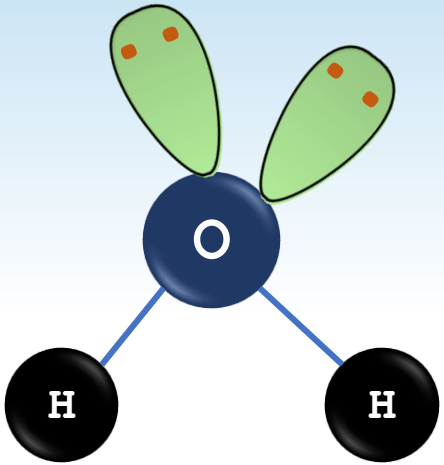
Attractive force which binds hydrogen atom of one molecule with the electronegative atom (F, O or N) of another molecule




Ammonia

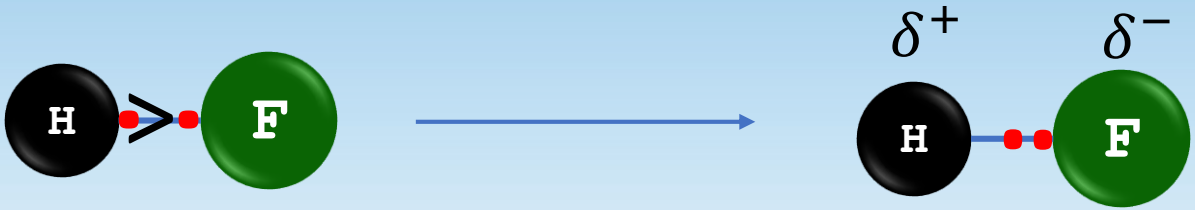



Hydrogen fluoride

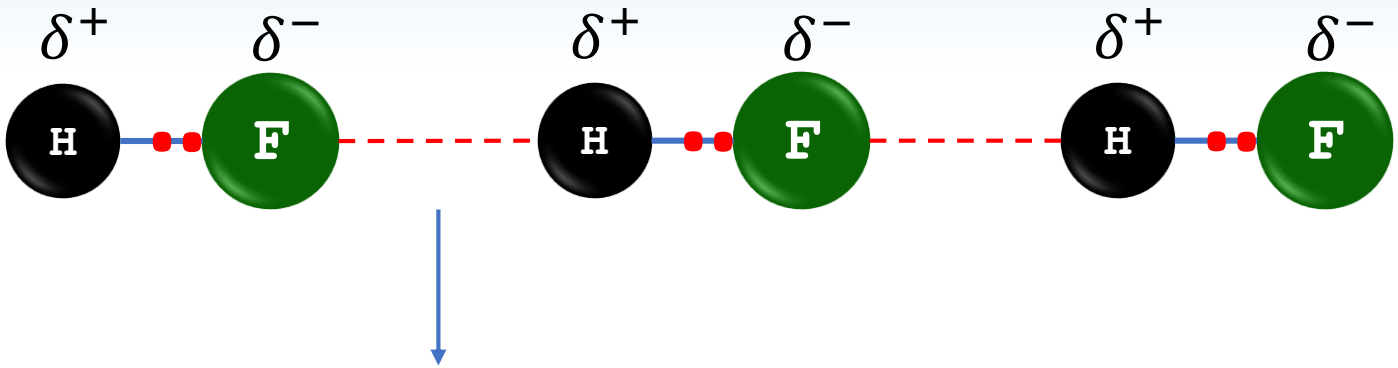



Water

# Cause of formation of Hydrogen Bond



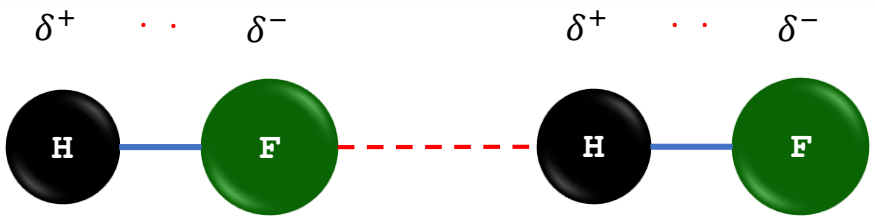

Highly electronegative



# Types of Hydrogen bonding

## Intermolecular Hydrogen bond

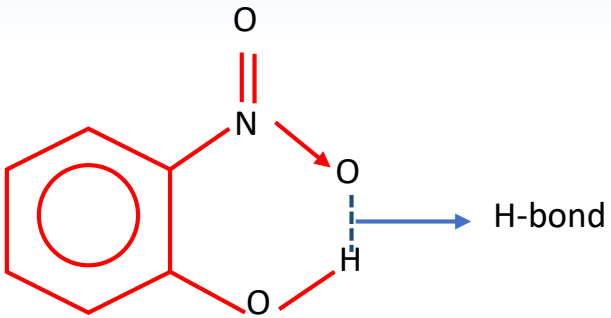
H-bond is formed between different molecules




Other e.g. alcohol and water

## Intramolecular Hydrogen bond

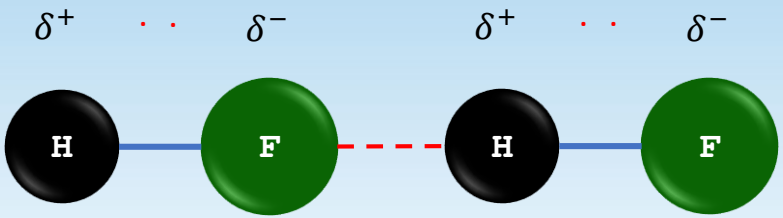
H-bond is formed within same molecules



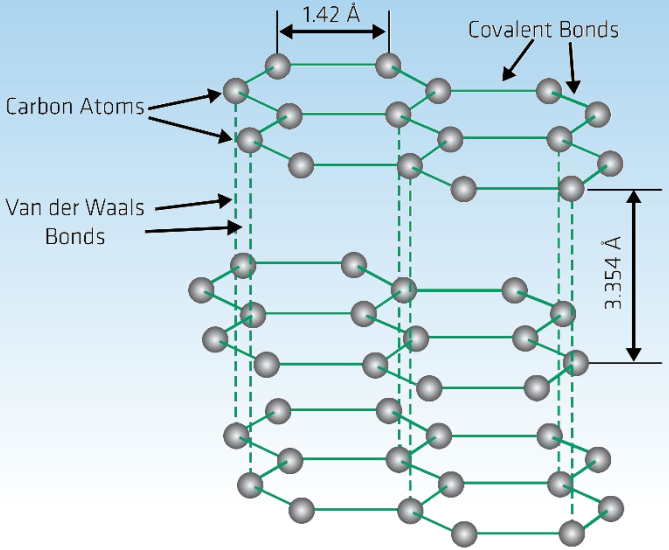

o-nitrophenol

**Q.** Is hydrogen bond weaker or stronger than the van der Waals forces?

**Sol.**




Extreme form of dipole-dipole interaction



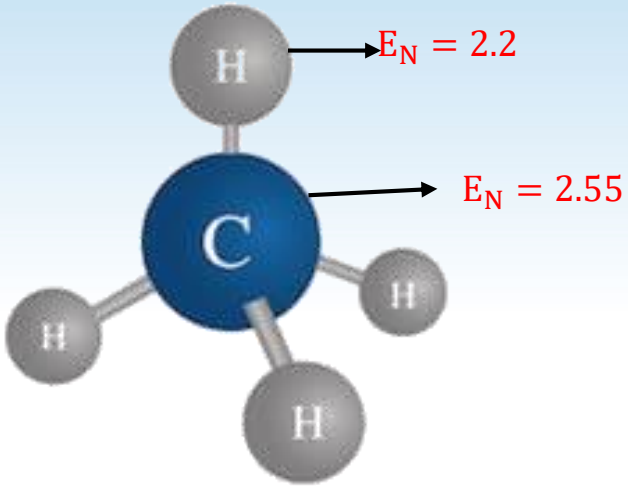

No need of dipole

Hydrogen bonds are stronger than van der Waals forces

**Q.** Which of the following molecules is not capable to form hydrogen bond ?

i. $NH_3$
ii. $H_2O$
iii. $CH_4$
iv. HF
v. $CH_3COOH$

**Sol.** $NH_3, H_2O, HF, CH_3COOH$




This makes methane a nonpolar molecule.

As a result, it is incapable of hydrogen bonding with other molecules.

## Summary

- Hydrogen molecule is diamagnetic in nature
- Helium molecule is unstable and does not exist
- Oxygen molecule has double bond and is paramagnetic in nature
- Hydrogen bond can be defined as the attractive force which binds hydrogen atom of one molecule with the electronegative atom (F, O or N) of another molecule